



Predicting Forex Pair Movements: Integrating Sentiment Analysis, Technical, and Fundamental Indicators using Machine Learning and Deep Learning Models

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Abstract

The foreign exchange (forex) market is one of the largest and most liquid financial markets globally, making accurate price prediction crucial for traders, policymakers, and financial institutions. This research focuses on improving forex price prediction by integrating sentiment analysis with technical and fundamental indicators. Historical forex data, economic indicators, and sentiment data extracted from financial news were employed to develop predictive models. The primary models used include Long Short-Term Memory (LSTM) networks, Extreme Gradient Boosting (XGBoost), ensemble models combining these approaches, and Transformer models. Large Language Models (LLMs), such as GPT-4 and GEMINI Advanced, were also utilised for sentiment analysis. The performance of these models was evaluated using comprehensive metrics, including Mean Absolute Error, Mean Absolute Percentage Error, Root Mean Square Error, and R-squared. The analysis was conducted on two different time frames: Four-hour and one-day data for forex pairs EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD. Key findings indicate that integrating sentiment analysis improves the predictive accuracy of models, particularly in the Four-hour time frame. The XGBoost model, when combined with technical, fundamental, and sentiment data, achieved the lowest Root Mean Square Error across various currency pairs, outperforming other configurations. In contrast, technical indicators alone delivered the most accurate predictions for the one-day time frame, although combining technical and fundamental indicators also resulted in solid performance. These results highlight the importance of integrating diverse data types, particularly sentiment analysis, for accurate and timely forex price prediction. The findings have important implications for traders, analysts, and policymakers, suggesting that incorporating sentiment analysis into forecasting models can enhance accuracy. Future research should explore further refinements of these models and investigate the integration of additional data sources to continue improving predictive capabilities.

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Glossary

Adam: An optimisation algorithm for training machine learning models

ASX200: Australian Securities Exchange 200 Index

ATR: Average True Range

AUD/NZD: Australian Dollar to New Zealand Dollar

AUBOT: Australian Balance of Trade

AUUR: Australian Unemployment Rate

BOT: Balance of Trade

CNN: Convolutional Neural Network

DX: US Dollar Index

EMA: Exponential Moving Average

Ensemble-LSTM-XGB FM: Ensemble model combining LSTM and XGBoost with Fundamental Indicators

Euro 50: Euro Stoxx 50 Index

EUGDP: Eurozone Gross Domestic Product

EUR/NZD: Euro to New Zealand Dollar

EUR/USD: Euro to United States Dollar

EURBOT: Eurozone Balance of Trade

EUM2: Eurozone Money Supply M2

EUM3: Eurozone Money Supply M3

EUUR: Eurozone Unemployment Rate

FA-LSTM: Fractional-Order Adaptive Long Short-Term Memory

FLF-LSTM: Forex Loss Function Long Short-Term Memory

FM: Fundamental Model

Forex: Foreign Exchange

GDP: Gross Domestic Product

GEMINI Advanced: A state-of-the-art language model from Google

GOLD CLOSE: Gold Close Price

GPT-4: Generative Pre-trained Transformer 4

GRU: Gated Recurrent Unit

GridSearchCV: A tool for hyperparameter tuning using cross-validation

LLM: Large Language Model

LLMs: Large Language Models

LSTM: Long Short-Term Memory

LSTM FM: Long Short-Term Memory with Fundamental Indicators

L2 regularisation: A technique to prevent overfitting by penalising large weights

MACD: Moving Average Convergence Divergence, a trend-following momentum indicator that shows the relationship between two moving averages of a security's price.

MAE: Mean Absolute Error

MAPE: Mean Absolute Percentage Error

ML: Machine Learning

MLP: Multilayer Perceptron

MSE: Mean Squared Error

NN: Neural Network

NLP: Natural Language Processing

NZBOT: New Zealand Balance of Trade

NZD/USD: New Zealand Dollar to United States Dollar

NZGDTPI: New Zealand Global Dairy Trade Price Index

NZGDP: New Zealand Gross Domestic Product

NZM3: New Zealand Money Supply M3

NZTA: New Zealand Tourist Arrivals

NZUR: New Zealand Unemployment Rate

NZX50: New Zealand Stock Exchange 50 Index

OHLC: Open, High, Low, Close

PSO: Particle Swarm Optimisation

R²: Coefficient of Determination

ReduceLROnPlateau: A callback in machine learning to reduce the learning rate when validation loss plateaus

RMSE: Root Mean Square Error

ROC: Rate of Change

RSI: Relative Strength Index

SA: Sentiment Analysis

S&P: Standard & Poor's

SMA: Simple Moving Average

sklearn: A machine learning library for Python

Spearman: A measure of rank correlation

StandardScaler: A standardisation tool from the sklearn library

SVM: Support Vector Machine

TensorFlow Keras: A deep learning library for Python

TM: Technical Model

train_test_split: A function from sklearn to split data into training and test sets

TRANSFORMER DE: Transformer Decoder

TRANSFORMER DE FM: Transformer Decoder with Fundamental Indicators

TRANSFORMER EN DE: Transformer Encoder-Decoder

TRANSFORMER EN DE FM: Transformer Encoder-Decoder with Fundamental Indicators

USD: United States Dollar

USD/IDR: United States Dollar to Indonesian Rupiah

USD/JPY: United States Dollar to Japanese Yen

USD Inflation Rate: United States Inflation Rate

USD Interest Rate: United States Interest Rate

USBOAT: United States Balance of Trade

USBOT: United States Balance of Trade

USM2: United States Money Supply M2

USUR: United States Unemployment Rate

XGBRegressor: A regression model from the XGBoost library

XGBoost: Extreme Gradient Boosting

XGBoost FM: Extreme Gradient Boosting with Fundamental Indicators

1 Introduction

1.1 Background Information

The foreign exchange (forex) market is one of the world's largest and most liquid financial markets, with immense daily trading activity. This market facilitates the global exchange of currencies, enabling businesses, governments, and individuals to conduct international trade and investment. The forex market's significance extends beyond currency transactions; it plays a crucial role in global economic stability, influencing everything from import and export prices to inflation rates and monetary policies. Forex trading operates around the clock, five days a week, allowing for constant participation from global financial centers such as London, New York, Tokyo, and Sydney. This continuous trading environment contributes to the market's high liquidity and volatility, presenting opportunities and risks for market participants. The decentralised nature of the forex market, without a centralised exchange, allows for a vast and diverse range of participants, including central banks, financial institutions, corporations, hedge funds, and individual investors. Accurate forecasting of forex prices is paramount for various stakeholders. For traders, precise predictions offer opportunities to maximise profits and minimise risks in a highly volatile market. Effective forex forecasting allows traders to make informed decisions about buying or selling currencies, leveraging anticipated price movements to their advantage. Policymakers rely on forex predictions to formulate economic policies that can stabilise their domestic economies. By understanding potential currency fluctuations, they can implement measures to mitigate adverse economic impacts, such as adjusting interest rates or intervening in the currency markets.

Financial institutions use these forecasts to manage their currency exposure and hedge against potential losses. Banks, investment firms, and multinational corporations often engage in forex transactions to protect their investments and operations from unfavorable currency movements. Precise forex predictions enable these institutions to devise strategies that safeguard their assets and ensure financial stability. Overall, precise forex predictions contribute to more informed decision-making and economic stability. The integration of advanced machine learning (ML) and deep learning (DL) models, combined with traditional economic indicators and sentiment analysis, enhances the ability to predict forex price movements accurately. These technological advancements provide a deeper understanding of market dynamics, allowing for more reliable forecasts and better financial planning. Furthermore, adopting sentiment analysis using large language models (LLMs) like GPT-4 and GEMINI Advanced represents a significant leap in forex forecasting. These models analyse vast amounts of unstructured data from financial news and social media, capturing market sentiment and its potential impact on currency prices. This holistic approach to forex prediction underscores the importance of integrating various data sources and analytical techniques to achieve a comprehensive and accurate forecasting model, ultimately benefiting traders, policymakers, and financial institutions.

alike. The importance of accurate forex price predictions must be balanced. Integrating sophisticated ML and DL models and advanced sentiment analysis techniques can significantly enhance forecasting accuracy, supporting more effective decision-making and global economic stability. The ability to anticipate market movements with greater precision not only aids individual traders and financial institutions but also supports broader economic policy-making and financial stability efforts.

1.2 Motivation

Forex price forecasting has witnessed significant advancements in recent years, with researchers exploring the integration of fundamental and technical analysis techniques. This has led to the development of various ML and DL models to improve the understanding and forecasting of market trends[1], [2]. Forex price forecasting has seen notable progress recently, with researchers integrating fundamental and technical analysis techniques. For instance, a study presented a deep recurrent neural network model incorporating technical indicators for forex exchange rate forecasting[3]. Another research proposed a deep learning ensemble model for financial time series forecasting that integrates fundamental data[4]. Despite these advancements, challenges in forex price forecasting still need to be addressed, such as inherent noise and volatility in forex data, robust and adaptable models, and the necessity to produce accurate predictions under diverse market conditions. The primary motivation for this research is to tackle these challenges. These studies highlight the growing interest in merging fundamental and technical analysis to improve forex price predictions. This research seeks to address the limitations of using either fundamental or technical analysis alone for forex price forecasting. The hybrid LSTM ensemble model is developed to capture a more complete view of market dynamics and provide more accurate predictions by integrating fundamental and technical analysis. One study employed a hybrid LSTM model with technical and macroeconomic indicators to forecast the directional movement of forex data[5]. Ahmed et al.[6], introduced an innovative prediction system using a Forex Loss Function to integrate fundamental and technical factors. These studies highlight the potential benefits of combining multiple elements to improve forecasting accuracy.

Despite extensive research in forex prediction, there still needs to be more, particularly in integrating multiple data sources such as technical indicators, fundamental analysis, and sentiment analysis. Existing studies often focus on more than one type of data, which limits their predictive accuracy and comprehensiveness. Predicting forex prices is inherently complex due to the market's high volatility and sensitivity to various economic indicators. Traditional methods often need to capture the nuanced influences of market sentiment and fundamental economic conditions. Integrating sentiment analysis adds another layer of complexity and holds the potential for more accurate and holistic predictions.

1.3 Goals

This research aims to improve forex price prediction by developing and assessing models that incorporate different data types: technical, fundamental, and sentiment data. The study seeks to evaluate the performance of machine learning (ML) and deep learning (DL) models using these data types both individually and in combination. By analysing how each data type affects prediction accuracy, this research will provide insights into the optimal way to integrate multiple data sources for better forex price prediction.

1.4 Research Questions

The primary research questions guiding this study are:

1. How do technical indicators compare with models, including fundamental indicators and sentiment analysis, in predicting forex prices?
2. How does sentiment analysis affect the accuracy of forex price predictions?
3. Which machine learning or deep learning models most effectively integrate these different data types for forex prediction?

1.5 Research Challenges

Predicting forex prices involves several challenges due to the market's high volatility, the influence of various economic indicators, and the integration of sentiment analysis. The inherent noise and volatility in forex data make it difficult to generate accurate predictions. Robust and adaptable models are required to handle diverse market conditions and integrate multiple data sources effectively. One significant challenge is overfitting, where models perform well on training data but fail to generalise to unseen data. Overfitting is especially problematic in financial markets due to its dynamic nature and the potential for models to learn noise rather than signal. Additionally, advanced natural language processing (NLP) techniques are necessary to derive meaningful sentiment scores from unlabelled financial news data. Addressing these challenges is crucial for developing reliable and accurate forex prediction models.

Key research challenges include:

- **High Volatility:** Forex markets are highly volatile, making it difficult to predict price movements accurately.
- **Economic Indicators:** Various economic indicators influence forex prices, and their integration into predictive models can be complex.

- **Sentiment Analysis Integration:** Combining sentiment analysis with traditional indicators poses challenges in data integration and model accuracy.
- **Noise in Data:** The inherent noise in financial data can obscure significant patterns, complicating model training and prediction.
- **Overfitting:** Ensuring models generalise well to new data is crucial, as overfitting can lead to poor performance on unseen data.

By addressing these challenges, developing more robust and accurate forex prediction models becomes achievable.

1.6 Structure of the Thesis

This thesis is structured as follows:

- **Chapter 1: Introduction** - Provides an overview of the research, outlining the objectives, significance, and methodology. It introduces integrating sentiment analysis, technical indicators, and fundamental analysis in forex prediction.
- **Chapter 2: Literature Review** - Reviews relevant literature to frame the research within the existing academic discourse. It covers previous studies on forex prediction, sentiment analysis, and the application of machine learning and deep learning models in financial forecasting.
- **Chapter 3: Methodology** - Details the methodology, including data collection, sentiment analysis using large language models (LLMs) such as GPT-4 and GEMINI Advanced, and the development of predictive models. It describes the preprocessing steps, model architectures, and evaluation metrics used.
- **Chapter 4: Results and Analysis** - Presents the results of the models on various forex pairs and timeframes. It includes performance metrics and comparative analysis, highlighting the effectiveness of different model configurations.
- **Chapter 5: Conclusion** - Summarises the findings, discusses the results implications, and suggests future research directions.

2 Literature Review

This section provides an in-depth review of the literature relevant to forex price prediction. It covers the foundational concepts and recent advancements in machine learning, deep learning, and ensemble methods, all critical to developing robust predictive models. The first subsection is into machine learning techniques, highlighting key algorithms and their applications in financial forecasting. The subsequent subsection focuses on deep learning, exploring various neural network architectures and their efficacy in handling complex time series data. The third subsection discusses ensemble methods, highlighting the benefits of combining multiple models to enhance predictive accuracy and robustness. Finally, the section on related works reviews significant studies in the field, summarising methodologies, findings, and the current state of research, thus setting the stage for the proposed methods in this study.

2.1 Literature Review Approach

- The literature was selected using a systematic approach from databases such as IEEE Xplore and Google Scholar.
- Keywords such as “forex prediction,” “machine learning in financial markets,” “deep learning for financial forecasting,” and “ensemble models in trading” were used to identify relevant studies.
- **Inclusion criteria:** Peer-reviewed articles published between 2016 and 2024, focusing on forex prediction using ML and DL, and studies that integrate technical, fundamental, and sentiment data.
- **Exclusion criteria:** Studies focusing exclusively on commodities markets, articles lacking quantitative evaluation metrics, or non-peer-reviewed sources were excluded from the review.

2.2 Machine Learning

The application of machine learning techniques in forex prediction has gained substantial attention due to their potential in handling complex and high-dimensional financial data. Various ML algorithms have been employed to enhance the accuracy of forex price forecasts, leveraging their ability to model non-linear relationships and patterns that traditional statistical methods may miss. This section reviews critical studies that have utilised machine learning approaches in the context of forex and other financial product prediction, highlighting their methodologies, results, and contributions to the field.

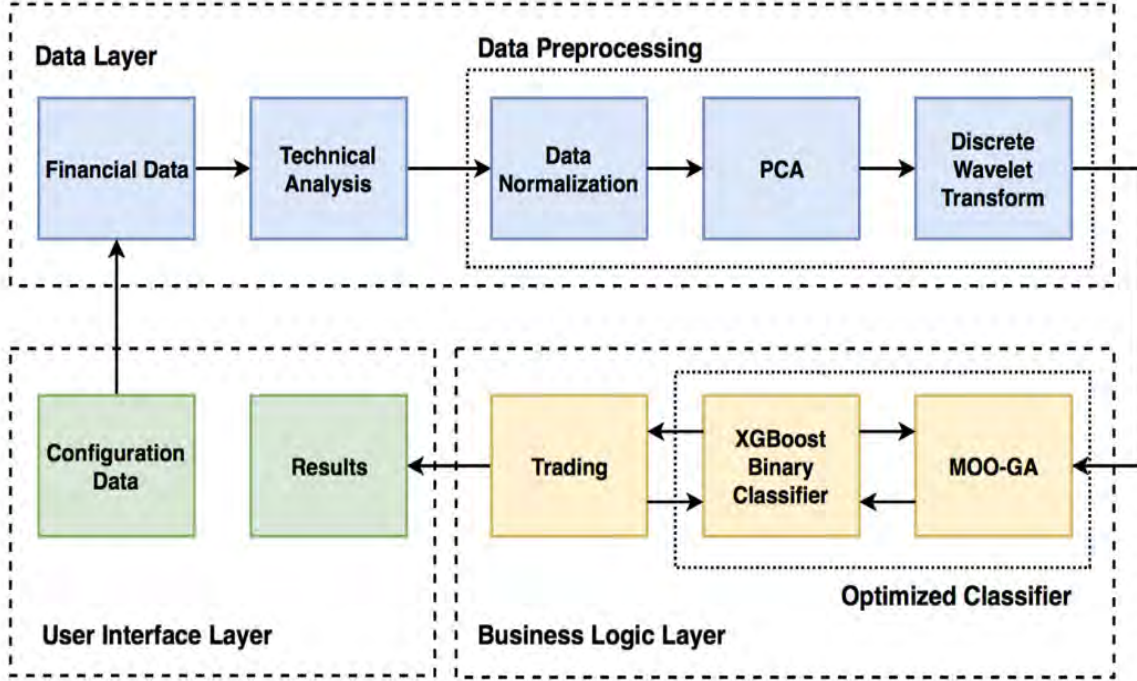


Figure 1: Framework combining PCA, DWT, and XGBoost for financial trading used by Nobre and Neves[7]).

Nobre and Neves present an innovative approach to financial trading by combining Principal Component Analysis (PCA), Discrete Wavelet Transform (DWT), and the XGBoost algorithm to create a robust trading strategy [7]. Their study focuses on optimizing trading performance in financial markets through systematic integration of these advanced data processing and machine learning techniques. By applying PCA, the authors reduce the dimensionality of market data, enhancing the model's efficiency. The DWT decomposes financial time series into a form that captures both frequency and location in time, providing a more precise signal for predictive modeling. The XGBoost algorithm, known for its high performance in classification tasks, is employed as a binary classifier to predict market trends. Their results demonstrate significant improvements in trading outcomes, suggesting that this combined approach can lead to more informed and strategic trading decisions in volatile financial environments.

Sudimanto et al. [8] conducted a pilot study demonstrating the effectiveness of ML algorithms in predicting exchange rates. Their study employed several ML techniques to forecast forex prices, including SVM and random forests. While their findings suggested significant enhancements in prediction accuracy compared to conventional methods, a notable disadvantage was the model's complexity and computational intensity, which can be challenging to implement in real-time trading environments. The framework used in their study, as depicted in Figure 2, includes several key steps: starting with foreign exchange data for XAU/USD, followed by data preprocessing and the application of the MACD indicator, and finally, the deployment of various machine learning models (TREE, SVM, Ensemble Tree, K-NN) for prediction and evaluation.

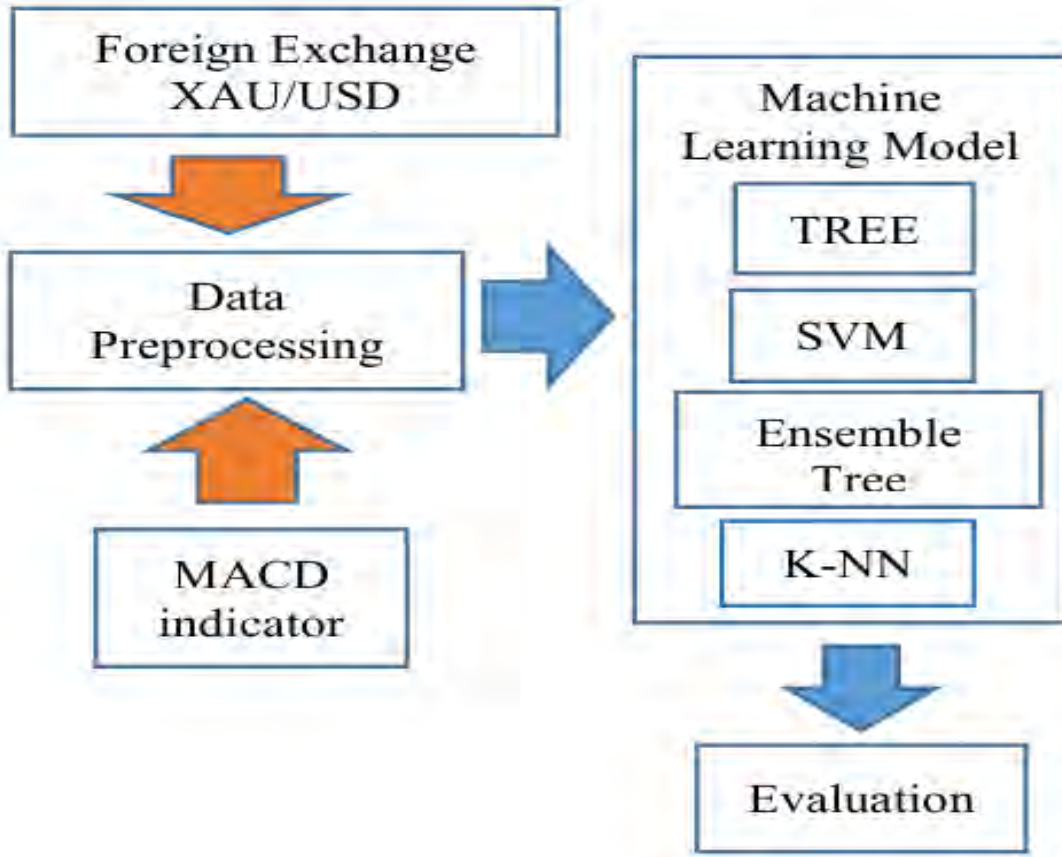


Figure 2: Framework of Forex Prediction Model Employed from Sudimanto et al. [8]

In another study, McNally et al. [9] explored the use of ML for predicting Bitcoin prices, offering insights into the applicability of ML techniques in financial forecasting beyond traditional forex markets. They utilised a range of algorithms, such as linear regression, decision trees, and neural networks, to predict Bitcoin prices. Their results indicated that machine learning models outperformed standard forecasting methods, especially those combining multiple algorithms. However, the study faced limitations in terms of overfitting and the need for extensive parameter tuning, which could limit the model's robustness in volatile market conditions.

Basak et al. [10] examined the direction of stock market prices using tree-based classifiers. Although their focus was on stock markets, their methodologies and findings are relevant to forex prediction. The study employed decision trees, random forests, and gradient boosting machines (GBM) to classify the direction of price movements. The results highlighted the robustness of tree-based classifiers in capturing market trends and making accurate predictions. Despite these strengths, the primary disadvantage was the model's sensitivity to noisy data and tendency to overfit when applied to highly volatile financial markets like forex. Thu and Xuan [11] investigated the use of SVM in forex prediction. Their research demonstrated that SVM could effectively handle forex data's non-linear and high-dimensional nature, improving prediction accuracy. The study emphasised the importance of selecting appropriate

kernel functions and tuning hyperparameters to enhance the model's performance. However, a significant drawback was the SVM's computational expense and the complexity of optimising hyperparameters, which can hinder its practicality for high-frequency trading applications.

Neghab et al. [12] integrated macroeconomic fundamentals with machine learning to explain exchange rate forecasts. The study provided a deeper understanding of how macroeconomic indicators influence exchange rates by employing interpretive ML techniques. Their research underscored the potential of combining fundamental analysis with machine learning to improve the interpretability and accuracy of forex predictions. Despite the promising results, the main disadvantage was the reliance on accurate and timely macroeconomic data, which can be challenging to obtain and may introduce latency in prediction updates.

Chantarakasemchit and Nuchitprasitchai [13] enhanced forex rate prediction by integrating machine learning with moving average methods and financial factors. Their research demonstrated improved predictive accuracy for the EUR/USD pair, highlighting the effectiveness of combining technical analysis with ML. However, the study's limitation was the potential for overfitting due to the complex model configurations required. Chen et al. [14] applied machine learning models to predict the Japanese Yen to US Dollar exchange rate. They used various techniques, including neural networks and support vector machines, demonstrating that these models could effectively forecast forex rates. The primary disadvantage noted was the high computational cost associated with training and validating these models.

De Prado [15] discussed advances in financial machine learning, highlighting various sophisticated techniques for improving predictive performance in financial markets. His work emphasises the importance of feature engineering, model selection, and validation techniques to enhance the robustness and accuracy of machine learning models. De Prado introduced novel methodologies, such as fractional differentiation, to make financial time series stationary while preserving memory, which is crucial for practical model training. He also explored ensemble methods and advanced techniques like meta-labeling to improve the predictive power of trading strategies.

The research consistently shows that XGBoost models deliver promising results compared to other machine learning models due to their ability to handle large datasets and capture complex patterns effectively. However, machine learning models can need help with capturing long-term dependencies and non-linear relationships inherent in financial time series data. Moreover, ML models often require extensive feature engineering and can be prone to overfitting, especially in highly volatile markets. To overcome these limitations, deep learning techniques offer several advantages. Deep learning models, particularly neural networks like LSTM and GRU, excel in handling sequential data and can automatically learn complex features without manual feature engineering. These models can more effectively capture long-term dependencies and temporal patterns, providing better predictive accuracy for financial time series. Additionally, deep learning models can integrate various types of data, such as technical indicators, fundamental factors, and sentiment analysis, to provide a more comprehensive forex

market analysis. The following section is on deep learning techniques and their application in forex prediction.

2.3 Deep Learning

Deep learning has become a potent tool in financial forecasting, especially for predicting forex prices. Its ability to model complex, non-linear relationships and capture long-term dependencies makes it ideal for managing the volatile and dynamic nature of financial markets. This section reviews significant studies that have utilised deep learning techniques for forex prediction, detailing their methodologies, results, and contributions. A comprehensive survey on forex and stock price prediction using deep learning techniques highlighted various DL models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), including LSTM and Gated Recurrent Unit (GRU) networks [16]. The advantages of deep learning over traditional machine learning methods, particularly in capturing temporal dependencies and complex patterns in financial time series data, were discussed. Another study demonstrated the application of LSTM networks for financial market predictions, explicitly targeting the forex market. The research revealed that LSTMs could effectively learn from sequential data, providing superior prediction accuracy compared to traditional models [17].

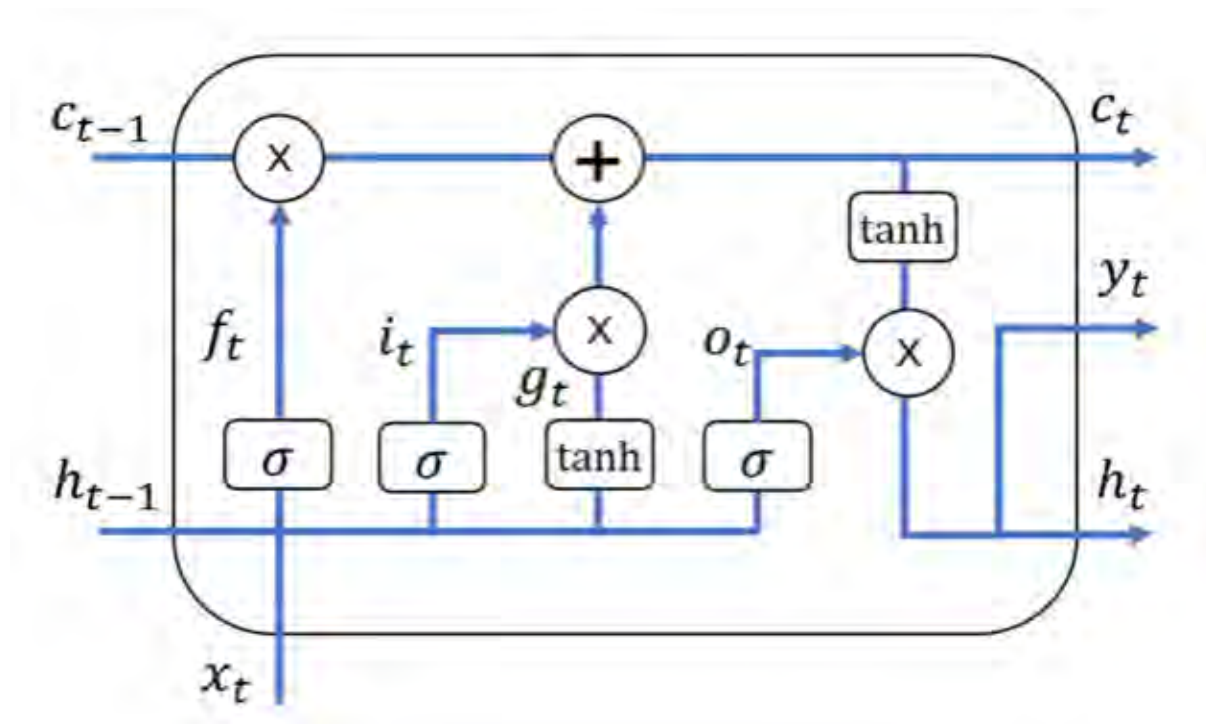


Figure 3: LSTM Cell Architecture (obtained from [18]).

The LSTM was trained on historical forex data and performed well in forecasting future price movements, validating the model's robustness and effectiveness in financial predictions. Research exploring the use of GRU networks for financial sequence predictions, including forex

prices, showed that GRU networks, with their simplified architecture compared to LSTM, could still capture essential temporal dependencies and deliver high prediction accuracy [19]. The study highlighted the efficiency of GRUs in training time and computational resources while maintaining predictive performance, making them a valuable alternative in financial forecasting. Another study focused on share price prediction using time-series data and deep learning approaches, extending the methodology to the forex market. The LSTM networks employed in this research demonstrated the model's capability to handle financial data's noisy and volatile nature, outperforming traditional forecasting techniques in terms of accuracy and robustness [20].

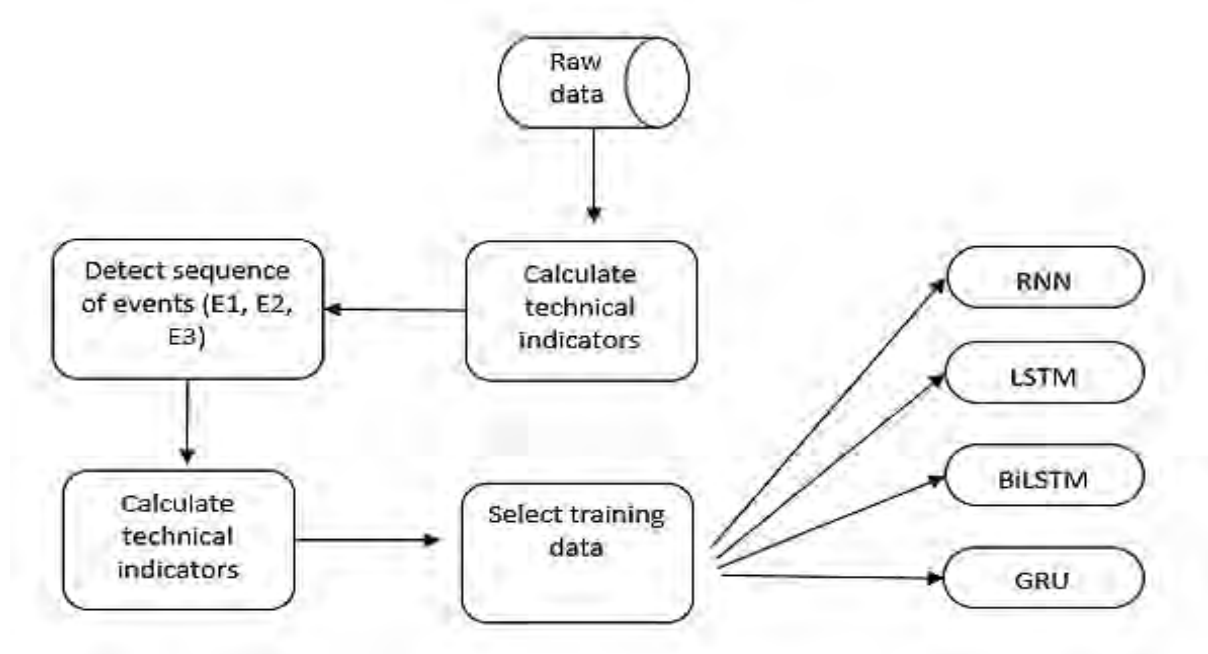


Figure 4: Process of prediction system from Qi et al. [21]

As Figure 4 shows, an Event-Driven LSTM model for forex price prediction was introduced, incorporating external events into the prediction process. The study highlighted the importance of considering market-moving events in forex forecasting, such as economic announcements and geopolitical developments [21]. The Event-Driven LSTM model showed improved prediction accuracy by integrating these external factors, showcasing the adaptability of deep learning models to incorporate diverse data sources for enhanced financial predictions. Researchers have significantly improved prediction accuracy by leveraging advanced neural network architectures such as LSTM and GRU. These models capture the intricate patterns and long-term dependencies in financial time series data. Moreover, integrating event-driven data further enhances the model's ability to respond to external market influences, providing a more comprehensive and nuanced approach to forex price prediction.

The research by Bilokon et al. compares the performance of Transformer models and LSTM networks in predicting financial time series data [22]. The study focuses on three pri-

mary tasks: mid-price prediction, mid-price difference prediction, and mid-price movement prediction using Limit Order Book (LOB) data. The results indicate that while LSTMs have been widely used and tested across various datasets, Transformers show promising potential in financial applications due to their ability to handle longer sequences and capture complex patterns more effectively. Another study by Fisichella et al. examines the application of deep learning models, specifically LSTMs, in enhancing the accuracy of forex price predictions [23]. The study demonstrates that LSTMs can effectively capture the temporal dependencies and complex patterns inherent in forex data, resulting in superior predictive performance compared to traditional technical analysis methods. This research underscores the importance of incorporating advanced neural network architectures to improve the reliability and accuracy of financial forecasting.

Markova [24] explored using CNNs for forex time series forecasting. The study found that CNNs could capture intricate patterns in financial data, leading to accurate predictions. However, the significant drawback was the substantial data preprocessing required, which can be time-consuming and complex. Zhao and Khushi [25] proposed a method combining wavelet denoised-ResNet CNN and LightGBM to predict forex rate changes. Their approach showed substantial improvements in prediction accuracy. However, integrating these complex models required extensive computational resources and fine-tuning, which could limit practical applications. Naeem et al. [26] used sentiment analysis of Twitter data to forecast the USD/PKR exchange rate with machine learning. The study demonstrated that incorporating sentiment analysis could enhance prediction accuracy. The main disadvantage was the reliance on social media data, which can be noisy and may only sometimes reflect actual market conditions.

The following section will explore the Ensemble models, discussing their methodologies, applications, and advantages in improving predictive accuracy in forex price forecasting.

2.4 Ensemble

The integration of multiple ML and DL models into an ensemble has shown promising results in financial forecasting. Ensemble models combine the strengths of different algorithms to enhance predictive performance and generalisability, addressing the limitations of individual models. Ji et al. [27] introduced a novel hybrid model combining Particle Swarm Optimisation (PSO) with LSTM networks to forecast stock indices. The PSO algorithm optimises the LSTM hyperparameters, improving the model's predictive capabilities. This hybrid approach leverages PSO's efficiency in searching the solution space and LSTM's ability to capture temporal dependencies in financial time series data. Their findings demonstrated that the PSO-LSTM model significantly outperforms traditional and standalone LSTM networks, indicating its potential for superior accuracy in stock market predictions.

Ghosh et al. [28] investigate the effectiveness of using a hybrid model that combines LSTM networks and random forests for predicting the directional movements of stock prices in

intraday trading. The authors developed a model that leverages the sequential learning capabilities of LSTM networks to capture temporal dependencies in stock price data and the robustness of random forests to handle non-linear relationships and interactions between features. This hybrid approach aims to improve the accuracy and reliability of stock price direction predictions. The study found that the hybrid LSTM-random forest model outperformed traditional methods in predicting the direction of stock price movements. This improvement is attributed to the model's ability to capture temporal patterns and complex feature interactions. The model is particularly suited for intraday trading, where accurate short-term predictions are crucial for profitable trading decisions.

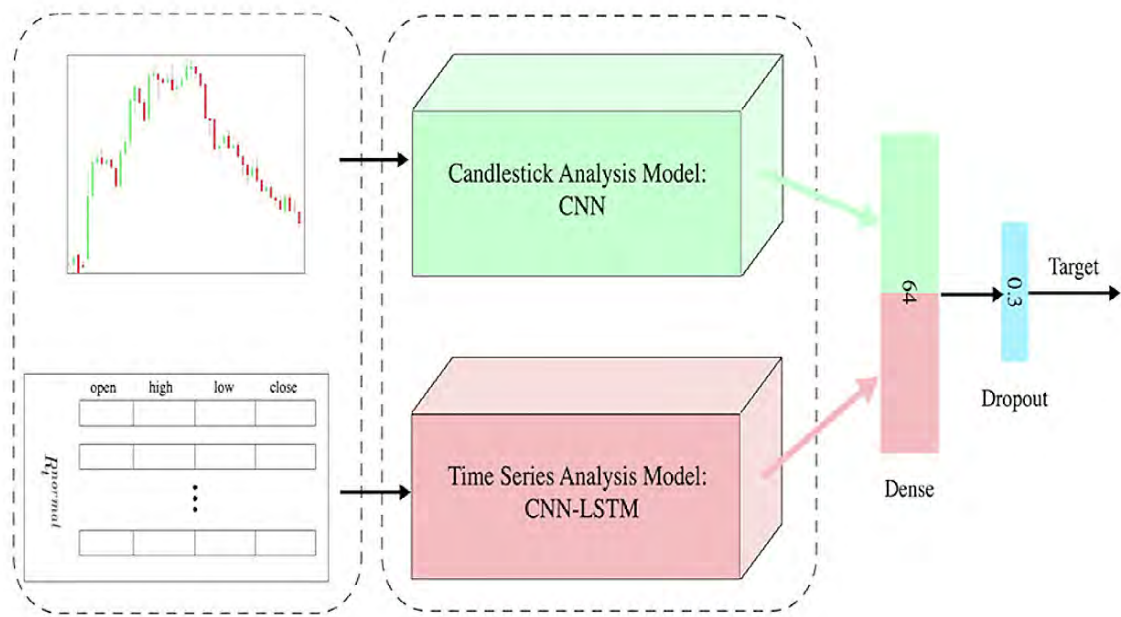


Figure 5: The architecture of the proposed hybrid model by Moghaddam et al. [29]

Moghaddam and Momtazi [29] proposed a hybrid model that integrates image processing techniques with time series analysis to predict profitable technical pattern positions in the forex market. Their approach leverages the strengths of both CNNs and LSTM networks. The CNN analyses candlestick chart patterns, while the CNN-LSTM combination captures temporal dependencies in the time series data. The architecture of this hybrid model is depicted in Figure 5. In the model, raw data is preprocessed and then fed into two separate analysis paths. The candlestick analysis model uses CNN to extract features from the candlestick charts. Simultaneously, the time series analysis model, which combines CNN and LSTM, processes the sequential data. Both model's outputs are concatenated and passed through a dense layer followed by a dropout layer to prevent overfitting, ultimately providing the target predictions. This innovative approach demonstrates the potential of combining image processing with deep learning for enhanced predictive accuracy in financial markets.

Escudero et al. [18] compared RNN and Autoregressive Integrated Moving Average

(ARIMA) models for forecasting the Euro/Dollar exchange rate. They developed a hybrid model integrating ARIMA for trend and seasonality detection with RNN to capture non-linear temporal dynamics. The combined ARIMA-RNN model exhibited enhanced predictive accuracy over standalone models, suggesting its effectiveness in forex exchange rate forecasting. Wang and Guo [30] proposed a mixed model combining ARIMA and XGBoost to forecast stock market volatility. The ARIMA component captures linear trends, while XGBoost models the non-linear aspects of the time series data. Their hybrid approach demonstrated superior forecasting accuracy and robustness compared to traditional models, highlighting its potential for improved stock market volatility predictions.

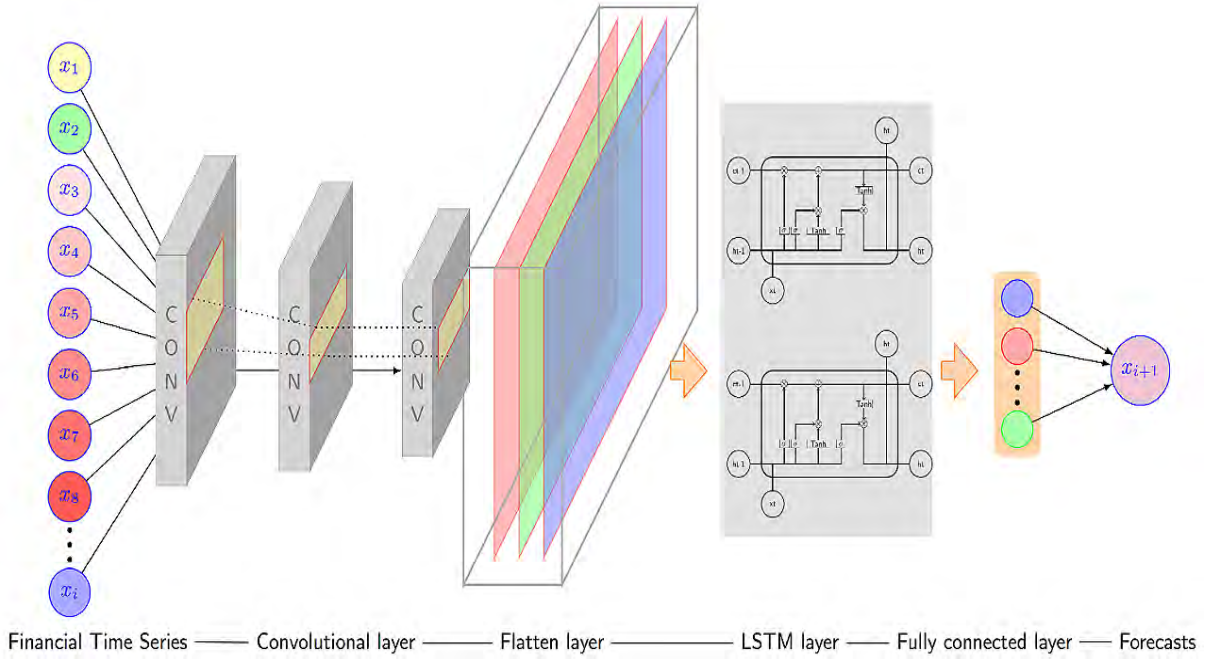


Figure 6: CNN-LSTM network structure used by He et al. [4]

He et al. [4] propose a deep learning ensemble model for financial time series forecasting, integrating fundamental and technical data. The authors use a dataset comprising historical financial time series data, including forex pairs like EUR/USD and GBP/USD, sourced from financial databases and market data providers. The ensemble approach combines multiple deep-learning techniques to enhance predictive accuracy and robustness. The study shows that the ensemble model outperforms individual models regarding prediction accuracy and generalisation to various market conditions. The research validates the effectiveness of using ensemble strategies in financial forecasting by leveraging diverse data sources and model architectures. The network structure used by the authors involves a CNN-LSTM architecture shown in Figure 6, where the CNN layers extract features from the input data, and the LSTM layers capture the temporal dependencies. This combination leverages the strengths of CNNs in handling spatial features and LSTMs in modeling sequential data, leading to improved forecasting performance.

These studies underscore the advantages of using ensemble models in financial fore-

casting. By integrating the strengths of different algorithms, ensemble models can achieve higher accuracy and better generalisation than individual models.

2.5 Related Research

Forex prediction has seen extensive research efforts to improve the forecasting model's accuracy and reliability. These efforts span various methodologies, including technical analysis, fundamental analysis, and the integration of sentiment analysis. By utilising ML and DL techniques, researchers have aim to identify complex patterns and dependencies in financial data. This offers an overview of important related research, emphasising key studies that have advanced forex prediction. The review covers works employing machine learning algorithms, deep learning models, ensemble methods, and hybrid approaches, illustrating the diverse strategies used to address the challenges in forex forecasting. The studies reviewed here highlight the importance of combining different analytical methods to enhance the predictive performance of forex models, reflecting the multi-faceted nature of financial markets.

Dautel et al. [3] explore the application of deep RNNs, such as LSTM networks and GRUs, for forecasting foreign exchange rates. The authors systematically compare these advanced architectures with traditional recurrent and feedforward networks. The dataset includes historical forex data for major currency pairs like EUR/USD, GBP/USD, and USD/JPY, collected from reputable financial databases. The study finds that deep learning models, particularly LSTMs and GRUs, show superior directional forecasting accuracy and trading profitability performance. However, the authors note the challenges in implementing and tuning these complex architectures, suggesting that simpler neural networks can sometimes achieve comparable results. The research underscores the potential of deep learning in forex forecasting while acknowledging the practical difficulties in model deployment.

This paper by Ahmed et al. [6] introduces a novel prediction system called FLF-LSTM, which incorporates a specialised Forex Loss Function to improve the accuracy of Forex price predictions. The dataset includes historical forex data for pairs like EUR/USD, GBP/USD, and AUD/USD obtained from financial data providers. By integrating fundamental and technical indicators, the model leverages the strengths of LSTM networks in handling sequential data and capturing complex temporal dependencies. The Forex Loss Function is designed to enhance the model's sensitivity to forex market dynamics, resulting in improved predictive performance compared to traditional loss functions.

Qi et al. [21] present an Event-Driven LSTM model for forex price prediction that incorporates external market-moving events into the forecasting process. The dataset includes historical forex price data for pairs such as EUR/USD and USD/JPY and event data like economic announcements and geopolitical developments sourced from financial news outlets and economic calendars. By integrating these event data into the LSTM framework, the model enhances its ability to predict forex price movements. The study demonstrates that considering

external events dramatically improves the prediction accuracy of LSTM models, highlighting the importance of incorporating diverse data sources in financial forecasting.

Sudimanto et al. [8] conducted a pilot study demonstrating the effectiveness of machine learning algorithms in predicting exchange rates. Their study employed machine learning techniques to forecast forex prices, including SVM and random forests. The findings indicated that these models could significantly enhance prediction accuracy compared to conventional methods.

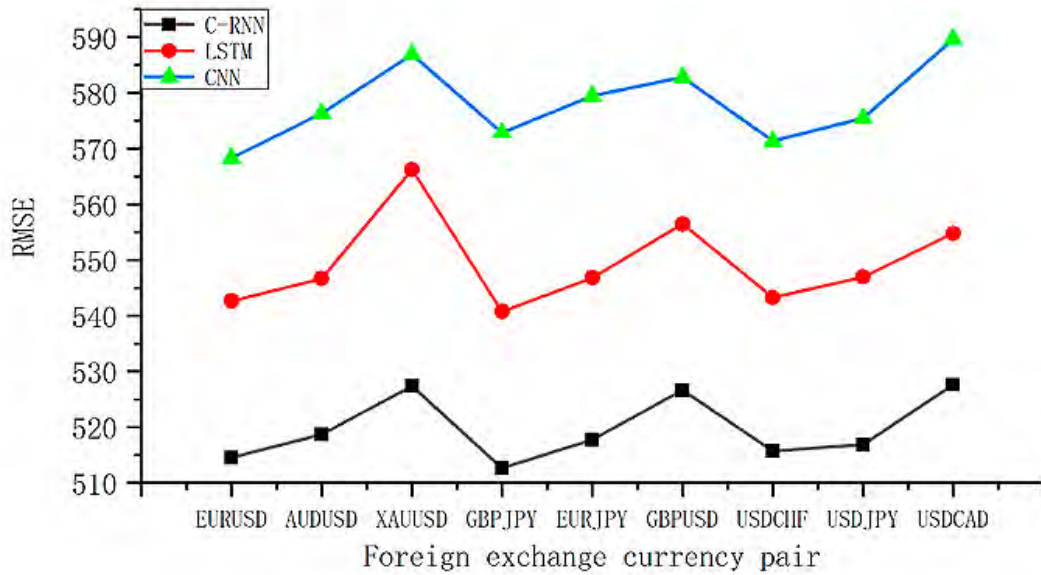


Figure 7: Comparison of RMSE of predicted values by different forecasting algorithms [31].

Ni et al. [31] explore the application of deep learning techniques for forecasting forex time series data. The study specifically focuses on using LSTM networks to predict future forex prices. The researchers collected historical data for major forex pairs and utilised the LSTM model to capture temporal dependencies in the data. Their findings indicate that LSTM networks significantly outperform traditional statistical models regarding predictive accuracy and the ability to handle non-linear relationships in the data. However, the primary limitation of this study is the computational complexity associated with training LSTM networks, which requires extensive hyperparameter tuning and significant computational resources. The image in Figure 7 provides a visual comparison of the RMSE values for different forecasting algorithms—C-RNN, LSTM, and CNN—across various foreign exchange currency pairs, including EUR/USD, AUD/USD, XAU/USD, GBP/JPY, EUR/JPY, GBP/USD, USD/CHF, USD/JPY, and USD/CAD. The RMSE values for each algorithm are plotted on the y-axis, with lower RMSE values indicating better predictive performance. The LSTM model (represented by red dots) consistently demonstrates lower RMSE values compared to CNN (green triangles) and C-RNN (black squares), highlighting its superior accuracy in forecasting forex prices. In the

research conducted by Vuong et al. [32], the authors investigate the combination of XGBoost and LSTM models for stock price forecasting. The study leverages the strengths of XGBoost in handling structured data and the sequential learning capabilities of LSTM networks. The hybrid model aims to improve the accuracy of stock price predictions by capturing both linear and non-linear relationships in the data. The results show that the hybrid XGBoost-LSTM model outperforms the individual model's prediction accuracy and robustness. However, one of the main limitations is the increased complexity and computational cost associated with combining two different machine learning models, which requires careful parameter tuning and significant computational resources.

Islam and Hossain [33] propose a hybrid network combining GRU and LSTM networks for predicting foreign exchange currency rates. The study aims to leverage the strengths of both GRU and LSTM in capturing temporal dependencies and non-linear patterns in the data. The hybrid GRU-LSTM network is trained on historical forex data, and the results indicate significant improvements in prediction accuracy compared to traditional models. The main limitation is the increased computational complexity and training time due to combining two deep learning models, necessitating extensive hyperparameter tuning. Wang et al. [34] develop a prediction model using a combination of CNN and TLSTM (Time-variant LSTM) for forecasting the USD/CNY exchange rate. The CNN component extracts features from the input data, while the TLSTM network captures temporal dependencies. The study demonstrates that the CNN-TLSTM model outperforms traditional and standalone deep learning models in terms of predictive accuracy. The main limitation of the CNN-TLSTM model is its complexity and the computational resources required for training, which necessitates extensive hyperparameter tuning and careful selection of input features.

Tripathi et al. [35] examine the use of ensemble modeling for exchange rate forecasting. The study combines multiple machine learning models to improve prediction accuracy and robustness. The ensemble approach leverages the strengths of different models and mitigates individual model weaknesses. The results indicate that the ensemble model provides better predictive performance and is more resilient to market volatility. The primary limitation of ensemble modeling is the increased computational cost and complexity associated with combining multiple models, which requires significant computational resources and time for training and validation. Meanwhile, Alonso-Monsalve et al. [36] investigate using CNN for high-frequency trend prediction of cryptocurrency exchange rates using technical indicators. The study leverages the feature extraction capabilities of CNNs to identify patterns in high-frequency trading data. The results show that the CNN model outperforms traditional models and provides accurate predictions for cryptocurrency trends. The main limitation of the CNN model is the computational complexity and resources required for training, especially for high-frequency data, necessitating extensive hyperparameter tuning.

The paper by Peng and Lee [37] focuses on mitigating overfitting in forex trading models. The authors propose a data selection strategy to enhance model robustness by carefully

choosing relevant data points that reflect genuine market conditions rather than noise. This approach aims to improve the predictive performance of machine learning models in the highly volatile forex market. Their methodology was tested using intraday trading data, demonstrating significant model accuracy and reliability improvements. The paper by Dassanayake [38] provides an in-depth analysis of different forecasting models. Dassanayake compares traditional statistical methods with modern deep learning techniques to predict the movements of the New Zealand Stock Market Index. The study highlights the strengths and weaknesses of each approach, showing that deep learning models, especially those designed to capture temporal dependencies, outperform statistical models in terms of accuracy and handling market volatility. The research uses historical stock market data from New Zealand to validate the models, underscoring the practical applications of advanced forecasting techniques in financial markets.

A survey conducted by Hu et al. [39] explores various deep learning models applied to forex and stock price prediction. The study analyses deep learning techniques, including CNNs, LSTM networks, Deep Neural Networks (DNNs), RNNs, and Reinforcement Learning. The survey classifies these models based on their methodologies and performance metrics such as RMSE, MAPE, MAE, MSE, accuracy, Sharpe ratio, and return rate. The findings indicate that combining LSTM with other models, such as DNN, shows promising results. Reinforcement learning also demonstrated substantial returns and performance in financial modeling, reflecting an increasing trend in employing deep learning methods for financial predictions. The paper by Lu [40] examines the application of LSTM networks to forecast the GBP/CNY exchange rate. The study highlights the advantages of LSTM in capturing long-term dependencies and non-linear relationships in financial time series data. By training the LSTM model on historical exchange rate data, the research demonstrates its ability to produce accurate and reliable forecasts, emphasising its robustness in handling the volatile nature of forex markets.

Research by Mesleh and Kiranyaz [41] compares the effectiveness of Multilayer Perceptron (MLP) and LSTM models in predicting future forex prices. The study involves training both models on historical forex data and evaluating their performance based on prediction accuracy. The findings suggest that LSTM models, with their capability to understand sequential data, outperform MLP models, providing more accurate forecasts for forex price movements. The paper by Ulina et al. [42] explores the use of Complementary Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) and an Improved Fractional-Order Adaptive LSTM (FA-LSTM) model for forex prediction. The study focuses on the USD/IDR currency pair and demonstrates that integrating CEEMDAN with FA-LSTM enhances the model's ability to handle non-stationary and noisy financial data, leading to improved prediction accuracy. The study by Ranjit et al. [43] compares various algorithms for predicting foreign exchange rates, including traditional statistical methods and advanced machine learning models. The research evaluates the performance of these algorithms based on their accuracy and computational efficiency, highlighting the strengths and weaknesses of each approach. The findings

provide valuable insights into selecting appropriate models for forex prediction under different market conditions.

Ni et al. [44] explored the effectiveness of deep learning models for forecasting forex time series data. Their study utilised LSTM networks to capture the intricate patterns in financial time series data. The researchers collected data from various forex pairs, including EUR/USD and GBP/USD, and demonstrated that LSTM models significantly outperformed traditional forecasting methods, providing more accurate and reliable predictions of future forex prices. Shiao et al. [45] conducted a case study using various machine learning models to predict forex prices. They compared the performance of Multiple Linear Regression (MLR), Decision Trees (DT), and Neural Networks (NN) using forex data from multiple currency pairs. The study concluded that neural networks, particularly LSTM models, provided the best predictive accuracy due to their ability to model complex temporal dependencies in the data. Neghab et al. [46] investigated using machine learning models to explain exchange rate forecasts with macroeconomic fundamentals. The study applied interpretable machine learning techniques to a dataset comprising key macroeconomic indicators from significant economies. The findings highlighted the importance of incorporating economic fundamentals into predictive models to improve the accuracy and interpretability of forex forecasts.

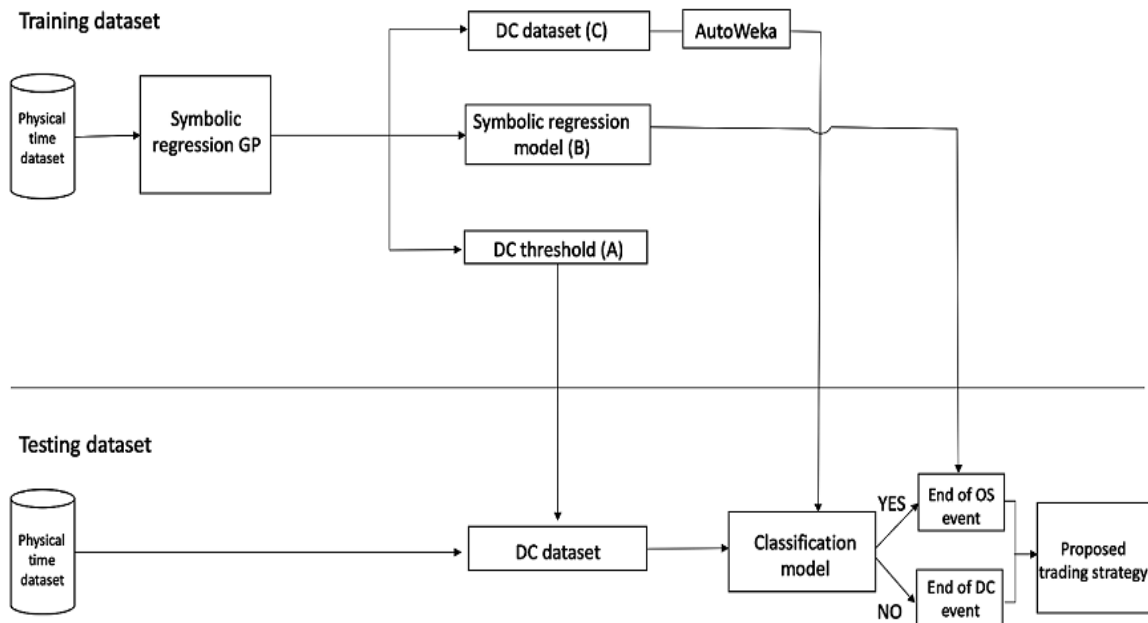


Figure 8: Proposed methodology by Adegboye and Kampouridis [47].

Figure 8 describes the proposed methodology used by Adegboye and Kampouridis for their forex prediction model. The methodology is divided into two main phases: training and testing. The physical time dataset is first input into a symbolic regression genetic programming (GP) model in the training phase. This model generates a symbolic regression model (B) and determines the directional change (DC) threshold (A). The physical time dataset is

also transformed into a DC dataset (C). AutoWeka is then employed to optimize the parameters and select the best classification model. The physical time dataset is again transformed into a DC dataset in the testing phase. This dataset is fed into the optimized classification model. The model predicts whether the end of an overshoot (OS) event or a DC event has been reached. Based on these predictions, a proposed trading strategy is developed, guiding trading decisions. Adegboye and Kampouridis [47] examined various machine learning models, including classification and regression techniques, to predict directional changes and trend reversals in forex markets. Their research focused on identifying key features that influence market movements and demonstrated that ensemble learning methods, such as Random Forests and Gradient Boosting, significantly enhanced predictive performance compared to individual models. Filippou et al. [48] analysed the impact of media sentiment on currency reversals. The study utilised sentiment analysis on a vast corpus of financial news articles and correlated the sentiment scores with currency exchange rate movements. The results indicated that negative media sentiment often preceded currency reversals, suggesting that integrating sentiment analysis into forex prediction models can improve their accuracy and responsiveness to market events.

The study by Rognone et al. [49] examines the impact of news sentiment on the cryptocurrency market compared to the forex market. Using sentiment analysis on news articles, they discovered significant differences in how sentiment affects these markets. The research highlighted that while news sentiment strongly influences both markets, the cryptocurrency market showed higher sensitivity due to its speculative nature. Farimani et al. [50] delved into leveraging latent economic concepts and news sentiments for market prediction. They employed advanced sentiment analysis and machine learning techniques to forecast market movements. Their findings indicated that incorporating latent economic concepts alongside sentiment scores significantly improved predictive accuracy across various financial markets, including forex. Semiromi et al. [51] analysed the predictive power of news story events listed in economic calendars on forex rates. This research utilised a dataset of economic news events and applied machine learning models to forecast forex rates. The results demonstrated that news story events have substantial predictive power, particularly when unexpected or economically significant. Song and Choi [52] compared various hybrid models for forecasting stock market indices, including CNN-LSTM, GRU-CNN, and ensemble models. Their study revealed that these hybrid models, which combine CNNs and RNNs, outperformed traditional models in capturing complex patterns and temporal dependencies in financial time series data.

In their research, Livieris et al. [53] proposed a CNN-LSTM model for forecasting gold prices. This model integrates CNN's feature extraction capabilities with LSTM's sequence modeling strengths, demonstrating superior performance in predicting gold price movements compared to traditional forecasting methods. The study utilised historical gold price data and validated the model's effectiveness in forecasting financial time series. Paliari et al. [54] conducted a comparison of optimised LSTM, XGBoost, and ARIMA models in time series fore-

casting. Using various datasets, including forex data, they found that LSTM and XGBoost models generally outperformed ARIMA in capturing complex temporal dependencies, with LSTM showing particular strength in long-term predictions. This study highlights the advantages of using advanced machine learning techniques over traditional statistical models for time series forecasting. Pornwattanaichai et al. [55] introduced BERTFOREX, a cascading model for forex market forecasting that integrates fundamental and technical indicators using the Bidirectional Encoder Representations from Transformers (BERT) framework. The study demonstrated that BERTFOREX effectively captures contextual dependencies within the data, improving prediction accuracy for forex markets. The model outperformed traditional methods, showcasing the potential of transformer-based architectures in financial forecasting. Wood et al. [56] presented the Momentum Transformer, an intelligent and interpretable architecture for trading. This model leverages the transformer framework to capture momentum signals in financial time series, providing enhanced interpretability and performance in trading applications. The study highlighted the model's ability to learn complex patterns and its applicability across financial instruments, including forex. The Momentum Transformer demonstrated superior predictive capabilities compared to traditional models.

Tsantekidis et al. [57] explore the application of deep learning for price prediction by exploiting stationary limit order book features. Their study employs a deep learning model to capture complex patterns in the limit order book data, which are then used to predict future price movements. The results demonstrate that the deep learning approach significantly outperforms traditional methods regarding predictive accuracy. However, the high computational cost associated with processing and training on large volumes of order book data remains a notable limitation, which may only be feasible for some trading environments.

Weerathunga and Silva [58] introduce a hybrid DRNN-ARIMA approach for short-term trend forecasting in the forex market. The model combines DRNN with the ARIMA model to enhance short-term prediction accuracy. Leveraging the strengths of both methods, the hybrid model shows improved forecasting performance. Despite these advantages, the model's complexity and the need for extensive parameter tuning are significant drawbacks, making it time-consuming and computationally expensive. Maté [59] investigates forecasting forex spot price intervals using a linear regression model based on interval arithmetic. This study analyses seven major currency pairs and employs interval arithmetic to predict price ranges for the following day. Although the approach offers a unique perspective on interval prediction, its main limitation lies in the simplistic linear regression model, which may not effectively capture the complexities and non-linear relationships inherent in forex data. Talebi et al. [60] present a multi-scale ensemble model designed for classifying trends in the forex market. The model combines predictions from multiple scales to improve trend classification accuracy. Findings indicate that the multi-scale ensemble approach enhances predictive performance compared to single-scale models. Nevertheless, the increased computational complexity and the challenge of integrating multiple scales effectively are significant limitations.

Panda et al. [61] conduct a systematic review of exchange rate prediction using ANN and deep learning methodologies. This review highlights various ANN and deep learning models applied to forex prediction, discussing their advantages and limitations. A significant limitation noted is the overfitting problem commonly encountered in deep learning models, which can negatively impact their generalisability to new data. Juszczuk and Kruś [62] develop a soft multicriteria computing approach to support decision-making in the forex market. Their model integrates multiple criteria to improve decision-making accuracy. While the approach is innovative, its primary limitation is the complexity of accurately weighting and integrating multiple criteria, which can be subjective and challenging to implement effectively. Rundo [63] investigates the integration of a deep LSTM with a reinforcement learning layer for financial trend prediction in high-frequency trading systems. The model combines the sequence modeling capabilities of LSTM with reinforcement learning to enhance predictive accuracy and decision-making. Despite the promising results, the computational intensity and the need for substantial data to train the reinforcement learning component are significant challenges.

2.6 Summary

This subsection summarises the problems and challenges identified in the reviewed literature, explaining the selection of XGBoost, LSTM, Transformer, and ensemble models for this research.

- **Problems and Challenges in ML Models:**

- Sudimanto et al. [8] highlighted the complexity and computational intensity of ML models like SVM and random forests, which limit their efficiency in financial forecasting.
- McNally et al. [9] noted issues of overfitting and the extensive parameter tuning required for ML models, reducing robustness in volatile market conditions.
- Basak et al. [10] discussed the sensitivity of tree-based classifiers to noisy data and their tendency to overfit in highly volatile markets.
- Thu and Xuan [11] emphasised the computational expense and complexity of optimising SVMs, making them less practical for frequent prediction updates.
- Neghab et al. [12] highlighted the reliance on accurate and timely macroeconomic data, which can cause delays in prediction updates.

- **Selection of XGBoost:**

- XGBoost models deliver promising results due to their ability to handle large datasets and capture complex patterns effectively [8].

- They offer robust performance by integrating technical, fundamental, and sentiment data for comprehensive market analysis.
- **Problems and Challenges in DL Models:**
 - Ni et al. [31] found that LSTM models outperform traditional statistical models but require extensive hyperparameter tuning and significant computational resources.
 - Vuong et al. [32] highlighted the increased complexity and computational cost of combining models like XGBoost and LSTM.
 - Islam and Hossain [33] discussed the increased training time and computational complexity due to combining GRU and LSTM models.
 - Wang et al. [34] demonstrated that CNN-TLSTM models outperform traditional models but require substantial computational resources for training.
- **Selection of LSTM and Transformer Models:**
 - LSTM models excel in capturing long-term dependencies and temporal patterns, providing better predictive accuracy for financial time series [31].
 - Transformer models are effective in handling longer sequences and capturing complex patterns, making them promising for financial applications [22].
- **Advantages of Ensemble Models:**
 - Ensemble models combine the strengths of various algorithms to enhance predictive performance and generalisability, addressing the limitations of individual models [27].
 - Ghosh et al. [28] showed that hybrid models combining LSTM networks and random forests improve the accuracy and reliability of stock price direction predictions.
 - He et al. [4] validated the effectiveness of using ensemble strategies in financial forecasting by leveraging diverse data sources and model architectures.
- **Combining LSTM and XGBoost:**
 - LSTM and XGBoost models were combined in this research to leverage their complementary strengths in time-series forecasting. LSTM is well-suited for capturing long-term temporal dependencies in sequential data, while XGBoost is effective in modeling non-linear relationships and feature interactions within structured data.
 - The novelty of this approach lies in using LSTM to model long-term trends in forex data and XGBoost to capture short-term fluctuations by integrating technical, fundamental, and sentiment indicators.

- By combining these models, the research addresses limitations of existing approaches where single models either focus solely on time-series patterns (LSTM) or feature-based prediction (XGBoost). This hybrid model enhances predictive accuracy by capturing both sequential dependencies and complex relationships between features.
- This research also stands out by applying the hybrid model to multiple currency pairs and two different time intervals (four-hour and one-day), making the approach more robust and generalizable across different market conditions.

• **Identified Gaps in the Literature:**

- While significant progress has been made in applying ML and DL models to financial forecasting, the integration of sentiment analysis with technical and fundamental indicators remains underexplored. Most studies focus solely on technical indicators, overlooking the potential benefits of using multiple data sources for improving prediction accuracy.
- Many of the existing studies, such as Ghosh et al. [28], focus on a single currency pair, limiting the generalisability of their findings. This research expands on previous work by using multiple currency pairs and two different time intervals, thereby providing a broader and more comprehensive analysis of forex market prediction.
- Few studies systematically compare the performance of ML, DL, and hybrid ensemble models in forex prediction using diverse data types. This leaves a gap in understanding which models offer the best predictive accuracy and robustness in financial markets.

• **Conclusion:**

- The challenges and gaps identified in the literature support the selection of XGBoost, LSTM, Transformer, and ensemble models in this research. By addressing the limitations of traditional ML and DL models and focusing on the integration of diverse data sources, including multiple currency pairs and time intervals, this study aims to enhance the predictive accuracy and robustness of forex price forecasts.

3 Methodology

This chapter outlines the comprehensive methodologies employed to investigate forex price predictions utilising a multi-modal approach that integrates technical indicators, fundamental analysis, and sentiment analysis derived from financial news. It meticulously describes the data collection process, detailing the types and sources used, including historical forex rates, economic indicators, and sentiment data. Furthermore, it elaborates on the model development process, illustrating the deployment of various advanced machine learning and deep learning models tailored to analyse and predict forex market movements. Additionally, this section specifies the evaluation metrics used to assess the models' accuracy and efficacy, ensuring the research findings' validity and reliability. Through a clear exposition of these methods, this chapter aims to provide a robust framework supporting replicability and scrutiny of the research outcomes.

3.1 Data Collection

The data collection phase of this research forms the foundation upon which the predictive models are built, encompassing a diverse array of datasets to ensure a comprehensive analysis of forex market dynamics. This section delineates the methodologies employed to gather and prepare three distinct types of data: sentiment analysis data, technical trading data, and fundamental economic indicators. Each data type uniquely contributes to the model's ability to predict forex price movements, providing a multifaceted view of the factors influencing currency values. The sources from which data was collected, the criteria for its selection, and the processes involved in its preparation are detailed, aiming to furnish a robust dataset that supports the sophisticated analytical techniques used in this study.

Significantly, two different timeframes of market data were utilised to capture varying market dynamics: One-Day OHLC data, and Four-Hour OHLC data. The Four-hour data was paired explicitly with sentiment analysis data and technical and fundamental indicators, aiming to capture the immediate impact of market sentiment on forex prices within shorter intervals. This combination was mainly selected to assess the influence of rapidly changing market sentiments alongside economic and technical indicators, providing a detailed temporal resolution suitable for high-frequency trading strategies.

Conversely, the One-Day OHLC data was used exclusively with technical and fundamental indicators. This broader timeframe allowed for examining longer-term trends and economic impacts without the immediate noise of market sentiment, offering insights into the more stable, underlying movements in forex prices. This strategic differentiation in data utilisation enabled a comparative analysis of model performance across four key forex pairs (EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD) under different architectural setups for each dataset. This approach highlights each model configuration's relative strengths and weaknesses and

showcases how different data integrations can impact predictive accuracy in forex market prediction.

3.1.1 Sentiment Analysis

Sentiment analysis is crucial in understanding market trends and predicting forex price movements. This section details the methodology employed to collect and process sentiment data from financial news headlines, highlighting the tools and techniques used to ensure a comprehensive and accurate dataset.

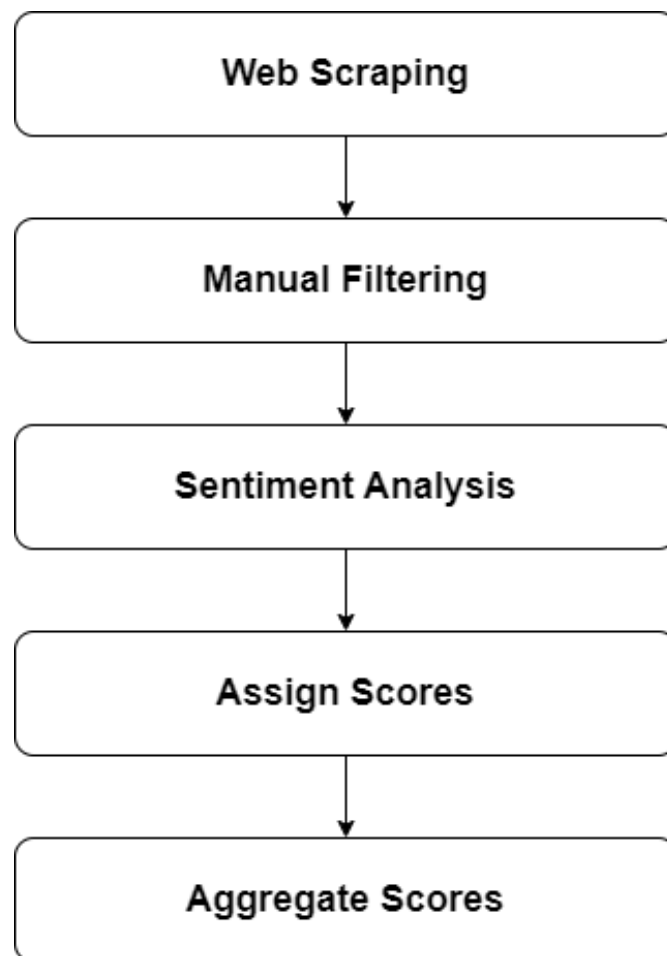


Figure 9: The Sentiment Analysis Workflow for Forex News Headlines.

Advanced web scraping tools and libraries were utilised to collect sentiment data. Selenium, an automation tool that simulates user interactions with web pages, was essential for navigating dynamic content. BeautifulSoup, a Python library for parsing HTML and XML documents, was used to extract specific elements such as news headlines and publication dates[64], [65]. Pandas, a data manipulation and analysis library, organised and stored the extracted data in structured formats. These tools were chosen for their robustness and ability to handle complex web scraping tasks, ensuring the reliability and comprehensiveness of the collected data.

The sentiment data was sourced from three prominent financial news websites, each known for its extensive coverage of forex-related news. Investing.com is a widely used platform that offers real-time data, news, and analysis on financial markets. It is popular among traders and investors for its comprehensive coverage of market trends and events [66]. This platform provided a substantial portion of the sentiment data used in this study.

DailyForex.com focuses on forex news, analysis, and educational content. It offers in-depth market insights and trading strategies, making it a valuable resource for novice and experienced traders [67]. The site's detailed analysis and up-to-date information on market movements contributed significantly to the sentiment data collected for this research.

Interest.co.nz is a New Zealand-based financial news website covering various economic and market topics. It provides localised news and analysis on the NZD currency, offering unique insights into the New Zealand forex market [68]. This site was beneficial for gathering sentiment data related to the NZD currency pairs.

From these sources, a total of 2048 news headlines were collected over the period from January 1, 2023, to January 31, 2024. The distribution of headlines among the different forex pairs was as follows: EUR/NZD with 246 headlines, NZD/USD with 585 headlines, EUR/USD with 466 headlines, and AUD/NZD with 192 headlines. This diverse data set from multiple reputable sources ensured a comprehensive analysis of market sentiment, improving the robustness of the predictive models.

The process involved several steps. First, web scraping with Selenium automated the opening of web pages and waited for dynamic content to load, which is crucial for handling websites with JavaScript-driven content. BeautifulSoup then parsed the HTML content of the web pages and extracted necessary elements, such as news titles and publication dates, ensuring only relevant information was captured. After collecting data from each website, the datasets were combined into a single comprehensive dataset using Pandas, ensuring consistency in the data format.

Once collected, the news headlines were processed using state-of-the-art Large Language Models (LLMs): GPT-4, and GEMINI Advanced. GPT-4, utilised for its advanced zero-shot learning capabilities, interprets context-rich financial news without explicit prior training on financial sentiment. GEMINI Advanced, designed for complex narrative understanding, provides deep insights from financial texts even without labeled data. Each headline was assigned a sentiment score (positive, neutral, or negative) based on the output of these models.

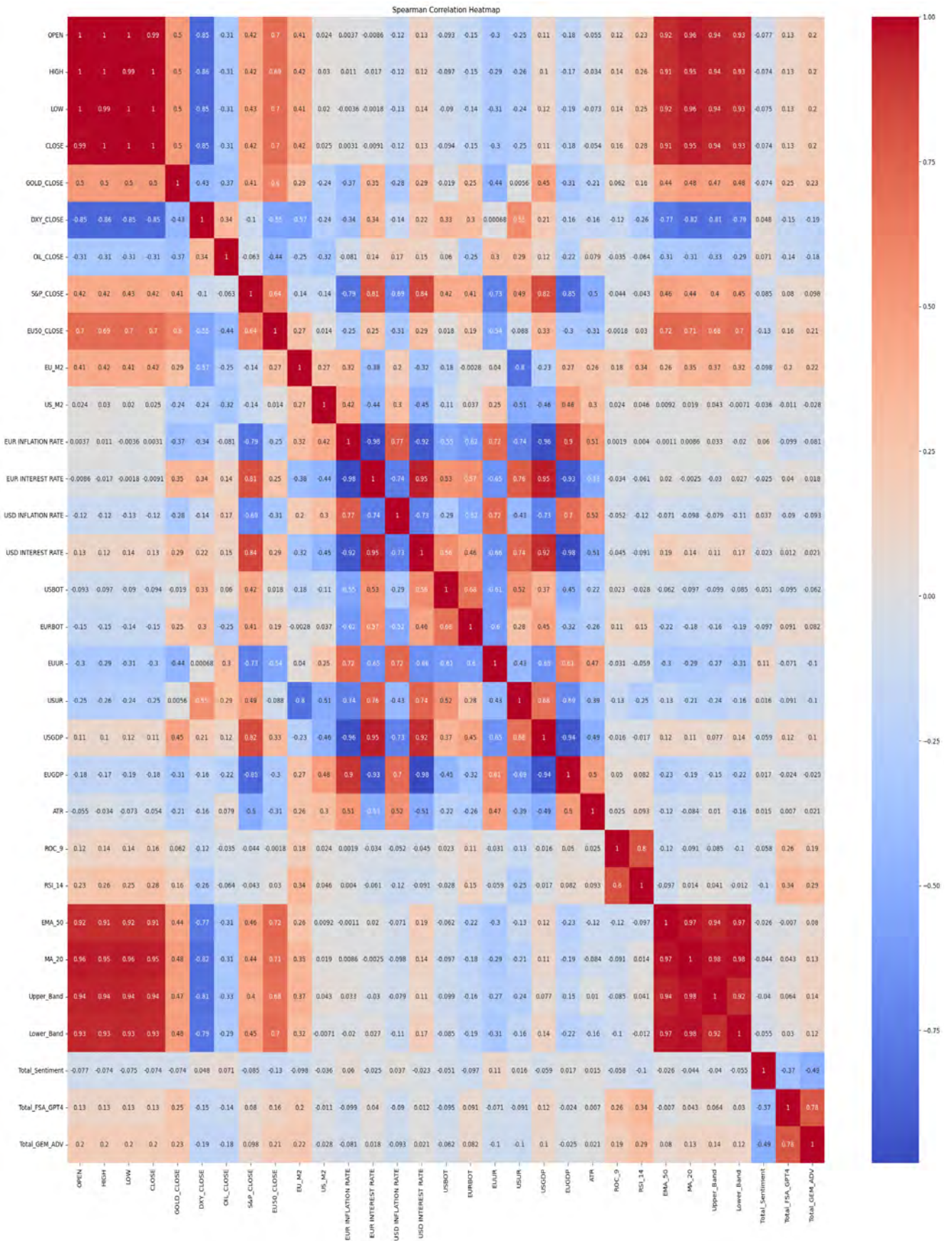


Figure 10: Heatmap showing Spearman correlation coefficients for EUR/USD data.

The sentiment classification process involved assigning scores of 1 for positive sentiment, 0 for neutral sentiment, and -1 for negative sentiment to each headline. These daily sentiment scores were aggregated to derive a comprehensive sentiment analysis score for each day. Given that the primary dataset operates on a Fourhour timeframe, Figure 10 shows the ROC_9 indicator—which represents the rate of change over a one-and-a-half-day period—provides a meaningful way to correlate the aggregated daily sentiment scores with forex price movements. This methodology allows for a nuanced understanding of how daily market sentiment influences price trends over subsequent days, capturing the temporal dynamics between sentiment shifts and their market effects. The comprehensive sentiment analysis process is illustrated in Figure 9, which outlines the sequential steps from web scraping to data extraction, sentiment classification, and sentiment score aggregation. This structured approach ensures that the sentiment data collected is relevant, timely, and accurately reflects market conditions, providing a solid foundation for subsequent predictive modeling. For the sentiment analysis, the following zero-shot prompt was used: *"Provide a sentiment analysis for the following headlines regarding the specified currency pair. Indicate whether each headline is positive, negative, or neutral for the specified currency pair."* This approach enabled us to effectively categorise the headlines without the need for pre-labeled data, leveraging the advanced capabilities of the LLMs [69], [70]. As shown in Figure 10, the Spearman correlation heatmap indicates that the GPT-4 sentiment score has a higher correlation with the close price compared to other sentiment models. Therefore, for the EUR/USD pair, the GPT-4 sentiment analysis score is used for model training to enhance the predictive accuracy of the EUR/USD models.

Following these steps, a robust dataset of financial news headlines was prepared for sentiment analysis, offering valuable insights into market sentiment and its impact on forex price movements. This comprehensive approach ensures that the sentiment analysis component is grounded in a well-curated and diverse dataset, enhancing the overall reliability and accuracy of the predictive models[71].

3.1.2 Technical Analysis

Technical analysis is critical in forex market prediction, as it utilises historical price data and statistical indicators to forecast future price movements. This section delves into the various types of technical indicators used in this research, the rationale behind their selection, and the formulas used for their calculation. The primary indicators included in the dataset are derived from the Open, High, Low, and Close (OHLC) prices, calculated using the TA-Lib library, a comprehensive library for technical analysis [72, 73].

Open, High, Low, and Close (OHLC) Prices: OHLC prices form the foundational data from which various technical indicators are computed. Each of these prices represents different aspects of market behavior within a specific timeframe, providing a granular view of price action:

- **Open Price:** The price at which the forex pair begins trading at the start of a given period.
- **High Price:** The maximum price reached during the trading period.
- **Low Price:** The minimum price reached during the trading period.
- **Close Price:** The price at which trading ends for the period.

Average True Range (ATR): ATR measures market volatility by considering the range of price movements. It is calculated based on the greatest of the following: the current high minus the current low, the absolute value of the current high minus the previous close, and the absolute value of the current low minus the previous close. The formula is presented below (Eq. 1):

$$ATR = \frac{1}{n} \sum_{i=1}^n \text{True Range}_i \quad (1)$$

Here, True Range is the maximum of (Eq. 2):

$$\text{True Range} = \max(\text{High} - \text{Low}, |\text{High} - \text{Close}_{\text{prev}}|, |\text{Low} - \text{Close}_{\text{prev}}|) \quad (2)$$

Rate of Change (ROC): ROC is a momentum oscillator that measures the percentage change in price between the current price and a price n periods ago. It indicates the speed at which the price is changing, which helps in identifying overbought or oversold conditions (Eq. 3):

$$ROC_n = \frac{\text{Close} - \text{Close}_{n \text{ periods ago}}}{\text{Close}_{n \text{ periods ago}}} \times 100 \quad (3)$$

- **ROC_7:** Used in the One-day timeframe, it calculates the rate of change over seven days.
- **ROC_9:** Used in the Four-hour timeframe, it calculates the rate of change over 36 hours (nine Four-hour periods).

Relative Strength Index (RSI): RSI is a momentum oscillator that measures the speed and change of price movements. It oscillates between 0 and 100 and is typically used to identify overbought or oversold conditions. The formula is presented below (Eq. 4):

$$RSI = 100 - \frac{100}{1 + RS} \quad (4)$$

where:

$$RS = \frac{\text{Average Gain over } n \text{ periods}}{\text{Average Loss over } n \text{ periods}} \quad (5)$$

- **RSI_14:** Calculated using 14 days, it helps understand the momentum of price changes and potential reversal points.

Exponential Moving Average (EMA): EMA is a moving average with greater weight and significance on the most recent data points. It reacts more quickly to recent price changes than the simple moving average (SMA). The formula is given in (Eq. 6):

$$EMA_t = \alpha \times Price_t + (1 - \alpha) \times EMA_{t-1} \quad (6)$$

where:

$$\alpha = \frac{2}{n + 1} \quad (7)$$

- **EMA_50:** A 50-period EMA that smooths out price data to identify the trend direction over a medium-term period.

Bollinger Bands: Bollinger Bands consist of a middle band (usually a 20-period SMA) and two outer bands (upper and lower), standard deviations from the middle band. They measure market volatility and identify potential overbought or oversold conditions. The formulas are shown below (Eq. 8, 9, 10):

$$\text{Middle Band} = MA_{20} \quad (8)$$

$$\text{Upper Band} = MA_{20} + (2 \times \text{Standard Deviation}_{20}) \quad (9)$$

$$\text{Lower Band} = MA_{20} - (2 \times \text{Standard Deviation}_{20}) \quad (10)$$

Selection of Technical Indicators

The selection of the above technical indicators is rooted in their proven effectiveness in capturing various aspects of market behavior:

- ATR (Eq. 1) is essential for understanding market volatility, helping traders adjust their strategies based on the extent of price movements.
- ROC indicators (ROC_7 and ROC_9) (Eq. 3) are crucial for identifying the momentum of price changes over different timeframes, which is critical for short-term and medium-term trading strategies.
- RSI (Eq. 4) provides insights into the momentum and potential reversal points, which are critical for timing entries and exits in the market.
- EMA (Eq. 6) helps smooth out price data to identify trends more accurately, reducing the noise from random price fluctuations.
- Bollinger Bands (Eq. 8, 9, 10) are invaluable for assessing market volatility and identifying overbought or oversold conditions, which can signal potential reversals.

These indicators offer a comprehensive view of market dynamics, enabling a nuanced analysis of price movements. By integrating these technical indicators, the models developed in this research aim to leverage the strengths of each indicator to enhance the accuracy of

forex price predictions. The combination of these indicators provides a robust framework for understanding and forecasting market behaviour across different timeframes, as detailed in subsequent sections of this research.

3.1.3 Fundamental Analysis

As the previous section shows, technical analysis indicators provide a robust framework for understanding market dynamics based on historical price data. This section focuses on fundamental analysis, which complements technical analysis by incorporating economic and financial indicators that reflect the broader economic environment. Fundamental indicators offer deeper insights into the factors influencing currency values, providing a comprehensive understanding of market movements. This section details the fundamental indicators used in this research, explaining their significance and impact on forex trading.

Table 1: Fundamental Indicators Used for Each Currency Pair in the Research [74, 75, 76, 77, 78, 79, 80, 81].

Currency Pair	Fundamental Indicators
EUR/USD	GOLD_CLOSE, DXY_CLOSE, OIL_CLOSE, S&P_CLOSE, EU50_CLOSE, EU_M2, US_M2, EUR INFLATION RATE, EUR INTEREST RATE, USD INFLATION RATE, USD INTEREST RATE, USBOAT, EURBOT, EUUR, USUR, USGDP, EUGDP
NZD/USD	GOLD_CLOSE, DXY_CLOSE, OIL_CLOSE, NZX50_CLOSE, S&P_CLOSE, USD INFLATION RATE, USD INTEREST RATE, NZD INFLATION RATE, NZD INTEREST RATE, NZGDTPI, NZTA, USBOT, NZBOT, NZUR, USUR, USGDP, NZGDP
EUR/NZD	GOLD_CLOSE, DXY_CLOSE, OIL_CLOSE, NZX50_CLOSE, EU50_CLOSE, EU_M3, NZD_M3, EUR INFLATION RATE, EUR INTEREST RATE, NZD INFLATION RATE, NZD INTEREST RATE, NZTA, NZBOT, EURBOT, NZUR, EUUR, NZGDP, EUGDP
AUD/NZD	GOLD_CLOSE, DXY_CLOSE, OIL_CLOSE, NZX50_CLOSE, ASX200_CLOSE, AUD_M3, NZD_M3, AU INFLATION RATE, AU INTEREST RATE, NZD INFLATION RATE, NZD INTEREST RATE, NZTA, NZBOT, AUBOT, NZUR, AUUR, NZGDP, AUGDP

Gold (GOLD_CLOSE): Gold prices are often considered a safe-haven asset, meaning they tend to hold or increase in value during economic uncertainty. Investors flock to gold during financial instability, leading to price increases. Rising gold prices indicate a lack of confidence in fiat currencies, as gold is seen as a more stable store of value. This can impact forex pairs such as EUR/USD and NZD/USD, as movements in gold prices reflect broader economic sentiments that can influence currency valuations.

US Dollar Index (DXY_CLOSE): The US Dollar Index (DXY) measures the value of the USD against a basket of foreign currencies, including the euro, Japanese yen, British pound, Canadian dollar, Swedish krona, and Swiss franc. Movements in the DXY can have significant implications for currency pairs involving the USD. A rising DXY indicates a strengthening USD, which can affect the EUR/USD and NZD/USD pairs by making the USD more valuable relative to these currencies.

Oil (OIL_CLOSE): Oil prices influence many aspects of the global economy, including inflation and trade balances. Countries that are major oil exporters or importers are particularly affected by fluctuations in oil prices. For example, higher oil prices can increase inflationary pressures in oil-importing countries, impacting their currency value. Conversely, oil-exporting countries may see their currency strengthen with rising oil prices. Therefore, oil price movements can affect forex pairs like EUR/USD and NZD/USD.

S&P 500 (S&P_CLOSE): The S&P 500 is a broad measure of the US stock market's performance and is an essential indicator of US economic health. A strong S&P 500 typically signals a robust economy, which can strengthen the USD. Consequently, currency pairs tied to the USD, such as EUR/USD and NZD/USD, are influenced by the performance of the S&P 500.

Euro 50 (EU50_CLOSE): The Euro Stoxx 50 Index represents 50 of the Eurozone's most extensive and most liquid stocks. It serves as a critical indicator for the EUR, reflecting the Eurozone's economic health and investor sentiment. Movements in the Euro Stoxx 50 can impact pairs like EUR/USD and EUR/NZD, as changes in the index reflect shifts in economic stability and investor confidence.

NZX 50 (NZX50_CLOSE): The NZX 50 Index includes the top 50 companies listed on the New Zealand Stock Exchange, serving as a barometer for the New Zealand economy. The performance of the NZX 50 affects the NZD, influencing currency pairs such as NZD/USD and EUR/NZD. A strong NZX 50 indicates economic growth and investor confidence in New Zealand, leading to a stronger NZD.

ASX 200 (ASX200_CLOSE): The ASX 200 Index is a benchmark for the Australian equity market. It provides insights into the Australian economy, influencing the AUD. The performance of the ASX 200 affects the AUD/NZD pair, as it reflects Australia's economic conditions and investor sentiment.

Money Supply (M2 and M3): Money supply measures like M2 and M3 reflect the liquidity in the economy. M2 includes cash, checking deposits, and easily convertible near money, while M3 includes M2 plus large-time deposits, institutional money market funds, and other more considerable liquid assets. Changes in money supply can influence inflation and economic growth. For instance: EU_M2 and US_M2: These indicators reflect the Eurozone and US money supply, impacting EUR/USD. An increasing money supply can lead to inflationary pressures, affecting currency value. NZD_M3 and AUD_M3: These measures are crucial for understanding monetary conditions in New Zealand and Australia, influencing NZD/USD and AUD/NZD pairs.

Inflation Rates (EUR, USD, NZD, AU): Inflation rates indicate the rate at which the general price level of goods and services rises, eroding purchasing power. High inflation typically leads to higher interest rates as central banks try to control inflation, affecting currency value. For example: EUR Inflation Rate: Understanding Eurozone inflation is vital for predicting EUR movements. High inflation may lead to higher interest rates, strengthening the EUR. USD Inflation Rate: US inflation impacts the USD's value against the NZD in the NZD/USD pair. NZD and AU Inflation Rates: These rates affect the NZD and AUD, influencing pairs like NZD/USD and AUD/NZD. Central New Zealand and Australian banks may adjust interest rates in response to inflation, impacting currency value.

Interest Rates (EUR, USD, NZD, AU): Central banks use interest rates to control inflation and stabilise the currency. Changes in interest rates impact currency value by affecting investment flows. Higher interest rates attract foreign investment, strengthening the currency. For instance: EUR and USD Interest Rates: These rates affect the EUR/USD pair. Higher Eurozone or US interest rates can attract investment, strengthening the EUR or USD. NZD Interest Rates: Influence the NZD/USD pair. Higher interest rates in New Zealand can attract investment, strengthening the NZD. AU Interest Rates: Affect the AUD/NZD pair. Changes in Australian interest rates influence the AUD, impacting the pair.

Gross Domestic Product (GDP): GDP measures a country's economic performance. Higher GDP growth typically strengthens a currency by signaling economic stability and growth. For example: US GDP and EUR/USD: US economic performance influences the USD. Strong GDP growth in the US can strengthen the USD, affecting the EUR/USD pair. NZ GDP and NZD/USD: New Zealand's economic health affects the NZD. Strong GDP growth in New Zealand can strengthen the NZD, impacting the pair. EU GDP and EUR/USD: Eurozone economic performance impacts the EUR. Strong GDP growth in the Eurozone can strengthen the EUR, affecting the EUR/USD pair. AU GDP and AUD/NZD: Australia's GDP influences the AUD. Strong GDP growth in Australia can strengthen the AUD, impacting the AUD/NZD pair.

Unemployment Rates (EUUR, USUR, NZUR, AUUR): Unemployment rates indicate economic health. Lower unemployment generally strengthens a currency by signaling economic stability. For example: US Unemployment Rate (USUR): Affects the USD in EUR/USD and NZD/USD pairs. Lower unemployment in the US can strengthen the USD. EU Unemployment Rate (EUUR): Impacts the EUR in EUR/USD and EUR/NZD pairs. Lower unemployment in the Eurozone can strengthen the EUR. NZ Unemployment Rate (NZUR): Influences the NZD in NZD/USD and EUR/NZD pairs. Lower unemployment in New Zealand can strengthen the NZD. AU Unemployment Rate (AUUR): Affects the AUD in AUD/NZD pairs. Lower unemployment in Australia can strengthen the AUD.

Balance of Trade (BOT): The balance of trade (BOT) is the difference between a country's exports and imports. A positive BOT (trade surplus) typically strengthens the currency, while a negative BOT (trade deficit) can weaken it. For example: USBOAT and EURBOT: Reflect trade balances for the US and Eurozone, impacting EUR/USD. A trade surplus in the US or

Eurozone can strengthen the USD or EUR. USBOT and NZBOT: Influence the USD and NZD in NZD/USD. A US or New Zealand trade surplus can strengthen the USD or NZD. AUBOT: Affects the AUD in AUD/NZD. A trade surplus in Australia can strengthen the AUD.

Trade Indicators (NZGDTPI and NZTA): Specific to New Zealand, these indicators reflect trade terms and agreements, influencing the NZD. For example: NZGDTPI: The Global Dairy Trade Price Index affects the NZD in NZD/USD. Changes in dairy prices can influence New Zealand's export revenues, impacting the NZD. NZTA: New Zealand Tourist Arrivals impact the NZD in NZD/USD and EUR/NZD. Tourist arrivals can affect trade flows and economic stability, influencing the NZD.

These fundamental indicators comprehensively overview the economic factors influencing forex markets. The research aims to leverage the broader economic context to enhance forex price prediction accuracy by integrating these indicators into predictive models. The combination of technical and fundamental analysis offers a holistic approach to understanding and forecasting market movements.

3.2 Model Development

In the previous section, we discussed the detailed data collection process, encompassing sentiment analysis and technical and fundamental indicators. This comprehensive dataset forms the backbone of our research, providing the essential inputs needed for accurate forex price prediction. In this section, we will delve into the detailed structure of the five models employed in this study, illustrating how each model integrates the diverse data types to analyse and predict forex market movements. The models include LSTM, XGBoost, Transformer architectures, and Ensemble Models.

3.2.1 LSTM

The LSTM networks were employed to leverage their strength in capturing temporal patterns and dependencies in sequential data. The LSTM models were used with both One-day and Four-hour OHLC data, combined with technical indicators (TM), fundamental indicators (FM), and sentiment analysis (SA) scores. To ensure the data was suitable for LSTM input, preprocessing steps included standardizing the features and targets. The `StandardScaler` from the `sklearn` library was utilised for this purpose. Standardising the data ensured that all features had a mean of zero and a standard deviation of one, which is crucial for achieving stable and efficient training of the LSTM model. The data was structured into sequences using a time step, aligning with the chosen temporal resolution for the LSTM model. This involved creating sliding windows of data to capture the temporal dependencies effectively. Each window consisted of consecutive time steps of input features, with the target being the forex price at the subsequent time step.

The LSTM model was constructed using the TensorFlow Keras library. The architecture included two LSTM layers with Dropout and BatchNormalisation layers to improve generalisation and model stability. The first LSTM layer was followed by a Dropout layer and a BatchNormalisation layer. The second LSTM layer was followed by another Dropout layer and another BatchNormalisation layer. The final output layer was a Dense layer with a single unit, representing the predicted forex price. L2 regularisation was applied to all layers to prevent overfitting, which penalises large weights and encourages the model to learn simpler patterns. The Adam optimiser was employed with an initial learning rate. Adam is an adaptive learning rate optimisation algorithm well-suited for training deep learning models. Learning rate scheduling was also implemented using the ReduceLROnPlateau callback, which reduces the learning rate when the validation loss plateaued, ensuring that the model continued learning even when improvements slowed. Early stopping was also employed using the EarlyStopping callback, which monitored the validation loss and stopped training if there was no improvement for a specified number of epochs, restoring the best weights observed during training. The model was trained for up to many epochs with a moderate batch size. The training process included validation data to monitor the model's performance and prevent overfitting. The ReduceLROnPlateau and EarlyStopping callbacks ensured efficient training by dynamically adjusting the learning rate and preventing overfitting.

LSTM with Technical Indicators (LSTM_TM): This configuration utilises technical indicators derived from One-day OHLC data to predict forex prices. The technical indicators included are ATR, ROC_7, RSI_14, EMA_50, and Bollinger Bands.

LSTM with Fundamental Indicators (LSTM_FM): This model focuses on fundamental economic indicators such as GDP, inflation rates, interest rates, and trade balances to forecast forex price movements.

LSTM with Technical and Fundamental Indicators (LSTM_TM_FM): This comprehensive model combines technical and fundamental indicators, providing a holistic view of the market dynamics influencing forex prices.

LSTM with Technical, Fundamental, and Sentiment Analysis (LSTM_TM_FM_SA): This configuration integrates technical and fundamental indicators with sentiment scores derived from financial news headlines. This model captures the immediate impact of market sentiment on forex prices within shorter intervals, aiming to enhance predictive accuracy by leveraging real-time market sentiment. Each configuration was tested on the respective datasets for the four key forex pairs (EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD). The performance of these models was assessed using a comprehensive suite of metrics, including MAE, MAPE, RMSE, R^2 , and the Spearman correlation coefficient. These metrics provided a quantitative assessment of the model's accuracy and the value added by integrating sentiment analysis with technical and fundamental indicators.

3.2.2 XGBoost

The XGBoost model was implemented to leverage its robust gradient-boosting framework, which effectively handles structured data and captures complex patterns. The model's capability to handle large datasets and its flexibility in tuning parameters made it an ideal choice for this research. The XGBoost models were trained and evaluated using both One-day and Four-hour OHLC data, combined with technical indicators (TM), fundamental indicators (FM), and sentiment analysis (SA) scores.

To prepare the data for the XGBoost model, it was essential to standardise the features and targets to ensure that all features had a mean of zero and a standard deviation of one. This standardisation was performed using the `StandardScaler` from the `sklearn` library. Standardising the data helps achieve stable and efficient training of the XGBoost model by ensuring that the features are on a comparable scale.

The data was split into training and test sets using an 80-20 split, ensuring that 20% of the data was reserved for testing. This split was performed using the `train_test_split` function from `sklearn`, with the `shuffle` parameter set to `False` to maintain the time series order of the data. The target variable (CLOSE price) was also scaled to ensure consistent scaling across all features.

The XGBoost model was constructed using the `XGBRegressor` from the XGBoost library, with the objective function set to `reg:squarederror`, which is suitable for regression tasks. A comprehensive hyperparameter tuning process was undertaken using `GridSearchCV` to identify the optimal hyperparameters for the model. The hyperparameter grid included a range of values for `max_depth`, `learning_rate`, `n_estimators`, `subsample`, `colsample_bytree`, `gamma`, and `reg_lambda`. The `GridSearchCV` approach involved performing an exhaustive search over specified parameter values for the XGBoost estimator. This search was evaluated using 5-fold cross-validation to ensure robustness and generalisability of the results. The scoring metric used was `neg_mean_squared_error`, which minimises the squared error between the predicted and actual values.

The training process for the XGBoost model involved fitting the model on the scaled training data and validating it using cross-validation to prevent overfitting. The best hyperparameters identified from the grid search were used to fit the final model, ensuring that the model was optimised for predictive performance.

The XGBoost model's architecture included the following key parameters:

- **Max Depth:** This parameter controls the maximum depth of the trees. A higher value can increase the model's capacity to learn complex patterns but may also lead to overfitting.
- **Learning Rate:** This parameter determines the step size at each iteration while moving toward a minimum of the loss function. A lower learning rate requires more trees but can improve the model's performance.

- **Number of Estimators:** This parameter defines the number of boosting rounds. More rounds can lead to better performance but may also increase the risk of overfitting.
- **Subsample:** This parameter controls the sample fraction to fit the individual base learners. Subsampling can prevent overfitting by introducing randomness.
- **Colsample_bytree:** This parameter controls the fraction of features used for fitting each tree. Similar to the subsample, it helps reduce overfitting by adding randomness.
- **Gamma:** This parameter specifies the minimum loss reduction required to partition a leaf node further. A higher value leads to more conservative algorithms.
- **Reg_lambda:** This parameter adds a regularisation term to the loss function, which helps in preventing overfitting.

These parameters were fine-tuned to optimise the performance of the XGBoost model, ensuring that it effectively captured the underlying patterns in the forex data.

3.2.3 Ensemble Model

The ensemble model was developed to combine the strengths of LSTM and XGBoost models, aiming to improve predictive accuracy by leveraging the unique capabilities of each model. The ensemble approach involved training LSTM and XGBoost models separately and then combining their predictions using a weighted averaging technique.

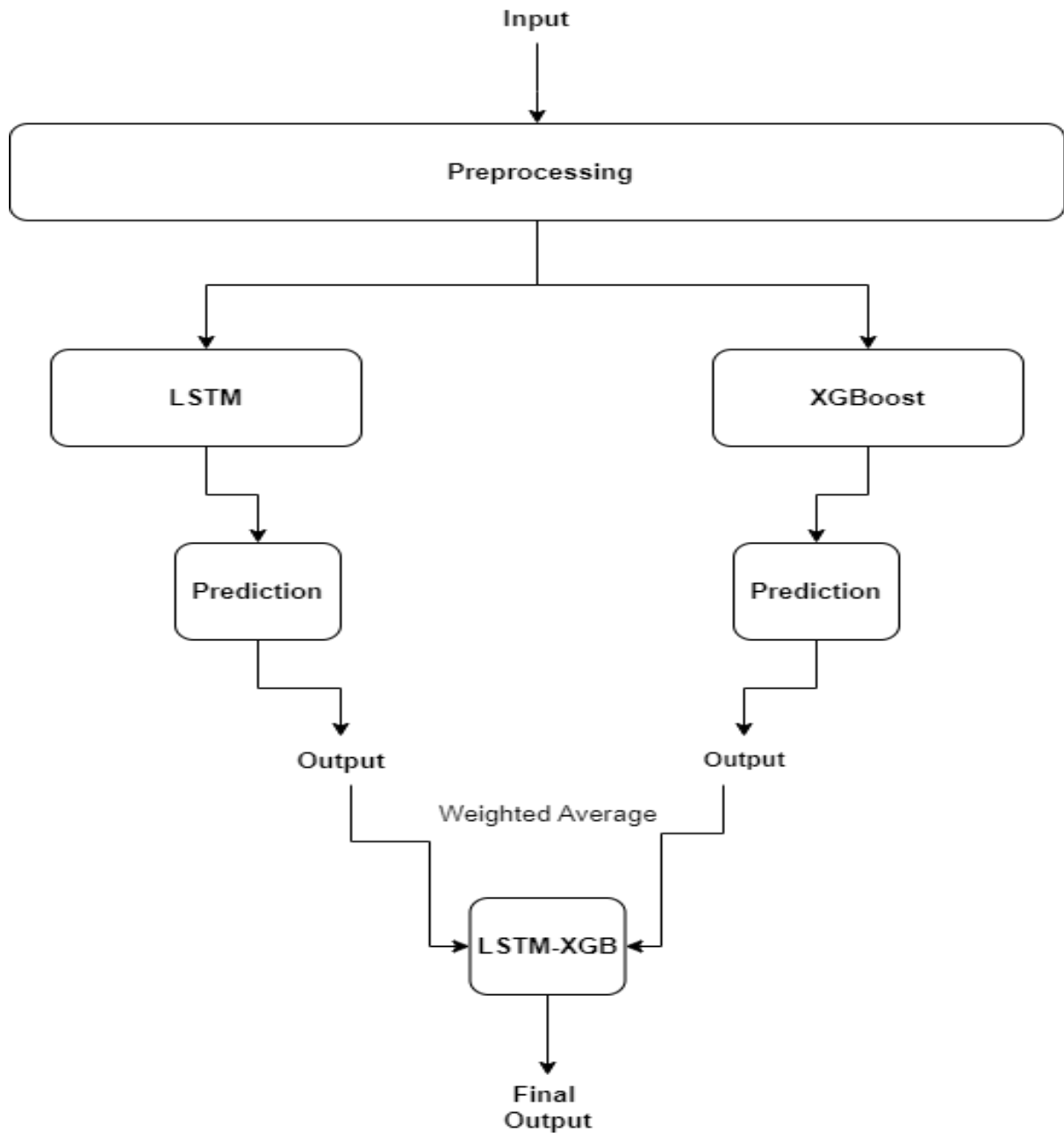


Figure 11: Architecture of the Ensemble Model Combining LSTM and XGBoost.

The LSTM model was trained using standardised One-day and Four-hour OHLC data and technical, fundamental, and sentiment indicators. The preprocessing steps involved standardising the features and targets using the `StandardScaler` from the `sklearn` library, ensuring that all features had a mean of zero and a standard deviation of one. The data was structured into sequences using a time step, capturing temporal dependencies effectively.

The XGBoost model was trained on the same standardised data, using a comprehensive hyperparameter tuning process with `GridSearchCV` to identify the optimal hyperparameters. The best hyperparameters were used to fit the final XGBoost model, ensuring it was optimised for predictive performance.

Once both models were trained, their predictions were combined using a weighted averaging technique. The weights were determined based on the performance of each model, with the LSTM model generally receiving a higher weight due to its superior performance in capturing temporal patterns. The ensemble model's predictions were then evaluated using a comprehensive suite of metrics, including MAE, MAPE, RMSE, R^2 , and the Spearman correlation coefficient.

As shown in Figure 11, the detailed architecture of the ensemble model combines LSTM and XGBoost. The ensemble model's architecture included the following steps:

- **Training LSTM and XGBoost Models:** Both models were trained separately using standardised data and optimised hyperparameters.
- **Making Predictions:** Each model made predictions on the test data, which were standardised to ensure consistency.
- **Combining Predictions:** The predictions from both models were combined using a weighted averaging technique, with weights assigned based on each model's performance.
- **Evaluating Performance:** The ensemble model's predictions were evaluated using a comprehensive suite of metrics to assess its predictive accuracy and the value added by combining LSTM and XGBoost models.

By integrating the strengths of LSTM and XGBoost models, the ensemble model aimed to provide more accurate and robust forex price predictions, leveraging the temporal patterns captured by LSTM and the complex patterns identified by XGBoost.

3.2.4 Transformer Model(TRANSFORMER_EN_DE)

The Transformer model captured complex dependencies within sequential forex data, leveraging its architecture's strengths in handling long-range dependencies and parallel processing capabilities. The Transformer model used in this research included encoder and decoder components designed to process historical forex data and predict future price movements. To prepare the data for the Transformer model, preprocessing steps involved standardising the features and targets to ensure all features had a mean of zero and a standard deviation of one. This standardisation was performed using the `StandardScaler` from the `sklearn` library. Standardising the data is crucial for achieving stable and efficient training of the Transformer model by ensuring that the features are on a comparable scale.

The data was structured into sequences using a sequence length of 10, aligning with the chosen temporal resolution for the Transformer model. This involved creating sliding windows of data to capture temporal dependencies effectively. Each window consisted of ten consecutive

time steps of input features, with the target being the forex price at the subsequent time step. The Transformer model was constructed using the TensorFlow Keras library, with an architecture that included both encoder and decoder layers. The encoder processed the input sequences, capturing dependencies within the data. The decoder used these processed sequences to predict future prices, considering the historical context provided by the encoder.

The architecture of the Transformer model included several key components:

- **Transformer Block:** Each block consisted of a multi-head attention mechanism, followed by a feed-forward network. Multi-head attention allowed the model to focus on different parts of the input sequence simultaneously, capturing various aspects of the temporal dependencies. Layer normalisation and dropout were applied to improve generalisation and stability.
- **Encoder:** The encoder comprises multiple Transformer blocks, processing the input sequences to extract meaningful representations. Each block in the encoder included multi-head attention and feed-forward layers.
- **Decoder:** The decoder also comprised multiple Transformer blocks, with each block focusing on the encoder outputs and the previous decoder outputs. The decoder's attention mechanism allowed it to focus on relevant parts of the encoder's output while generating predictions.
- **Output Layer:** A time-distributed dense layer was used as the final output layer, applying a linear activation function to predict the forex price at each time step.

L2 regularisation was applied to all dense layers to prevent overfitting, which penalises large weights and encourages the model to learn simpler patterns. The Adam optimiser was employed with an initial learning rate, and learning rate scheduling was implemented using the ReduceLROnPlateau callback. This callback reduced the learning rate by 0.5 when the validation loss plateaued, ensuring that the model continued learning even when improvements slowed. Early stopping was also employed using the EarlyStopping callback, which monitored the validation loss and stopped training if there was no improvement for 60 epochs, restoring the best weights observed during training. The training process for the Transformer model involved fitting the model on the training data and validating it using validation data to monitor the model's performance and prevent overfitting. The model was trained for up to 1000 epochs with a batch size tailored to the dataset. The performance of the Transformer model was assessed using a comprehensive suite of metrics, including MAE, MAPE, RMSE, R^2 , and the Spearman correlation coefficient. These metrics provided a quantitative assessment of the model's accuracy and the value added by integrating sentiment analysis with technical and fundamental indicators.

3.2.5 Transformer Model(TRANSFORMER_DE)

The Transformer Decoder model captured complex dependencies within sequential forex data, leveraging its architecture's strengths in handling long-range dependencies and parallel processing capabilities. This research utilised a Transformer model with a decoder-only architecture to process historical forex data and predict future price movements.

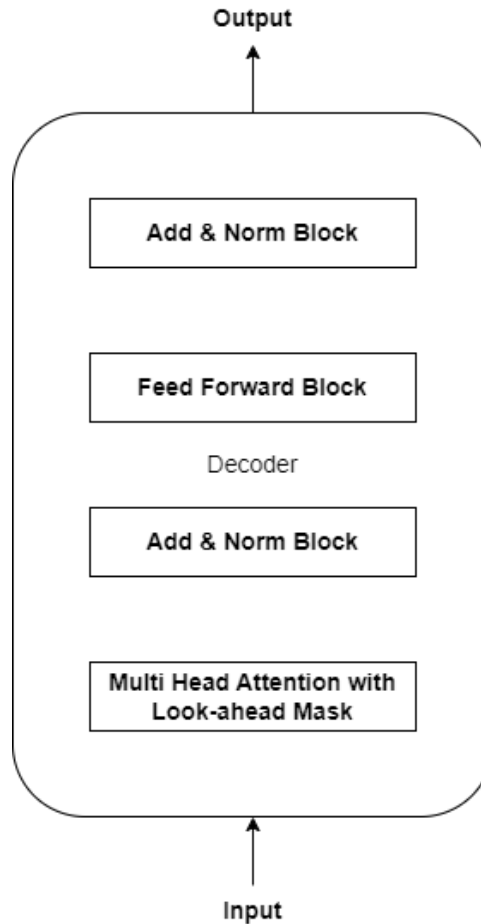


Figure 12: Decoder Architecture of the Transformer Model.

To prepare the data for the Transformer model, preprocessing steps involved standardising the features and targets to ensure all features had a mean of zero and a standard deviation of one. This standardisation was performed using the `StandardScaler` from the `sklearn` library. Standardising the data is crucial for achieving stable and efficient training of the Transformer model by ensuring that the features are on a comparable scale. The data was structured into sequences using a sequence length of 8, aligning with the chosen temporal resolution for the Transformer model. This involved creating sliding windows of data to capture temporal dependencies effectively. Each window consisted of eight consecutive time steps of input features, with the target being the forex price at the subsequent time step.

The architecture of the Transformer model included several key components, as illustrated in Figure 12:

- **Transformer Block:** Each block consisted of a multi-head attention mechanism with a look-ahead mask, followed by a feed-forward network. Multi-head attention allowed the model to focus on different parts of the input sequence simultaneously, capturing various aspects of the temporal dependencies. Layer normalisation and dropout were applied to improve generalisation and stability.
- **Decoder:** The decoder is comprised of multiple transformer blocks. Each block focused on the decoder's attention mechanism, allowing it to concentrate on relevant parts of the input sequence while generating predictions.
- **Output Layer:** A time-distributed dense layer was used as the final output layer, applying a linear activation function to predict the forex price at each time step.

The model's architecture included the following layers:

- **Input Layer:** Accepting the standardised input sequences.
- **Transformer Blocks:** Processing the input sequences through multiple layers containing multi-head attention and feed-forward layers.
- **Output Layer:** A dense layer with a linear activation function that produces the final price predictions.

The look-ahead mask was used in the attention mechanism to ensure that the model could only attend to previous and current positions in the sequence, preventing it from using future information during training. This is crucial for time-series forecasting tasks, which aim to predict future values based on past and current observations. The look-ahead mask creates a triangular matrix that masks out future time steps, allowing the model to maintain the autoregressive property necessary for sequential data prediction.

L2 regularisation was applied to all dense layers to prevent overfitting, which penalises large weights and encourages the model to learn simpler patterns. The Adam optimiser was employed with an initial learning rate, and learning rate scheduling was implemented using the `ReduceLROnPlateau` callback. This callback reduced the learning rate by a factor of 0.5 when the validation loss plateaued, ensuring that the model continued to learn even when improvements slowed down. Early stopping was also employed using the `EarlyStopping` callback, which monitored the validation loss and stopped training if there was no improvement for 60 epochs, restoring the best weights observed during training. The training process for the Transformer model involved fitting the model on the training data and validating it using validation data to monitor the model's performance and prevent overfitting. The model was trained for up to 1000 epochs with a batch size tailored to the dataset.

3.3 Evaluation Metrics

This section discusses various statistical measures to assess the predictive models efficacy. Each metric provides unique insights into the models performance during testing, allowing for a comprehensive evaluation of accuracy and reliability.

Mean Absolute Error (MAE): The MAE measures the average magnitude of the errors between the model's predictions and the actual values, disregarding the direction of the errors. It provides a straightforward interpretation of the average prediction error. The formula for calculating MAE is presented below (Eq. 11):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

Here, y_i represents the true value, $\hat{y}_i$ the predicted value, and n the total count of data points. A lower MAE indicates a more accurate model, as it reflects smaller average errors.

Root Mean Square Error (RMSE): The RMSE estimates the error magnitude, emphasising larger errors due to the squaring of each error term. This metric is beneficial for understanding the model's performance in scenarios with significant outliers. The calculation for RMSE is as follows (Eq. 12):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (12)$$

The RMSE penalises more significant errors than smaller ones, making it sensitive to outliers. A lower RMSE value indicates better predictive accuracy.

R-squared (R^2): The R^2 metric offers a statistical measure of the fit, indicating how much of the variability in the dependent variable is explained by the model. It measures how well the observed outcomes are replicated by the model based on the proportion of total variation. It is calculated using (Eq. 13):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (13)$$

In this equation, $\bar{y}$ denotes the average of actual values. An R^2 value closer to 1 indicates that the model explains a higher proportion of the variance in the dependent variable, thus demonstrating a better fit.

Mean Absolute Percentage Error (MAPE): The MAPE quantifies the precision of the model as a percentage of error, making the performance easily interpretable in relative terms. It is advantageous when comparing prediction accuracy across different datasets. The MAPE is defined as (Eq. 14):

$$\text{MAPE} = \left(\frac{100}{n} \right) \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (14)$$

MAPE expresses the error as a percentage, making understanding the average magnitude of errors about the actual values easier. A lower MAPE indicates better model accuracy,

especially in a business context with more meaningful percentage errors.

Spearman Correlation Coefficient: The Spearman correlation coefficient assessed the monotonic relationship between the predicted and actual values. This non-parametric measure helps determine how well the predictions preserve the rank order of the data. Given the absence of labeled sentiment data, the Spearman correlation was beneficial for evaluating the relationship between sentiment scores and forex price movements. Using the Spearman correlation coefficient, the study determined which large language model (LLM) for sentiment analysis had the highest correlation with forex price movements, thereby selecting the most effective sentiment analysis model for the predictive models.

The models performance on the testing data can be comprehensively evaluated by employing these metrics, encapsulated from Eq. 11 to Eq. 14. Each metric provides a different perspective on the models accuracy, robustness, and practical utility, enabling a thorough comparison to identify the most precise and reliable model configurations. The combination of these metrics ensures a well-rounded assessment of the predictive models, facilitating informed decisions on model improvements and future applications.

3.4 Addressing Key Research Challenges

This section outlines how the key challenges identified in predicting forex prices such as volatility, integration of economic indicators, sentiment analysis, noise in data, and overfitting were addressed in the research.

- **High Volatility:** Forex markets are known for their high volatility, which complicates accurate predictions. To address this, the research uses LSTM and Transformer models, which excel in capturing long-term dependencies and patterns in sequential data. Additionally, the models are trained on both short-term (four-hour) and long-term (one-day) time frames, ensuring that they can adapt to varying market conditions.
- **Economic Indicators Integration:** The integration of various economic indicators into predictive models can be complex due to the diverse nature of the data. In this research, XGBoost is used to handle structured data, such as economic indicators, by efficiently managing non-linear relationships between multiple features. By combining technical and fundamental data, the model captures both short-term and long-term factors influencing forex prices.
- **Sentiment Analysis Integration:** Sentiment analysis provides valuable insights into market sentiment but poses challenges in integrating unstructured data with traditional technical and fundamental indicators. This research employs LLMs such as GPT-4 and GEMINI Advanced for sentiment extraction from financial news. These sentiment scores are then incorporated into the prediction models, allowing the model to consider market sentiment alongside traditional indicators.

- **Noise in Data:** Financial data is often noisy, which can obscure meaningful patterns and lead to inaccurate predictions. To reduce the impact of noise, this research employs pre-processing techniques such as feature selection and data smoothing. Ensemble methods, such as the combination of XGBoost and LSTM, also help mitigate the effects of noisy data by leveraging multiple models for more robust predictions.
- **Overfitting:** Overfitting is a common issue in financial forecasting, where models perform well on training data but fail to generalize to unseen data. In this study, overfitting is mitigated through the use of regularization techniques, cross-validation, dropout and early stopping during model training. These techniques ensure that the model generalizes well to new data.

By addressing these challenges systematically, the research develops more robust and accurate forex prediction models, capable of adapting to the volatile and complex nature of the forex market.

4 Results and Analysis

This section presents the results and analysis of the predictive models, focusing on their performance across different configurations and forex pairs. Specifically, the analysis covers models that utilise only technical indicators, models based on fundamental indicators, models integrating both technical and fundamental indicators, and models that incorporate sentiment analysis alongside technical and fundamental indicators. For each pair, the predictive accuracy of the models is assessed using a comprehensive suite of evaluation metrics. This section aims to determine the most effective combination of indicators and model configurations for accurate forex price prediction, providing insights into each approach's practical applicability and robustness.

4.1 Technical Models (TM)

The performance of the technical models (TM) on test data using the OHLC one-day time-frame is summarised in Table 2. This table highlights the results for various models, including LSTM_TM, XGBoost_TM, Ensemble-LSTM-XGB_TM, TRANSFORMER_EN_DE_TM, and TRANSFORMER_DE_TM, evaluated across four currency pairs: EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD.

The LSTM_TM model shows strong performance across all currency pairs, particularly with AUD/NZD, which achieved an RMSE of 0.0037. The XGBoost_TM model demonstrated exceptional accuracy, especially for EUR/NZD and AUD/NZD, with RMSE values of 0.0048 and 0.0017, respectively, indicating a very high level of precision. The Ensemble-LSTM-XGB_TM model combines the strengths of both LSTM and XGBoost, resulting in improved performance metrics for EUR/USD and NZD/USD, achieving RMSE values of 0.0064 and 0.0039, respectively.

The Transformer-based models, both the encoder-decoder (TRANSFORMER_EN_DE_TM) and the decoder-only (TRANSFORMER_DE_TM) architectures, showed varied performance. While they did not outperform the LSTM or XGBoost models overall, they still provided valuable insights, particularly for EUR/NZD and AUD/NZD, with the encoder-decoder model achieving an RMSE of 0.0148 for EUR/NZD and the decoder-only model achieving an RMSE of 0.0051 for AUD/NZD.

Table 2: Model Performance on Test Data Using Technical Indicators with OHLC One-Day Time Frame

Model	Currency Pair	MAE	MAPE	RMSE	R ²
LSTM_TM	EUR/USD	0.0415	0.4635%	0.0065	0.9888
	NZD/USD	0.0592	0.6291%	0.0050	0.9738
	EUR/NZD	0.0341	0.4273%	0.0092	0.9531
	AUD/NZD	0.0318	0.2619%	0.0037	0.9796
XGBoost_TM	EUR/USD	0.0077	0.7379%	0.0157	0.9363
	NZD/USD	0.0025	0.4079%	0.0048	0.9872
	EUR/NZD	0.0032	0.1846%	0.0048	0.9920
	AUD/NZD	0.0013	0.1258%	0.0017	0.9943
Ensemble-LSTM-XGB_TM	EUR/USD	0.0045	0.4184%	0.0064	0.9893
	NZD/USD	0.0030	0.4605%	0.0039	0.9917
	EUR/NZD	0.0037	0.2154%	0.0051	0.9913
	AUD/NZD	0.0016	0.1519%	0.0021	0.9917
TRANSFORMER_EN_DE_TM	EUR/USD	0.0222	2.0422%	0.0255	0.8295
	NZD/USD	0.0121	2.0033%	0.0181	0.6555
	EUR/NZD	0.0114	0.6832%	0.0148	0.8771
	AUD/NZD	0.0039	0.3565%	0.0049	0.9644
TRANSFORMER_DE_TM	EUR/USD	0.0172	1.6682%	0.0259	0.8246
	NZD/USD	0.0126	2.0856%	0.0176	0.6757
	EUR/NZD	0.0103	0.6152%	0.0132	0.9032
	AUD/NZD	0.0041	0.3741%	0.0051	0.9615

In this section, the four top-performing models, as highlighted in Table 2, will be further analysed using figures that compare predicted and actual prices. These visualisations provide a deeper understanding of how each model performs over time and across different market conditions, highlighting the strengths and areas for improvement in forex price prediction.

EUR/USD: The performance of the Ensemble-LSTM-XGB_TM model on EUR/USD test data is displayed in Figure 13. This figure shows the close price predictions generated by the ensemble model compared to the actual close prices over a specified period. The green line represents the actual close prices, while the red dashed line indicates the predicted close prices.



Figure 13: Performance of the Ensemble-LSTM-XGB_TM model on EUR/USD Test Data.

The figure shows the ensemble model's ability to closely follow market trends, with minimal deviations between the predicted and actual values. This alignment underscores the model's robustness in capturing the underlying patterns and movements in the EUR/USD forex pair, providing high predictive accuracy. The close correspondence between the predicted and actual prices indicates the efficacy of integrating LSTM and XGBoost models, leveraging both temporal dependencies and gradient-boosting techniques to enhance predictive performance.

NZD/USD: The performance of the Ensemble-LSTM-XGB_TM model on NZD/USD test data is displayed in Figure 14. The figure shows the predicted close prices in the red dashed line against the actual close prices in the green line over the testing period. The close alignment between the two lines demonstrates the model's strong predictive capability, accurately capturing the market trends and price movements for the NZD/USD pair. This indicates the model's effectiveness in leveraging LSTM and XGBoost techniques to enhance forex price prediction accuracy.



Figure 14: Performance of the Ensemble-LSTM-XGB_TM model on NZD/USD Test Data.

EUR/NZD:



Figure 15: Performance of the XGBoost_TM model on EUR/NZD Test Data.

Figure 15 displays the performance of the XGBoost_TM model on EUR/NZD test data. Cross-validation helps prevent overfitting by validating the model's performance on multiple

subsets of the data. As a result, only the train and test periods are displayed in the figure, reflecting the model's application to previously unseen data. The close alignment of the predicted and actual prices during the testing phase demonstrates the XGBoost_TM model's proficiency in forecasting forex price movements. This accuracy indicates that the model effectively utilises technical indicators to capture market trends and predict future prices for the EUR/NZD pair.

AUD/NZD:



Figure 16: Performance of the XGBoost_TM model on AUD/NZD Test Data.

Figure 16 displays the performance of the XGBoost_TM model on the AUD/NZD test data. The model parameters were optimised using cross-validation to avoid overfitting. The close match between the predicted and actual prices during the testing phase indicates the model's efficacy in predicting AUD/NZD price movements based on technical indicators. The following section will examine the performance of the fundamental models. This will include evaluating how well these models predict forex price movements based on fundamental economic indicators for each currency pair.

4.2 Fundamental Models (FM)

This section assesses the performance of various models that leverage fundamental economic indicators to predict forex price movements. These fundamental indicators include GDP, inflation rates, interest rates, and trade balances, which provide insights into the broader economic conditions influencing currency values.

Table 3: Model Performance on Test Data Using Fundamental Indicators with OHLC One-Day Time Frame

Model	Currency Pair	MAE	MAPE	RMSE	R ²
LSTM_FM	EUR/USD	0.1097	1.2457%	0.0190	0.9079
	NZD/USD	0.0705	0.7480%	0.0059	0.9630
	EUR/NZD	0.0647	0.8150%	0.0165	0.8485
	AUD/NZD	0.0715	0.5889%	0.0075	0.9165
XGBoost_FM	EUR/USD	0.0077	0.7379%	0.0157	0.9363
	NZD/USD	0.0068	1.0679%	0.0093	0.9525
	EUR/NZD	0.0051	0.2982%	0.0068	0.9844
	AUD/NZD	0.0023	0.2124%	0.0029	0.9845
Ensemble-LSTM-XGB_FM	EUR/USD	0.0101	0.9396%	0.0155	0.9384
	NZD/USD	0.0046	0.7017%	0.0057	0.9825
	EUR/NZD	0.0061	0.3566%	0.0078	0.9796
	AUD/NZD	0.0029	0.2688%	0.0036	0.9751
TRANSFORMER_EN_DE_FM	EUR/USD	0.0290	2.7551%	0.0379	0.6250
	NZD/USD	0.0176	2.8499%	0.0226	0.4642
	EUR/NZD	0.0268	1.5957%	0.0345	0.3354
	AUD/NZD	0.0169	1.5886%	0.0204	0.3792
TRANSFORMER_DE_FM	EUR/USD	0.0253	2.3743%	0.0292	0.7794
	NZD/USD	0.0139	2.2515%	0.0178	0.6694
	EUR/NZD	0.0228	1.3748%	0.0293	0.5187
	AUD/NZD	0.0135	1.2685%	0.0171	0.5614

The LSTM_FM model demonstrated varied performance across different currency pairs, with the lowest RMSE observed for the NZD/USD pair. This suggests that the LSTM_FM model is particularly effective at capturing the fundamental economic factors influencing the NZD/USD exchange rate. The XGBoost model, when applied to fundamental indicators, outperformed the LSTM model in terms of RMSE for the EUR/NZD and AUD/NZD pairs. This indicates that the XGBoost model can capture the complex relationships inherent in the fundamental economic data of these pairs. Notably, the XGBoost_FM model achieved the lowest RMSE for EUR/NZD and AUD/NZD pairs, showcasing its robustness in handling fundamental data.

The Ensemble-LSTM-XGB_FM model, which combines the strengths of both LSTM and XGBoost, provided the best overall performance for EUR/USD and NZD/USD pairs. This ensemble approach leverages the temporal dependencies captured by LSTM and the gradient-boosting capabilities of XGBoost, resulting in superior predictive accuracy. Specifically, the Ensemble-LSTM-XGB_FM model achieved the lowest RMSE for EUR/USD and NZD/USD pairs, highlighting its effectiveness in integrating multiple data types. The Transformer models, including the encoder-decoder (TRANSFORMER_EN_DE_FM) and decoder-only (TRANSFORMER_DE_FM) architectures, showed lower performance than LSTM and XGBoost models. This suggests that while Transformer models excel in natural language processing tasks, they may be less effective for time series forecasting with fundamental economic indicators in this context. Evaluating these models highlights the importance of selecting appropriate model architectures and integrating various data types to improve predictive accuracy. The results indicate that combining different models and data types, such as in the Ensemble-LSTM-XGB_FM model, can significantly enhance forex price predictions based on fundamental indicators. The detailed performance metrics for these models are presented in Table 3.

EUR/USD:



Figure 17: Performance of the Ensemble-LSTM-XGB_FM model on EUR/USD Test Data.

The performance of the Ensemble-LSTM-XGB_FM model on the EUR/USD test data is displayed in Figure 17. The green line represents the close prices, while the red dashed line indicates the predicted prices. The plot demonstrates the model's ability to closely follow the actual price movements, capturing trends and fluctuations over the testing period. Using a

weighted averaging technique, the ensemble model integrates the predictions from both LSTM and XGBoost, improving the accuracy and reliability of the predictions. This approach allows the model to utilise the complementary strengths of each method, leading to improved predictive performance for the EUR/USD currency pair.

NZD/USD: Figure 18 shows the performance of the Ensemble-LSTM-XGB_FM model on NZD/USD test data. The model effectively tracks actual market movements, accurately capturing trends and fluctuations during testing. By employing weighted averaging, this ensemble technique combines predictions from LSTM and XGBoost models, leveraging their strengths to enhance overall forecasting performance for the NZD/USD currency pair.



Figure 18: Performance of the Ensemble-LSTM-XGB_FM model on NZD/USD Test Data.

EUR/NZD: The performance of the XGBoost_FM model on the EUR/NZD test data is depicted in Figure 19. This figure shows how the model predicts the closing prices of the EUR/NZD currency pair based on fundamental indicators.

The model was trained on data from 2004 to 2019 and tested on data from 2020 to 2024. The separation between the training and testing periods is visually marked, highlighting the model's performance in the unseen data during the testing phase. The figure illustrates the model's ability to capture the general trend and significant turning points in the EUR/NZD currency pair, showcasing its effectiveness in leveraging fundamental indicators to forecast price movements. Cross-validation during the training phase ensured the model was robust and generalised well to the testing data. The figure demonstrates the model's ability to predict close prices accurately, indicating that the fundamental indicators used were relevant and provided

valuable information for forecasting the EUR/NZD currency pair.

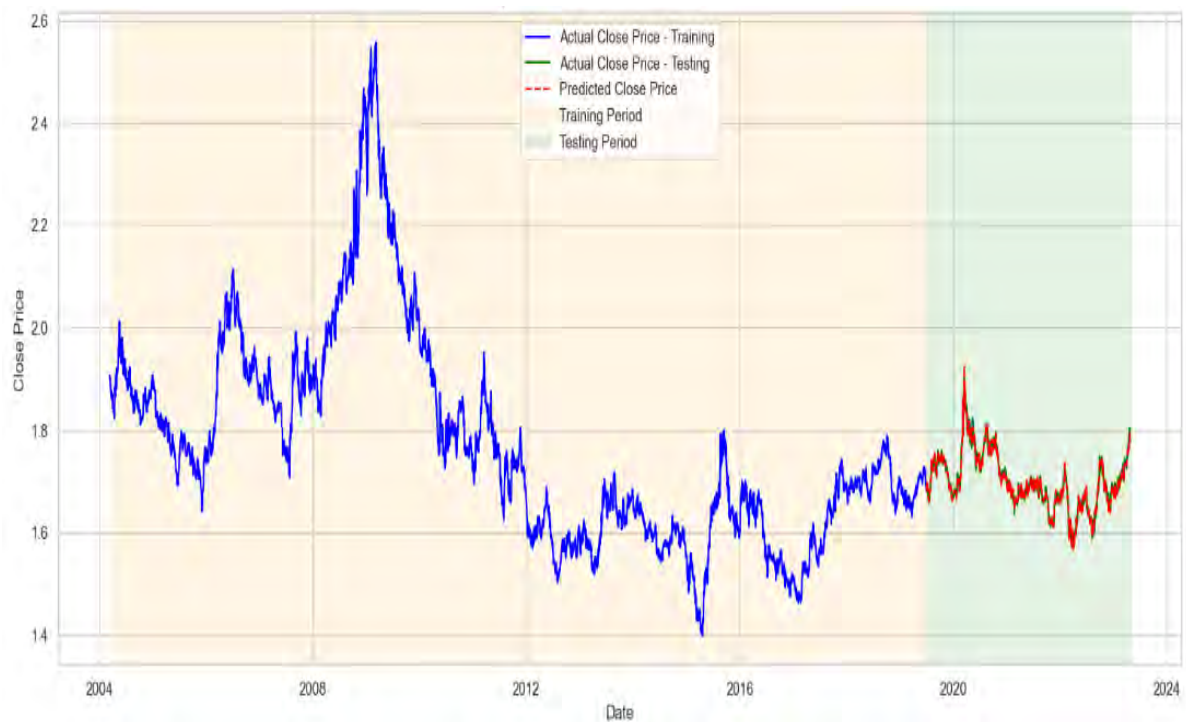


Figure 19: Performance of the XGBoost_FM model on EUR/NZD Test Data.

AUD/NZD:



Figure 20: Performance of the XGBoost_FM model on AUD/NZD Test Data.

The performance of the XGBoost_FM model on the AUD/NZD test data is shown in

Figure 20. This figure exhibits the model’s ability to predict the closing prices of the AUD/NZD currency pair using fundamental indicators. The blue line represents the closing prices during the training period, while the green line shows the closing prices during the testing period. The red dashed line indicates the predicted closing prices during the testing period.

Up to this stage, we have assessed the performances of the technical and fundamental models separately. The evaluation metrics and visualisations have offered insights into the contributions of each approach in predicting forex price movements. The technical models utilised indicators derived from OHLC data to capture market trends and price dynamics, while the fundamental models integrated economic indicators to reflect broader market influences and economic conditions. In the next section, we will explore the results of models that integrate both technical and fundamental indicators. By combining these two approaches, the models aim to harness the strengths of each methodology, providing a more comprehensive analysis of market behavior and improving predictive accuracy. This integrated approach offers a holistic view of the factors influencing forex prices, enhancing our understanding and forecasting capabilities.

4.3 Technical and Fundamental Models (TM_FM)

This section evaluates models combining technical and fundamental indicators (TM_FM) over a one-day timeframe. Table 4 presents the results for four forex pairs: EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD. The LSTM_TM_FM model performs well, especially for EUR/NZD, with an MAE of 0.0476 and an RMSE of 0.0123. For EUR/USD, this model achieves an MAE of 0.1140, a MAPE of 1.2640%, and an RMSE of 0.0159. Although it performs adequately for NZD/USD and AUD/NZD, its accuracy is somewhat lower than for EUR/NZD, indicating varying model performance across different currency pairs. The XGBoost_TM_FM often outperforms the LSTM model, particularly for EUR/NZD and AUD/NZD pairs. For EUR/NZD, it achieves an MAE of 0.0046, a MAPE of 0.2708%, and an RMSE of 0.0061, showing high accuracy. The XGBoost model also excels for AUD/NZD, with an MAE of 0.0020 and an RMSE of 0.0025, demonstrating its robustness in incorporating technical and fundamental indicators.

The Ensemble model, which combines LSTM and XGBoost (Ensemble-LSTM-XGB_TM_FM), best performs for EUR/USD and NZD/USD pairs. Using a weighted average of predictions from both models, an MAE of 0.0092 and an RMSE of 0.0118 for EUR/USD are achieved. For NZD/USD, it records an MAE of 0.0037 and an RMSE of 0.0052, highlighting the advantage of integrating multiple model architectures to improve predictive accuracy. Transformer models, TRANSFORMER_EN_DE_TM_FM, and TRANSFORMER_DE_TM_FM architectures perform less effectively than LSTM and XGBoost models. The TRANSFORMER_EN_DE_TM_FM model has an RMSE of 0.0306 for EUR/USD and 0.0184 for NZD/USD, indicating that while Transformer models can capture complex dependencies in sequential data, they are not as effective in this context as LSTM and XGBoost models.

Table 4: Model Performance on Test Data Using Technical and Fundamental Indicators with OHLC One-Day Time Frame

Model	Currency Pair	MAE	MAPE	RMSE	R ²
LSTM_TM_FM	EUR/USD	0.1140	1.2640%	0.0159	0.9341
	NZD/USD	0.0931	1.0096%	0.0081	0.9306
	EUR/NZD	0.0476	0.5979%	0.0123	0.9153
	AUD/NZD	0.0788	0.6518%	0.0082	0.9004
XGBoost_TM_FM	EUR/USD	0.0077	0.7379%	0.0157	0.9363
	NZD/USD	0.0042	0.6703%	0.0062	0.9794
	EUR/NZD	0.0046	0.2708%	0.0061	0.9868
	AUD/NZD	0.0020	0.1888%	0.0025	0.9879
Ensemble-LSTM-XGB_TM_FM	EUR/USD	0.0092	0.8477%	0.0118	0.9641
	NZD/USD	0.0037	0.5701%	0.0052	0.9856
	EUR/NZD	0.0049	0.2899%	0.0064	0.9859
	AUD/NZD	0.0026	0.2420%	0.0032	0.9805
TRANSFORMER_EN_DE_TM_FM	EUR/USD	0.0229	2.1760%	0.0306	0.7547
	NZD/USD	0.0127	2.0954%	0.0184	0.6440
	EUR/NZD	0.0280	1.6832%	0.0333	0.3814
	AUD/NZD	0.0126	1.1768%	0.0161	0.6108
TRANSFORMER_DE_TM_FM	EUR/USD	0.0201	1.8884%	0.0264	0.8200
	NZD/USD	0.0145	2.3681%	0.0179	0.6677
	EUR/NZD	0.0208	1.2425%	0.0250	0.6494
	AUD/NZD	0.0095	0.8929%	0.0130	0.7493

Table 4 shows that combining technical and fundamental indicators improves the model's predictive abilities, with XGBoost and ensemble models leading in accuracy. This combined approach effectively captures the complexity of forex market movements, offering valuable insights for forex price prediction.

EUR/USD: The performance of the Ensemble-LSTM-XGB_TM_FM model on EUR/USD test data is illustrated in Figure 21. This model combines the LSTM and XGBoost models' predictive strengths, leveraging technical and fundamental indicators. The green line represents the closing prices, while the red dashed line shows the predicted closing prices.



Figure 21: Performance of the Ensemble-LSTM-XGB_TM_FM model on EUR/USD Test Data.

The chart demonstrates the model's robust capability to capture the general trend and fluctuations in the EUR/USD closing prices over time. Notably, the ensemble model effectively tracks the peaks and troughs of the actual prices, indicating its strength in forecasting both upward and downward movements. By incorporating a weighted average technique, the ensemble model enhances prediction accuracy, balancing the LSTM's ability to capture temporal dependencies with XGBoost's proficiency in handling structured data. The close alignment between the actual and predicted prices underscores the model's effectiveness, particularly in capturing complex market dynamics and interactions between various technical and fundamental indicators.

NZD/USD: Figure 22 displays the performance of the Ensemble-LSTM-XGB_TM_FM model on the NZD/USD test data. This ensemble model synergistically combines the temporal pattern recognition of LSTM with the structured data handling capabilities of XGBoost, utilising both technical and fundamental indicators. The chart highlights the model's proficiency in accurately predicting price movements, especially during significant market shifts. This combined approach ensures a more robust and reliable forecast, as evidenced by the close tracking of the actual prices by the predicted values.



Figure 22: Performance of the Ensemble-LSTM-XGB_TM_FM model on NZD/USD Test Data.

EUR/NZD:

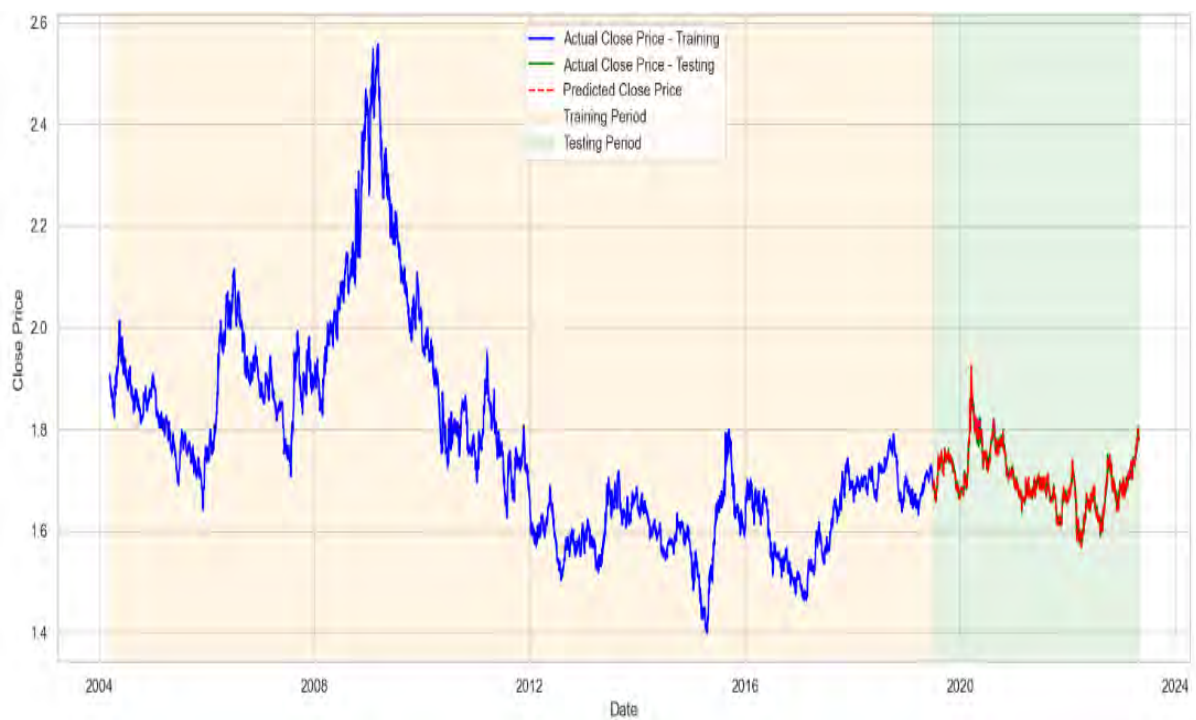


Figure 23: Performance of the XGBoost_TM_FM model on EUR/NZD Test Data.

Figure 23 showcases the performance of the XGBoost_TM_FM model on the EUR/NZD test data. This model combines technical and fundamental indicators using the robust gradient-boosting framework of XGBoost. The blue line represents the closing prices during the training

period, while the red line indicates the actual prices during the testing period. The dashed red line shows the model's predicted closing prices. The model effectively captures the price movements and trends in the EUR/NZD pair, closely aligning with the actual data.

AUD/NZD: Figure 24 displays the performance of the XGBoost_TM_FM model on the AUD/NZD test data, spanning from 2004 to 2024. In this figure, the model combines both technical and fundamental indicators using the XGBoost framework to predict the closing prices of the AUD/NZD pair.



Figure 24: Performance of the XGBoost_TM_FM model on AUD/NZD Test Data.

This close tracking of the actual prices by the predicted values highlights the robustness and reliability of the XGBoost_TM_FM model in forecasting forex prices. The training and testing periods are distinctly separated to ensure the model's performance is thoroughly evaluated on unseen data, thus preventing data leakage. Furthermore, cross-validation during the training phase optimised the model's parameters. This method ensures that the model's performance is not overly dependent on a particular subset of the data, promoting generalisability and robustness across different market conditions. The figure underscores the effectiveness of integrating both technical and fundamental indicators to improve the predictive accuracy of the AUD/NZD currency pair over an extended timeframe.

4.4 Technical, Fundamental, and Sentiment Models (TM_FM_SA)

This section presents the performance evaluation of models that integrate technical indicators, fundamental indicators, and sentiment analysis (TM_FM_SA) in a Four-hour timeframe. Table 5 summarises the results of these models across the four forex pairs. The LSTM_TM_FM_SA model shows varying performance across currency pairs. For EUR/USD, it achieves an MAE of 0.0906, a MAPE of 0.1364%, and an RMSE of 0.0020, with an R^2 of 0.8967. The model performs similarly well for EUR/NZD with an MAE of 0.0811 and an RMSE of 0.0043. However, the model's performance on NZD/USD and AUD/NZD pairs is comparatively lower, indicating that while LSTM can effectively integrate multiple data types, its accuracy can vary depending on the currency pair.

The XGBoost_TM_FM_SA model significantly outperforms other models in this integrated approach, particularly for EUR/USD and NZD/USD pairs. For EUR/USD, the model achieves a remarkably low MAE of 0.0007, a MAPE of 0.0707%, and an RMSE of 0.0010, with an R^2 of 0.9897. The performance is similarly vital for NZD/USD, EUR/NZD, and AUD/NZD, indicating that XGBoost excels in leveraging technical, fundamental, and sentiment data to enhance predictive accuracy. The Ensemble-LSTM-XGB_TM_FM_SA model also shows robust performance, particularly for EUR/USD and NZD/USD pairs. The ensemble approach, which uses weighted averaging to combine the predictions of both models, achieves a good balance between the strengths of LSTM and XGBoost. For EUR/USD, it achieves an MAE of 0.0008, a MAPE of 0.0759%, and an RMSE of 0.0010, with an R^2 of 0.9873.

The Transformer models, both TRANSFORMER_EN_DE_TM_FM_SA and TRANSFORMER_DE_TM_FM_SA architectures, exhibit lower performance compared to LSTM and XGBoost models. For instance, the TRANSFORMER_EN_DE_TM_FM_SA model achieves an RMSE of 0.0042 for EUR/USD, significantly higher than the XGBoost model. This suggests that while Transformer models are powerful in capturing complex dependencies in sequential data, their performance in this context is less effective than LSTM and XGBoost.

Table 5 clearly shows that integrating technical indicators, fundamental indicators, and sentiment analysis data enhances the predictive capabilities of the models, with XGBoost leading in accuracy and the ensemble model providing a balanced performance across different currency pairs. This comprehensive integration approach effectively captures the multifaceted nature of forex market movements, offering valuable insights for forex price prediction.

Table 5: Model Performance on Test Data with Technical, Fundamental, and Sentiment Analysis on a 4H Timeframe

Model	Currency Pair	MAE	MAPE	RMSE	R²
LSTM_TM_FM_SA	EUR/USD	0.0906	0.1364%	0.0020	0.8967
	NZD/USD	0.2492	0.6308%	0.0042	0.6833
	EUR/NZD	0.0811	0.2015%	0.0043	0.8896
	AUD/NZD	0.1403	0.1334%	0.0017	0.6719
XGBoost_TM_FM_SA	EUR/USD	0.0007	0.0707%	0.0010	0.9897
	NZD/USD	0.0007	0.1199%	0.0009	0.9942
	EUR/NZD	0.0017	0.0990%	0.0021	0.9922
	AUD/NZD	0.0004	0.0434%	0.0005	0.9837
Ensemble-LSTM-XGB_TM_FM_SA	EUR/USD	0.0008	0.0759%	0.0010	0.9873
	NZD/USD	0.0013	0.2155%	0.0016	0.9811
	EUR/NZD	0.0019	0.1086%	0.0024	0.9895
	AUD/NZD	0.0005	0.0524%	0.0007	0.9761
TRANSFORMER_EN_DE_TM_FM_SA	EUR/USD	0.0034	0.3117%	0.0042	0.5254
	NZD/USD	0.0030	0.4805%	0.0036	0.7657
	EUR/NZD	0.0080	0.4546%	0.0093	0.4854
	AUD/NZD	0.0019	0.1726%	0.0023	0.4068
TRANSFORMER_DE_TM_FM_SA	EUR/USD	0.0022	0.2015%	0.0028	0.7878
	NZD/USD	0.0019	0.3095%	0.0024	0.8936
	EUR/NZD	0.0058	0.3328%	0.0070	0.7075
	AUD/NZD	0.0016	0.1459%	0.0019	0.5935

EUR/USD: Bellow Figure 25 shows the performance of the XGBoost_TM_FM_SA model on the EUR/USD test data. This model integrates technical, fundamental, and sentiment analysis indicators using the XGBoost framework to predict the closing prices of the EUR/USD currency pair. The blue line in the figure represents the actual closing prices during the training period from January 2023 to October 2023, while the red line denotes the actual closing prices during the testing period from October 2023 to January 2024. The dashed red line signifies the predicted closing prices by the model during the testing phase.

The close alignment between the predicted and actual prices in the testing period highlights the model's strong predictive capabilities, accurately reflecting the market trends and

price movements. The use of cross-validation in the training phase ensured that the model was not overfitted to any specific subset of data, enhancing its generalisability and robustness across different periods and market conditions. The shaded areas distinguish the training period (light blue) and the testing period (light green), indicating that the model was evaluated on unseen data to validate its predictive performance. This separation ensures the reliability and robustness of the model's predictions. By incorporating sentiment analysis, the XGBoost_TM_FM_SA model leverages real-time market sentiment, further refining the predictive accuracy for the EUR/USD currency pair.

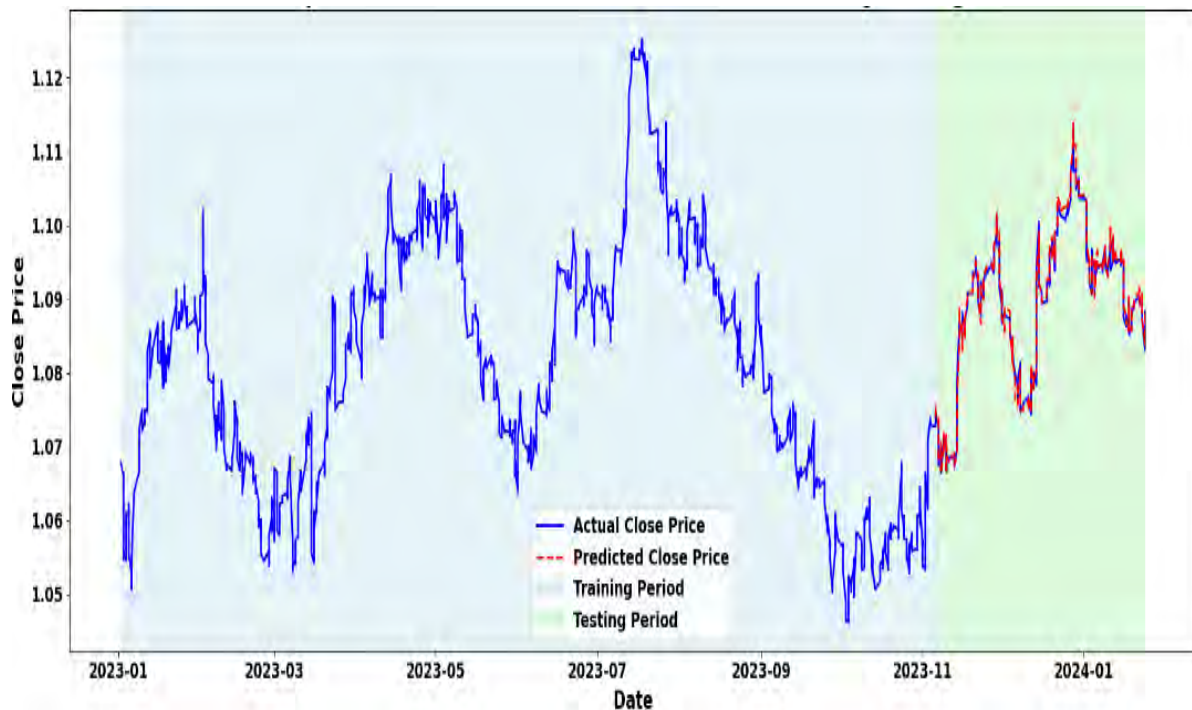


Figure 25: Performance of the XGBoost_TM_FM_SA model on EUR/USD Test Data.

NZD/USD: Figure 26 illustrates the performance of the XGBoost_TM_FM_SA on the NZD/USD test data. This model employs a combination of technical, fundamental, and sentiment analysis indicators within the XGBoost framework to predict the closing prices of the NZD/USD currency pair. The alignment between the predicted and actual prices during the testing period underscores the model's robust predictive capabilities, effectively capturing the trends and fluctuations in the market. Cross-validation was used during the training phase to prevent overfitting, ensuring the model's performance was consistent across various subsets of the data, thereby enhancing its generalisability.

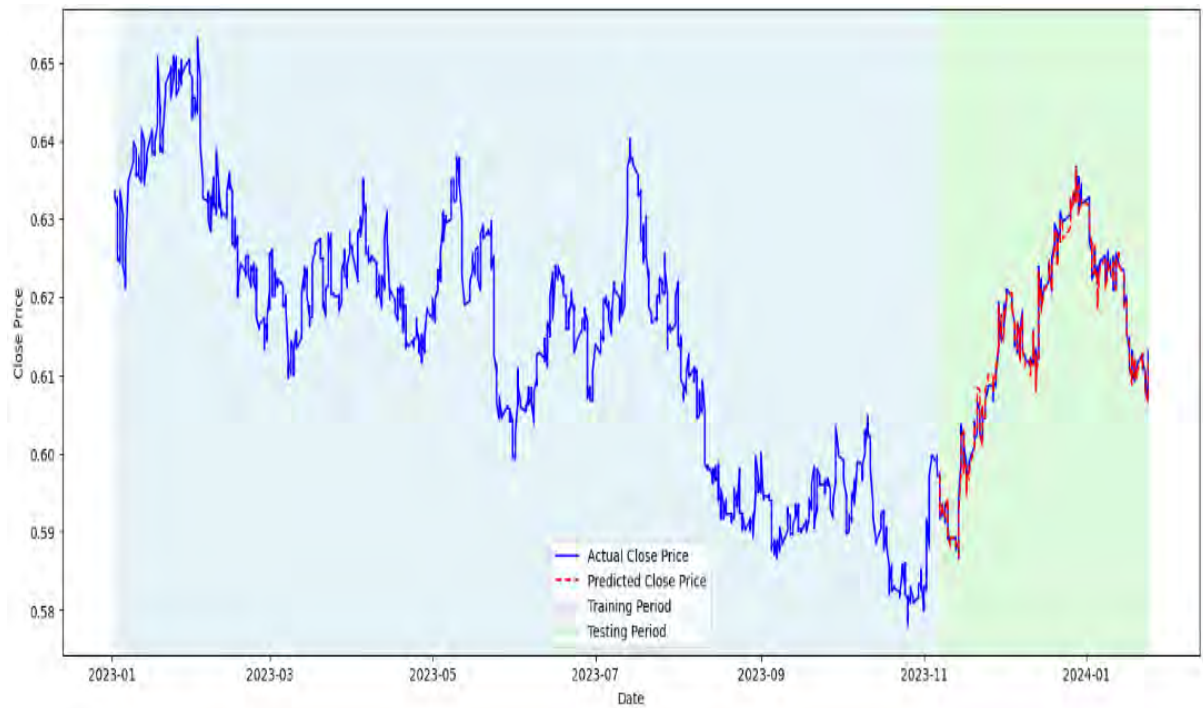


Figure 26: Performance of the XGBoost_TM_FM_SA model on NZD/USD Test Data.

EUR/NZD:

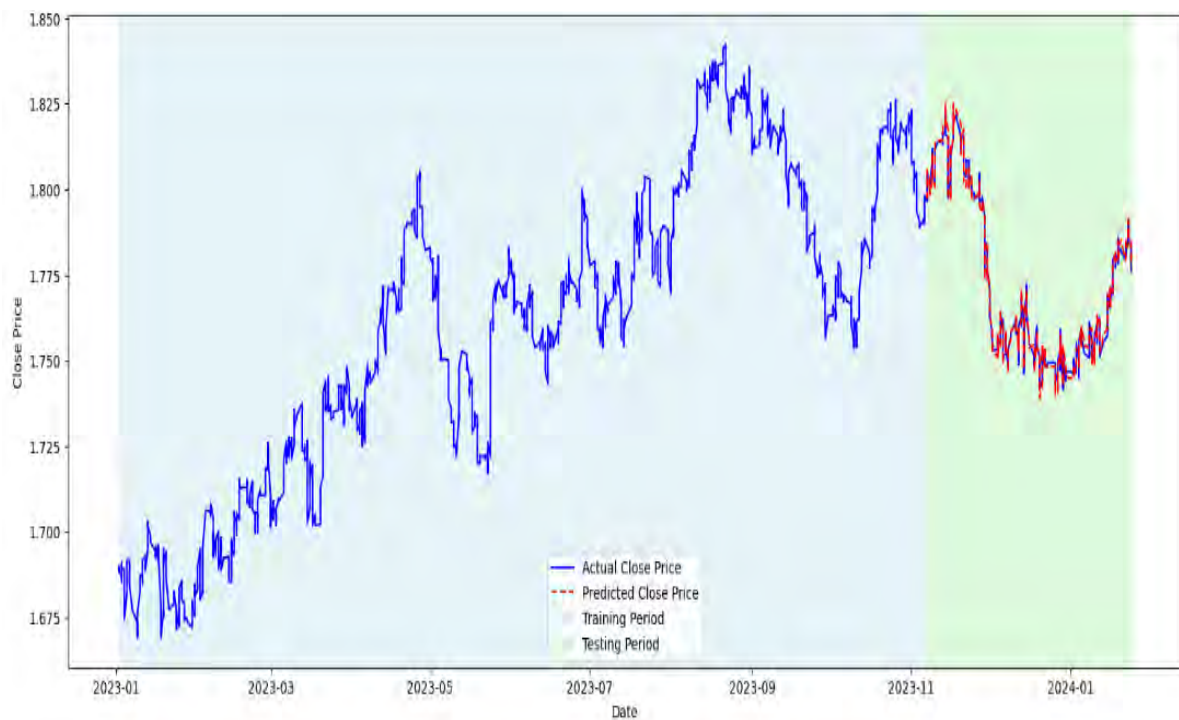


Figure 27: Performance of the XGBoost_TM_FM_SA model on EUR/NZD Test Data.

Figure 27 illustrates the performance of the XGBoost_TM_FM_SA model on EUR/NZD test data. This model combines technical indicators, fundamental data, and sentiment analysis

to predict closing prices. The blue line represents actual prices during the training period (January 2023 to October 2023), while the red line shows actual prices during the testing period (October 2023 to January 2024). The dashed red line indicates the model's predictions. The high alignment between predicted and actual prices in the testing period highlights the model's effectiveness. Cross-validation during training ensures a robust performance.

AUD/NZD:Figure 28 shows how the XGBoost_TM_FM_SA model performs on AUD/NZD test data. This model uses a combination of technical, fundamental, and sentiment analysis indicators to predict closing prices. In the figure, the blue line represents the actual closing prices during the training period, while the solid red line shows the actual prices during the testing period. The shaded regions distinguish between the training period (light blue) and the testing period (light green), showing that the model's effectiveness was validated on new, unseen data. The dashed red line indicates the prices predicted by the model during the testing period.

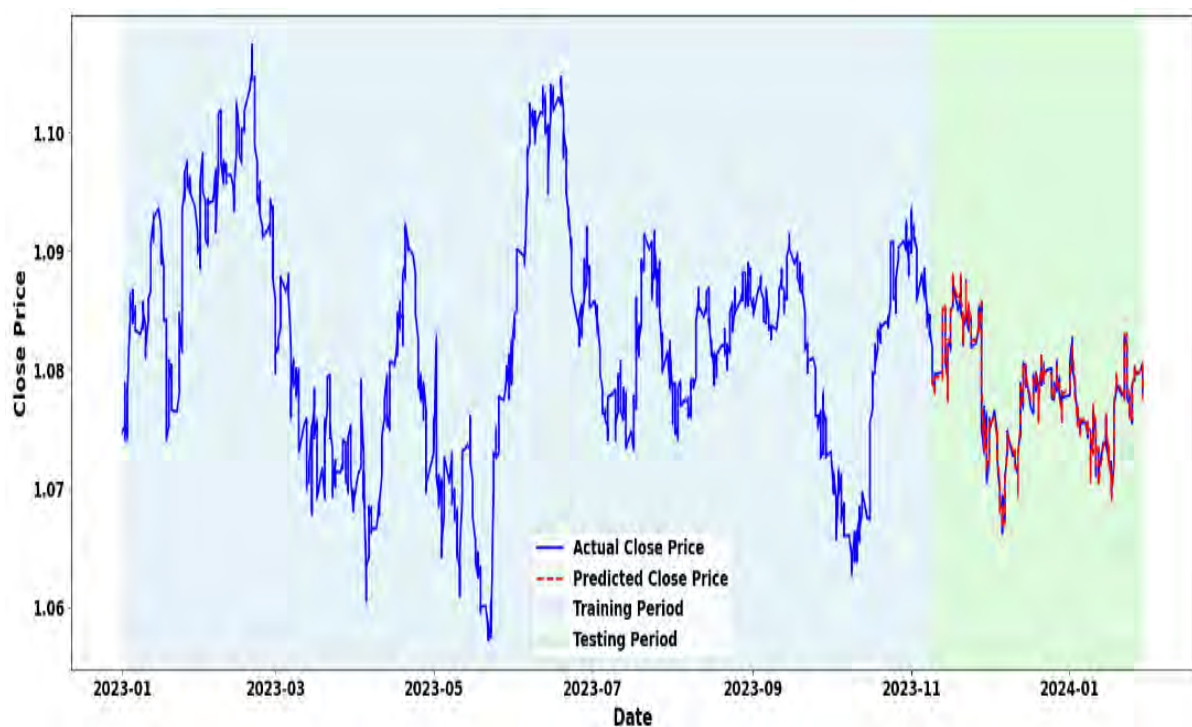


Figure 28: Performance of the XGBoost_TM_FM_SA model on AUD/NZD Test Data.

4.5 Analysis and Discussion

In the previous section, the performance results of various predictive models were presented, highlighting their effectiveness in forecasting forex prices based on technical, fundamental, and combined indicators. This section is a detailed analysis of these results. This analysis will help to identify the most effective approaches for forex price prediction and provide insights into how these models can be further improved for better generalisation and practical application in

real-world scenarios.

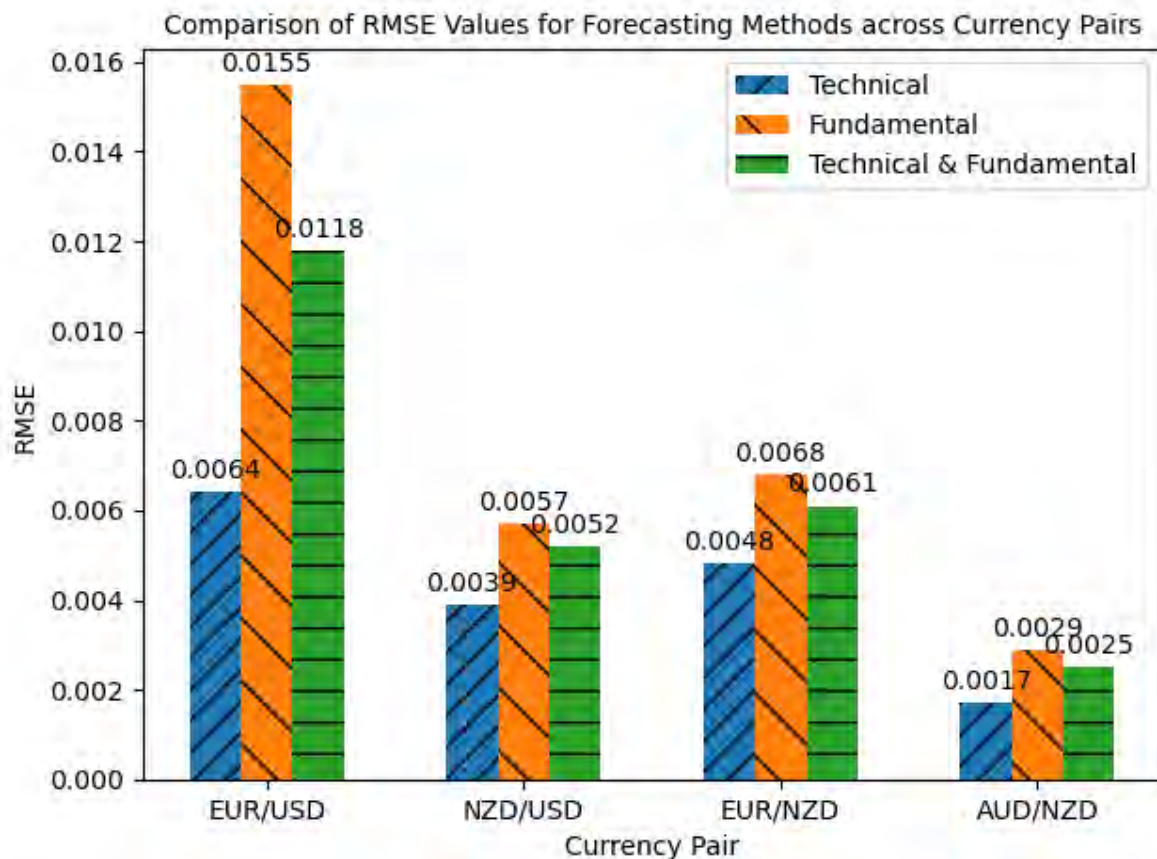


Figure 29: Comparative Analysis of RMSE Values Across Forex Currency Pairs Using Different Prediction Models for One Day Time Frame.

Figure 29 reveals that models using technical indicators alone consistently achieved the lowest RMSE values across all currency pairs when compared to models using fundamental indicators and those combining both types of indicators.

For the EUR/USD pair, the technical model attained an RMSE of 0.0064, which is significantly lower than the RMSE values for the fundamental model (0.0155) and the combined model (0.0118). Similarly, the technical model for NZD/USD had an RMSE of 0.0039, outperforming both the fundamental model (0.0057) and the combined model (0.0052). The EUR/NZD pair followed the same pattern, with the technical model achieving the lowest RMSE of 0.0048, compared to 0.0068 for the fundamental model and 0.0061 for the combined model. The AUD/NZD pair also demonstrated the superior performance of the technical model, with an RMSE of 0.0017, lower than both the fundamental model (0.0029) and the combined model (0.0025).

These results indicate that technical indicators alone provide the most accurate forecasts for forex prices in these scenarios. The technical indicators are derived directly from the OHLC data, which might explain their superior performance. The inclusion of fundamental indicators did not significantly improve the predictive accuracy and, in some cases, slightly increased

the RMSE. However, while the technical models show the lowest RMSE, the combined use of technical and fundamental indicators could still be valuable. Combining both approaches might help generalise the model better, potentially offering robustness against different market conditions and reducing overfitting. Therefore, while technical indicators appear to be the most effective for minimising RMSE in this study, the integration of both technical and fundamental indicators may provide a more generalised and potentially more resilient forecasting model in the long run.

Next, the discussion will focus on analysing the Four-hour time frame data, integrating technical, fundamental, and sentiment indicators to assess the performance of different models. By examining the correlation between sentiment scores and the ROC_9 indicator and evaluating the predictive accuracy through various metrics, the aim is to determine the most effective model configurations for short-term forex market forecasting.

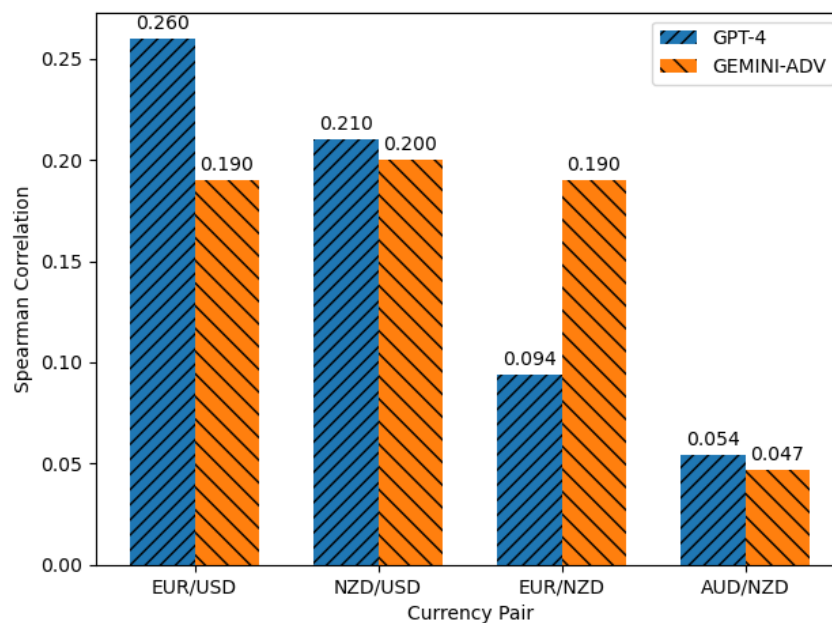


Figure 30: ROC_9 Spearman Correlation of Sentiment Analysis on Forex Pairs Using Four-Hour Time Frame Data

Figure 30 shows the Spearman correlation coefficients between sentiment scores and the ROC_9 indicator for the EUR/USD, NZD/USD, EUR/NZD, and AUD/NZD currency pairs. The GPT-4 model demonstrated a higher correlation for EUR/USD (0.260) and NZD/USD (0.210) than the GEMINI Advanced model. Conversely, GEMINI Advanced showed a stronger correlation for EUR/NZD (0.190), while both models performed similarly for AUD/NZD. These results guided the selection of sentiment analysis models, with GPT-4 chosen for EUR/USD and NZD/USD and GEMINI Advanced for EUR/NZD, enhancing the predictive accuracy of our models.

A key observation from the results is the consistently superior performance of the XG-Boost models, particularly noted for their lower RMSE values across most currency pairs. This

exceptional performance is detailed in Table 5:

- The XGBoost model combined with technical, fundamental, and sentiment analysis (XGBoost_TM_FM_SA) achieved the lowest RMSE of 0.001022 for the EUR/USD pair, indicating high predictive accuracy, as visually represented in Figure 25.
- Similarly, for the NZD/USD and EUR/NZD pairs, the same XGBoost configuration reported RMSE values of 0.000924 and 0.002145, respectively, outperforming other model setups.
- For the AUD/NZD pair, the XGBoost_TM_FM_SA model also demonstrated superior performance with the lowest RMSE of 0.000594, further confirming its robustness across different forex markets, as shown in Figure 28.

These results underscore the effectiveness of XGBoost in handling complex datasets where multiple data streams are integrated. The analysis also reveals varying levels of performance across different models and currency pairs, illustrating the complexity of forex market forecasting. The R^2 values complement the RMSE results by providing insights into the variance the models explain, with higher R^2 indicating better model fit to the data.



Figure 31: Performance of the XGB_TM model on EUR/USD Test Data.

The use of ensemble models in forex price prediction highlights the significant improvement in performance and generalisation of test data. As illustrated in Figure 31 and Figure 32, both XGBoost and LSTM models individually demonstrate robust predictive capabilities. The XGBoost model excels in handling complex datasets and capturing non-linear relationships,

while the LSTM model effectively leverages its sequential memory structure to model temporal dependencies.

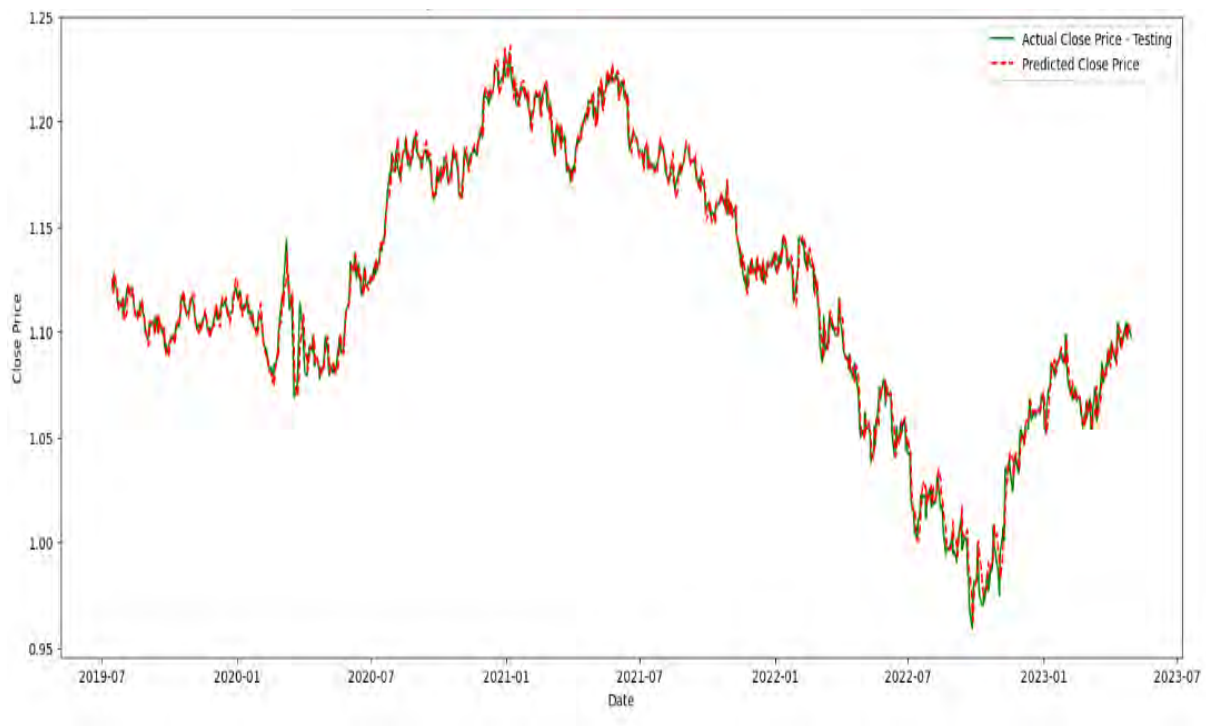


Figure 32: Performance of the LSTM_TM model on EUR/USD Test Data.

However, the ensemble model, which combines the strengths of both XGBoost and LSTM, further enhances predictive accuracy and generalisation, as shown in Figure 13. By integrating these models through a weighted average technique, the ensemble model mitigates the individual weaknesses of each model, leading to more accurate and reliable predictions. This approach harnesses the superior generalisation capability of ensemble learning, reducing overfitting and improving performance across various market conditions.

4.5.1 Addressing the Research Questions

Research Question 1: How do technical indicators compare with models, including fundamental indicators and sentiment analysis, in predicting forex prices?

As demonstrated in Figure 29, models using technical indicators alone consistently outperformed those incorporating both fundamental and sentiment indicators, particularly for the one-day time frame. For example, the technical model for EUR/USD achieved the lowest RMSE, indicating the effectiveness of technical models in capturing immediate market movements. Although combined models integrating fundamental and sentiment indicators offer broader market insights, their inclusion did not lead to lower RMSE values, suggesting that technical indicators are better suited for short-term forex price predictions.

Research Question 2: How does sentiment analysis affect the accuracy of forex price predictions?

The analysis in Figure 30 demonstrates how sentiment analysis improves prediction accuracy, particularly for the four-hour time frame. The positive correlation between sentiment scores and the ROC_9 indicator, especially for EUR/USD and NZD/USD pairs, suggests that sentiment analysis significantly enhances short-term model performance. The results indicate that sentiment analysis plays an important role in improving the accuracy of forex price predictions when combined with technical indicators.

Research Question 3: Which machine learning or deep learning models most effectively integrate these different data types for forex prediction?

The findings, as shown in Table 5, highlight the superior performance of XGBoost and Ensemble-LSTM-XGB models in integrating technical, fundamental, and sentiment data. XGBoost consistently delivered the lowest RMSE values across most currency pairs, demonstrating its ability to handle complex datasets with multiple data sources. Additionally, the Ensemble-LSTM-XGB model enhanced accuracy by leveraging the strengths of both LSTM and XGBoost, making it a robust approach for integrating diverse data types in forex prediction.

5 Conclusion

This research effectively demonstrated the improvement of forex prediction accuracy by integrating sentiment analysis derived from LLMs with technical and fundamental data analyses. By employing LLMs such as GPT-4 and GEMINI Advanced, this study could leverage unlabelled financial news data to generate sentiment scores combined with traditional forex market data. Using Spearman correlation coefficients to assess the alignment between sentiment scores and the Rate of Change (ROC_9) indicators was pivotal. This analysis helped determine the most effective LLM for each currency pair based on how closely sentiment changes mirrored market movements. For instance, the selected LLMs exhibited varying degrees of correlation with the ROC_9, guiding the choice of sentiment analysis model for integration into the forex prediction framework, as shown in Figure 30.

The findings, as evidenced in Tables 2, 3, 4, and 5, indicate that the decoder-only architecture of Transformers exhibits promising results in forex prediction, outperforming the encoder-decoder framework. Moreover, our comparative analysis, as depicted in Figure 29, has demonstrated that technical models outperform those based on fundamental indicators or a combination of both. This could be attributed to the direct reflection of market sentiments and patterns captured in technical data, which these models distill and interpret.

The Ensemble-LSTM-XGB model combined with technical indicators achieved the best RMSE values for most currency pairs for the one-hour data analysis. Specifically, the best-performing models and their RMSE values are as follows: For the EUR/USD pair, the Ensemble-LSTM-XGB with technical indicators had an RMSE of 0.0064, while the Transformer with an encoder-decoder architecture was the worst, with an RMSE of 0.0259, as seen in Table 2. For the NZD/USD pair, the Ensemble-LSTM-XGB with technical indicators had an RMSE of 0.0039, and the Transformer with an encoder-decoder architecture performed the worst with an RMSE of 0.0176. The EUR/NZD pair showed the XGBoost with technical indicators as the best performer with an RMSE of 0.0048, while the Transformer with an encoder-decoder architecture had the highest RMSE of 0.0148. For the AUD/NZD pair, the XGBoost with technical indicators was the best, with an RMSE of 0.0017, and the Transformer with an encoder-decoder architecture was the worst, with an RMSE of 0.0051, as detailed in Table 2.

For the four-hour data analysis, the XGBoost model combined with technical, fundamental, and sentiment indicators showed the lowest RMSE values across various currency pairs. For the EUR/USD pair, the XGBoost with technical, fundamental, and sentiment analysis had an RMSE of 0.0010, while the LSTM with technical, fundamental, and sentiment analysis had the highest RMSE of 0.0020, as shown in Table 5. For the NZD/USD pair, the XGBoost with technical, fundamental, and sentiment analysis had an RMSE of 0.0009, with the LSTM model performing the worst with an RMSE of 0.0042. The EUR/NZD pair performed best with the XGBoost model with an RMSE of 0.0021 and worst with the LSTM model with an RMSE

of 0.0043. For the AUD/NZD pair, the XGBoost with technical, fundamental, and sentiment analysis had the lowest RMSE of 0.0005, while the LSTM model had the highest RMSE of 0.0017.

Most compelling is the revelation from Figure 11 that employing ensemble methods, integrating the strengths of XGBoost and LSTM models through a weighted average technique, significantly improves model robustness and generalisability. Such ensemble models have shown superior performance in our study, underpinning their potential as a robust framework for future forex market analysis. The use of ensemble models in forex price prediction highlights the significant improvement in performance and generalisation of test data. As illustrated in Figure 31 and Figure 32, both XGBoost and LSTM models individually demonstrate robust predictive capabilities. The XGBoost model excels in handling complex datasets and capturing non-linear relationships, while the LSTM model effectively leverages its sequential memory structure to model temporal dependencies.

However, the ensemble model, which combines the strengths of both XGBoost and LSTM, further enhances predictive accuracy and generalisation, as shown in Figure 13. By integrating these models through a weighted average technique, the ensemble model mitigates the individual weaknesses of each model, leading to more accurate and reliable predictions. This approach harnesses the superior generalisation capability of ensemble learning, reducing overfitting and improving performance across various market conditions.

Research and Practical Implications

The results of this study have important implications for both academic research and financial practice. The significant improvement in forex prediction accuracy, particularly through the use of ensemble methods, adds to the growing body of evidence supporting the integration of machine learning and deep learning models in financial forecasting. For researchers, this study presents a robust framework for further exploration of hybrid and ensemble approaches in financial time-series prediction, highlighting the potential for advanced models like LSTM and XGBoost to work in concert for enhanced accuracy.

In practice, the integration of sentiment analysis with technical and fundamental indicators offers traders and financial institutions a more comprehensive toolkit for predicting forex market movements. The demonstrated effectiveness of combining these data sources can help practitioners develop more reliable models that better capture market complexities, enabling them to make more informed trading decisions. Additionally, the demonstrated robustness of ensemble models suggests their use in developing automated trading algorithms that can adapt to diverse market conditions and reduce the risks associated with overfitting.

Future Research Directions

While this study produced promising results, several avenues for future research remain. One area of interest is the inclusion of real-time data processing to assess how these models perform under dynamic market conditions. Furthermore, exploring more advanced sentiment analysis techniques such as those derived from social media or geopolitical analysis

could further enhance prediction accuracy. Future studies could also delve into the potential of combining more sophisticated ensemble techniques, such as stacking or blending, to improve performance.

Expanding the range of economic and geopolitical factors considered in the models would offer additional insights into forex price movements, potentially leading to even more precise predictions. This research lays the groundwork for future advancements in the field of financial forecasting, pushing the boundaries of machine learning and deep learning applications in trading and decision-making.

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Appendix

This section provides detailed code snippets used in the study to implement machine learning and deep learning models for predicting forex prices. These code snippets offer a comprehensive view of the technical implementation processes, including data preprocessing, model training, and evaluation.

Firstly, Figure 33 shows the code for splitting and scaling the data, which is crucial in preparing the dataset for model training. This is followed by implementing the LSTM model for time series forecasting, illustrated in Figure 34. The subsequent figure, Figure 35, details the code for implementing the XGBoost model for predicting forex prices.

The implementation of the Transformer Encoder and Decoder model, which is used for sequence modeling, is divided into two parts. Part one of the code is shown in Figure 36, and part two in Figure 37. Similarly, the Transformer Decoder model code is also split into two parts, with part one presented in Figure 38 and part two in Figure 39.

In addition, the appendix includes the code for the Ensemble model that combines LSTM and XGBoost. This implementation is also divided into two parts: Figure 40 shows part one, and Figure 41 shows part two.

Furthermore, the appendix contains figures that depict the training and validation loss over epochs for various models applied to the EUR/USD currency pair. Specifically, Figure 42 describes the training and validation loss for the LSTM_TM_FM model, Figure 43 shows the loss for the TRANSFORMER_EN_DE_TM_FM model, and Figure 44 presents the loss for the TRANSFORMER_DE_TM_FM model.

These figures collectively provide a clear and detailed overview of this study's coding processes and model performance evaluation metrics, demonstrating the robust methodologies applied in predicting forex prices.

LSTM_TM_FM (EUR/USD 1D)

```
In [221]: 1 # Calculate the number of rows for each split
2 total_rows = len(df)
3 train_size = int(0.75 * total_rows)
4 val_size = int(0.15 * total_rows)
5 test_size = total_rows - train_size - val_size
6
7 # Split the data
8 train_data = df.head(train_size)
9 val_data = df.iloc[train_size: train_size + val_size]
10 test_data = df.tail(test_size)
11
12 # Display the shape of each split
13 print("Training Data Shape:", train_data.shape)
14 print("Validation Data Shape:", val_data.shape)
15 print("Test Data Shape:", test_data.shape)
16
17
18 Training Data Shape: (3729, 28)
19 Validation Data Shape: (745, 28)
20 Test Data Shape: (499, 28)
```

```
In [222]: 1 from sklearn.preprocessing import StandardScaler
2
3 # Initialize scalers for input features and target
4 scaler_x = StandardScaler()
5 scaler_y = StandardScaler()
6
7 # Scale the features (all columns except 'CLOSE')
8 train_x_scaled = scaler_x.fit_transform(train_data.drop(columns=['CLOSE']))
9 val_x_scaled = scaler_x.transform(val_data.drop(columns=['CLOSE']))
10 test_x_scaled = scaler_x.transform(test_data.drop(columns=['CLOSE']))
11
12 # Scale the target ('CLOSE')
13 train_y_scaled = scaler_y.fit_transform(train_data[['CLOSE']])
14 val_y_scaled = scaler_y.transform(val_data[['CLOSE']])
15 test_y_scaled = scaler_y.transform(test_data[['CLOSE']])
```

Figure 33: Code for the split and scale of the data.

```
In [223]: 1 def create_dataset(X, y, time_step=10):
2     Xs, ys = [], []
3     for i in range(len(X) - time_step):
4         Xs.append(X[i:i+time_step])
5         ys.append(y[i + time_step])
6     return np.array(Xs), np.array(ys)
7
8 time_step = 10
9 X_train, y_train = create_dataset(train_x_scaled, train_y_scaled[:, 0], time_step)
10 X_val, y_val = create_dataset(val_x_scaled, val_y_scaled[:, 0], time_step)
11 X_test, y_test = create_dataset(test_x_scaled, test_y_scaled[:, 0], time_step)
```

```
In [224]: 1 model = Sequential([
2     LSTM(450, return_sequences=True, input_shape=(time_step, X_train.shape[2]), kernel_regularizer=l2(0.02)), # L2 regularization
3     Dropout(0.2),
4     BatchNormalization(),
5     LSTM(200, kernel_regularizer=l2(0.02)), # L2 regularization
6     Dropout(0.1),
7     BatchNormalization(),
8     Dense(1, kernel_regularizer=l2(0.02)) # L2 regularization
9 ])
10
```

Figure 34: Code for the implementation of the LSTM model used for time series forecasting

```

In [54]: 1 from sklearn.model_selection import train_test_split
2
3 # Split the data into training and test sets
4 X_train, X_test, y_train, y_test = train_test_split(
5     df.drop(columns=['CLOSE']),
6     df['CLOSE'],
7     test_size=0.20, # Allocating 20% of the data for testing
8     random_state=42, # For reproducibility
9     shuffle=False # set to False data is a time series
10 )
11
12

In [564]: 1 from sklearn.preprocessing import StandardScaler
2
3 # Initialize the scaler
4 scaler_x = StandardScaler()
5 scaler_y = StandardScaler()
6
7 # Fit the scaler on the training data and transform the training data
8 X_train_scaled = scaler_x.fit_transform(X_train)
9 y_train_scaled = scaler_y.fit_transform(y_train.values.reshape(-1, 1))
10
11 # Use the already fitted scaler to transform the test data
12 X_test_scaled = scaler_x.transform(X_test)
13 y_test_scaled = scaler_y.transform(y_test.values.reshape(-1, 1))

In [565]: 1 import xgboost as xgb
2 from sklearn.model_selection import GridSearchCV
3 from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error
4
5 # Initialize XGBoost regressor
6 xgb_model = xgb.XGBRegressor(objective='reg:squarederror')
7
8 # Define the parameter grid
9 param_grid = {
10     'max_depth': [3, 4, 5, 10],
11     'learning_rate': [0.0001, 0.001, 0.01, 0.2],
12     'n_estimators': [100, 200, 300],
13     'subsample': [0.8, 0.9, 1],
14     'colsample_bytree': [0.3, 0.7, 1],
15     'gamma': [0, 0.1, 0.2],
16     'reg_lambda': [1, 1.5, 2]
17 }
18
19 # Set up GridSearchCV
20 grid_search = GridSearchCV(
21     estimator=xgb_model,
22     param_grid=param_grid,
23     scoring='neg_mean_squared_error', # You can change this to another scoring metric
24     cv=5,
25     verbose=2,
26     n_jobs=-1
27 )

```

Figure 35: Code for the implementation of the XGBoost model used for predicting forex prices.

TRANSFORMERS_EN_DE_TM_FM (EURUSD_1D)

```
In [275]: 1 from sklearn.preprocessing import StandardScaler
2 import numpy as np
3
4 |
5 features = df.drop(columns=['CLOSE'])
6 target = df[['CLOSE']]
7
8 # Initialize the StandardScaler
9 scaler_x = StandardScaler()
10 scaler_y = StandardScaler()
11
12 # Fit and transform the features
13 scaled_features = scaler_x.fit_transform(features)
14
15 # Fit and transform the target
16 scaled_target = scaler_y.fit_transform(target)
17
18 # Convert scaled data back to DataFrame for easy handling
19 scaled_features_df = pd.DataFrame(scaled_features, columns=features.columns)
20 scaled_target_df = pd.DataFrame(scaled_target, columns=['CLOSE'])
21
22 # Split the scaled data
23 train_features = scaled_features_df.head(train_size)
24 val_features = scaled_features_df.iloc[train_size: train_size + val_size]
25 test_features = scaled_features_df.tail(test_size)
26
27 train_target = scaled_target_df.head(train_size)
28 val_target = scaled_target_df.iloc[train_size: train_size + val_size]
29 test_target = scaled_target_df.tail(test_size)
30
31 # Create sequences for the Transformer model
32 def create_sequences(features, target, sequence_length):
33     X, y = [], []
34     for i in range(len(features) - sequence_length):
35         X.append(features.iloc[i:i+sequence_length].values)
36         y.append(target.iloc[i:i+sequence_length].values)
37     return np.array(X), np.array(y)
38
39 sequence_length = 10 # Define the sequence length
40
41 # Prepare the sequences
42 X_train, y_train = create_sequences(train_features, train_target, sequence_length)
43 X_val, y_val = create_sequences(val_features, val_target, sequence_length)
44 X_test, y_test = create_sequences(test_features, test_target, sequence_length)
45
46 # Now, X_train, X_val, and X_test are ready to be used as inputs to your Transformer model
47 # y_train, y_val, and y_test are the corresponding targets
48
```

Figure 36: Code for the implementation of the Transformer Encoder and Decoder model used for sequence modeling part-1.


```

In [278]: 1 import tensorflow as tf
2 from tensorflow.keras.models import Model
3 from tensorflow.keras.layers import Input, Dense, LayerNormalization, MultiHeadAttention, Dropout, TimeDistributed, LeakyReLU
4 from tensorflow.keras.regularizers import l2
5
6 def transformer_block(inputs, head_size, num_heads, ff_dim, dropout, l2_reg):
7     # Normalization and Attention
8     attention_out = MultiHeadAttention(key_dim=head_size, num_heads=num_heads, dropout=dropout)(inputs, inputs)
9     attention_out = Dropout(dropout)(attention_out)
10    attention_out = LayerNormalization(epsilon=1e-6)(attention_out + inputs) # Add & Norm
11
12    # Feed Forward Part
13    ff_out = Dense(ff_dim, activation="relu", kernel_regularizer=l2(l2_reg))(attention_out)
14    ff_out = Dropout(dropout)(ff_out)
15    ff_out = Dense(inputs.shape[-1], kernel_regularizer=l2(l2_reg))(ff_out)
16    ff_out = LayerNormalization(epsilon=1e-6)(ff_out + attention_out) # Add & Norm
17    return ff_out
18
19 def transformer_encoder(inputs, head_size, num_heads, ff_dim, dropout, l2_reg, num_layers):
20    x = inputs
21    for _ in range(num_layers):
22        x = transformer_block(x, head_size, num_heads, ff_dim, dropout, l2_reg)
23    return x
24
25 def transformer_decoder(encoder_outputs, inputs, head_size, num_heads, ff_dim, dropout, l2_reg, num_layers):
26    x = inputs
27    for _ in range(num_layers):
28        # Decoder Attention Block
29        attention_out = MultiHeadAttention(key_dim=head_size, num_heads=num_heads, dropout=dropout)(x, encoder_outputs, encoder_outputs)
30        attention_out = Dropout(dropout)(attention_out)
31        attention_out = LayerNormalization(epsilon=1e-6)(attention_out + x) # Add & Norm
32
33        # Feed Forward Part
34        ff_out = Dense(ff_dim, activation="relu", kernel_regularizer=l2(l2_reg))(attention_out)
35        ff_out = Dropout(dropout)(ff_out)
36        ff_out = Dense(inputs.shape[-1], kernel_regularizer=l2(l2_reg))(ff_out)
37        x = LayerNormalization(epsilon=1e-6)(ff_out + attention_out) # Add & Norm
38    return x
39
40 def build_transformer_model(sequence_length, features, num_layers=3, l2_reg=0.04, dropout=0.4):
41    encoder_inputs = Input(shape=(sequence_length, features))
42    encoder_outputs = transformer_encoder(encoder_inputs, head_size=64, num_heads=8, ff_dim=64, dropout=dropout, l2_reg=l2_reg)
43
44    # Prepare decoder inputs, initialized to zeros
45    decoder_inputs = tf.zeros_like(encoder_inputs)
46
47    decoder_outputs = transformer_decoder(encoder_outputs, decoder_inputs, head_size=64, num_heads=6, ff_dim=64, dropout=dropout, l2_reg=l2_reg)
48
49    outputs = TimeDistributed(Dense(1, activation="linear", kernel_regularizer=l2(l2_reg)))(decoder_outputs)
50
51    model = Model(inputs=encoder_inputs, outputs=outputs)
52
53    model.compile(optimizer="adam", loss="mean_squared_error")
54
55    return model
56
57

```

Figure 37: Code for the implementation of the Transformer Encoder and Decoder model used for sequence modeling part-2.

TRANSFORMERS_DE_TM_FM (EURUSD_1D)

```
In [105]: 1 from sklearn.preprocessing import StandardScaler
2 import numpy as np
3
4 # Assuming 'df' is your DataFrame and 'CLOSE' is the target variable
5 features = df.drop(columns=['CLOSE'])
6 target = df[['CLOSE']]
7
8 # Initialize the StandardScaler
9 scaler_x = StandardScaler()
10 scaler_y = StandardScaler()
11
12 # Fit and transform the features
13 scaled_features = scaler_x.fit_transform(features)
14
15 # Fit and transform the target
16 scaled_target = scaler_y.fit_transform(target)
17
18 # Convert scaled data back to DataFrame for easy handling
19 scaled_features_df = pd.DataFrame(scaled_features, columns=features.columns)
20 scaled_target_df = pd.DataFrame(scaled_target, columns=['CLOSE'])
21
22 # Split the scaled data
23 train_features = scaled_features_df.head(train_size)
24 val_features = scaled_features_df.iloc[train_size: train_size + val_size]
25 test_features = scaled_features_df.tail(test_size)
26
27 train_target = scaled_target_df.head(train_size)
28 val_target = scaled_target_df.iloc[train_size: train_size + val_size]
29 test_target = scaled_target_df.tail(test_size)
30
31 # Create sequences for the Transformer model
32 def create_sequences(features, target, sequence_length):
33     X, y = [], []
34     for i in range(len(features) - sequence_length):
35         X.append(features.iloc[i:i+sequence_length].values)
36         y.append(target.iloc[i+sequence_length].values)
37     return np.array(X), np.array(y)
38
39 sequence_length = 8 # Define the sequence length
40
41 # Prepare the sequences
42 X_train, y_train = create_sequences(train_features, train_target, sequence_length)
43 X_val, y_val = create_sequences(val_features, val_target, sequence_length)
44 X_test, y_test = create_sequences(test_features, test_target, sequence_length)
45
46 # Now, X_train, X_val, and X_test are ready to be used as inputs to your Transformer model
47 # y_train, y_val, and y_test are the corresponding targets
48
```

Figure 38: Code for the implementation of the Transformer Decoder model used for sequence modeling part-1.

```

In [108]: 1 def transformer_block(inputs, head_size, num_heads, ff_dim, dropout, l2_reg, look_ahead_mask):
2         # Normalization and Attention with Look-ahead mask
3         attention_out = MultiHeadAttention(key_dim=head_size, num_heads=num_heads, dropout=dropout)(
4             inputs, inputs, inputs, attention_mask=look_ahead_mask)
5         attention_out = Dropout(dropout)(attention_out)
6         attention_out = LayerNormalization(epsilon=1e-6)(attention_out + inputs) # Add & Norm
7
8         # Feed Forward Part
9         ff_out = Dense(ff_dim, activation="relu", kernel_regularizer=l2(l2_reg))(attention_out)
10        ff_out = Dropout(dropout)(ff_out)
11        ff_out = Dense(inputs.shape[-1], kernel_regularizer=l2(l2_reg))(ff_out)
12        ff_out = LayerNormalization(epsilon=1e-6)(ff_out + attention_out) # Add & Norm
13        return ff_out
14
15    def build_transformer_model(sequence_length, features, num_layers=4, l2_reg=0.03, dropout=0.5):
16        encoder_inputs = Input(shape=(sequence_length, features))
17        look_ahead_mask = create_look_ahead_mask(sequence_length)
18
19        x = encoder_inputs
20        for _ in range(num_layers):
21            x = transformer_block(x, head_size=64, num_heads=6, ff_dim=32, dropout=dropout, l2_reg=l2_reg, look_ahead_mask=look_ahead_mask)
22
23        outputs = TimeDistributed(Dense(1, activation="linear", kernel_regularizer=l2(l2_reg)))(x)
24
25        model = Model(inputs=encoder_inputs, outputs=outputs)
26
27        model.compile(optimizer="adam", loss="mean_squared_error")
28
29        return model
30

```

```

In [109]: 1 import tensorflow as tf
2 from tensorflow.keras.models import Model
3 from tensorflow.keras.layers import Input, Dense, LayerNormalization, MultiHeadAttention, Dropout, TimeDistributed, LeakyReLU
4 from tensorflow.keras.regularizers import l2
5
6 # Define the initial Learning rate
7 initial_learning_rate = 0.0002
8
9 # Setup EarlyStopping
10 early_stopping = EarlyStopping(
11     monitor='val_loss', # Monitor the validation loss
12     patience=60, # Stop after 10 epochs with no improvement
13     restore_best_weights=True # Restore model weights from the epoch with the best value of the monitored metric
14 )
15
16 # Setup ReduceLROnPlateau
17 reduce_lr = ReduceLROnPlateau(
18     monitor='val_loss',
19     factor=0.5, # Reduce Learning rate by half
20     patience=20,
21     min_lr=0.0000001 # Set a minimum Learning rate
22 )

```

Figure 39: Code for the implementation of the Transformer Decoder model used for sequence modeling part-2.

```

In [116]: 1 import numpy as np
          2 from sklearn.model_selection import train_test_split
          3 from sklearn.preprocessing import StandardScaler, MinMaxScaler
          4 from keras.models import Sequential
          5 from keras.layers import LSTM, Dense, Dropout
          6 from xgboost import XGBRegressor
          7 from sklearn.metrics import mean_squared_error, r2_score
          8
          9 X = df_tm.drop('CLOSE', axis=1)
         10 y = df_tm['CLOSE']
         11
         12 # Split the data into training and testing sets
         13 X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42, test_size=0.2, shuffle=False)
         14

In [109]: 1 from sklearn.preprocessing import StandardScaler
          2 from sklearn.preprocessing import StandardScaler
          3
          4 # Initialize the scaler
          5 scaler_x = StandardScaler()
          6 scaler_y = StandardScaler()
          7
          8 # Fit the scaler on the training data and transform the training data
          9 X_train_scaled = scaler_x.fit_transform(X_train)
         10 y_train_scaled = scaler_y.fit_transform(y_train.values.reshape(-1, 1)) # Reshape y to a 2D array for the scaler
         11
         12 # Use the already fitted scaler to transform the test data
         13 X_test_scaled = scaler_x.transform(X_test)
         14 y_test_scaled = scaler_y.transform(y_test.values.reshape(-1, 1))
         15

In [110]: 1 from keras.models import load_model
          2
          3 # Load LSTM model
          4 lstm_model = load_model('LSTM_TM_EURUSD_1D.keras')
          5
          6 # Load XGBoost model
          7 import joblib
          8 xgb_model = joblib.load('tm_xgboost_model.joblib')
          9

In [111]: 1 # Make predictions with XGBoost
          2 y_pred_xgb = xgb_model.predict(X_test_scaled)
          3
          4
          5 y_pred_xgb_rescaled = scaler_y.inverse_transform(y_pred_xgb.reshape(-1, 1))
          6

```

Figure 40: Code for the implementation of the Ensemble model combining LSTM and XGBoost part-1.

```

In [114]: 1 def create_dataset(X, y, time_step=10):
2         Xs, ys = [], []
3         for i in range(len(X) - time_step):
4             Xs.append(X[i:(i+time_step)])
5             ys.append(y[i + time_step])
6         return np.array(Xs), np.array(ys)
7
8 time_step = 10
9 X_train, y_train = create_dataset(X_train_scaled, y_train_scaled[:, 0], time_step)
10 X_test, y_test = create_dataset(X_test_scaled, y_test_scaled[:, 0], time_step)
11

In [115]: 1 predicted_y_scaled = lstm_model.predict(X_test)
2 predicted_y = scaler_y.inverse_transform(predicted_y_scaled).flatten()
3
4
5 actual_y_full = scaler_y.inverse_transform(y_test_scaled).flatten() # Flatten to make 1D
6 actual_y = actual_y_full[10:]
7

31/31 [=====] - 1s 22ms/step

In [117]: 1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 y_test_adjusted = y_test[-len(predicted_y):]
5
6 plt.figure(figsize=(18, 7))
7
8 # Plot the actual close prices for the test period
9 plt.plot(y_test_adjusted.index, y_test_adjusted, label='Actual Close Price - Testing', color='green')
10
11 # Plot the predicted close prices for the test period
12 plt.plot(y_test_adjusted.index, predicted_y, label='Predicted Close Price', linestyle='--', color='red')
13
14 plt.title('Comparison of Actual and Predicted Close Prices for the Test Period')
15 plt.xlabel('Date')
16 plt.ylabel('Close Price')
17 plt.legend()
18 plt.show()
19

```

Figure 41: Code for the implementation of the Ensemble model combining LSTM and XGBoost part-2.

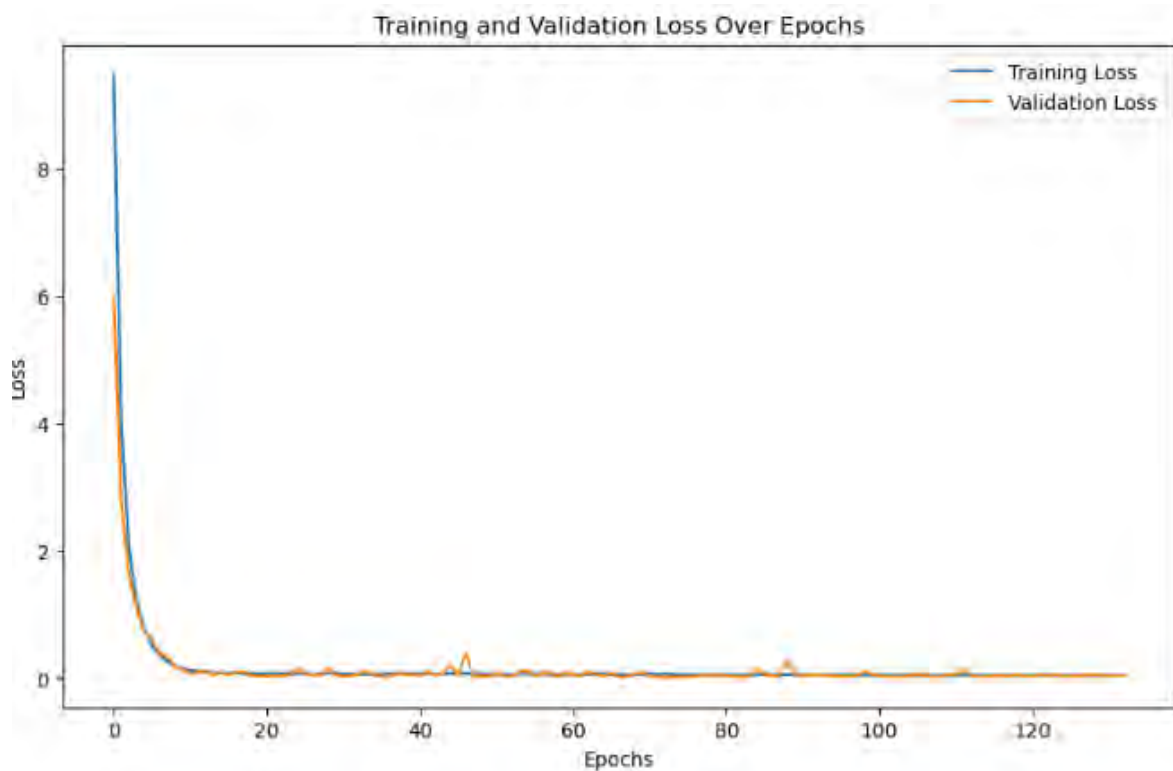


Figure 42: Training and validation loss over epochs for the LSTM_TM_FM model for EUR/USD.

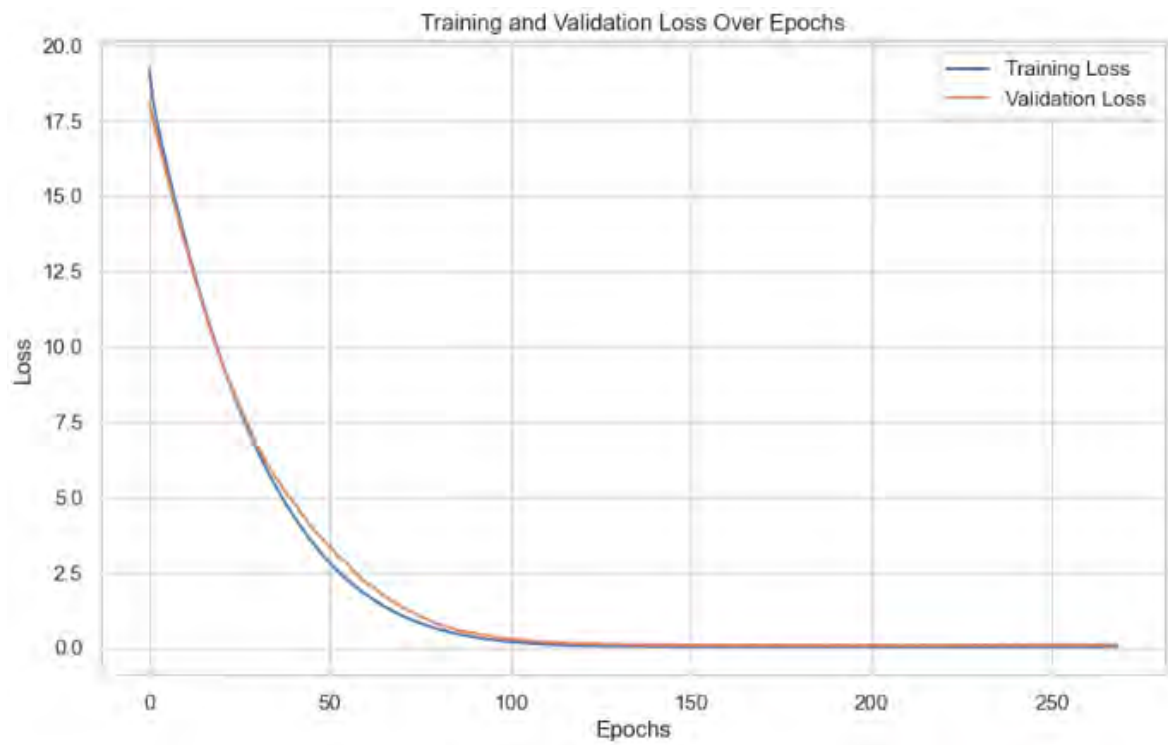


Figure 43: Training and validation loss over epochs for the TRANSFORMER_EN_DE_TM_FM model for EUR/USD.

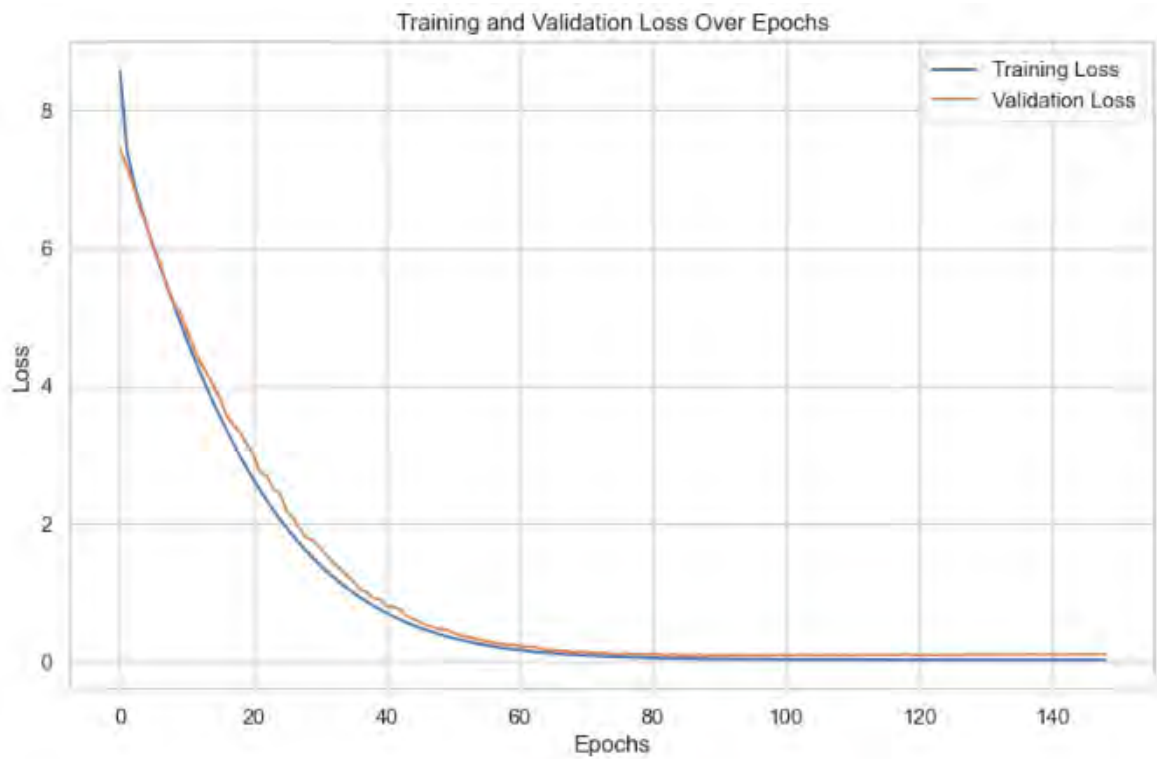


Figure 44: Training and validation loss over epochs for the TRANSFORMER_DE_TM_FM model for EUR/USD.